# Properties of Matter (Fluid Mechanics)

**Fluid:** Fluid is a substance which can flow.

- liquids → incompressible
- gases → compressible

* Fluids cannot support "shearing force".

*

Fluid Mechanics

- **Static** (Rest)
  - we can study fluid properties by knowing
    1. pressure
    2. pascal's law
    3. Archimedi's principle
- **dynamics** (motion)
  - we can study by knowing
    1. eqn of continuity
    2. Bernouli's theorem

**Density of fluids**

$$\text{Density } (\rho) = \frac{\text{mass}}{\text{volume}} = \frac{m}{V}$$

Units: → $gm/cm^3$ (C.G.S)

$kg/m^3$ (S.I)

$$1 gm/cm^3 = \frac{10^{-3} kg}{(10^{-2})^3 m^3} = 10^3 kg/m^3.$$

* For immisible liquids,

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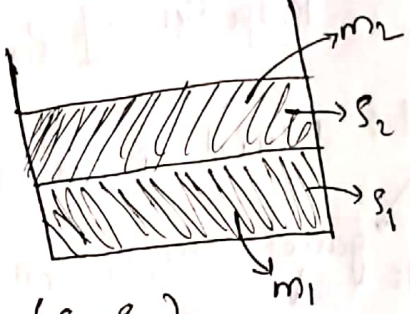


Density of mixture :-

$$\rho_{mix} = \frac{\text{total mass}}{\text{total volume}}$$

$$= \frac{m_1+m_2}{V_1+V_2} \quad \text{— (1)}$$

$$\Rightarrow \boxed{\rho_{mix} = \frac{V_1\rho_1 + V_2\rho_2}{V_1+V_2}}$$

Case-(1) :

If $V_1 = V_2$ then $\boxed{\rho_{mix} = \frac{\rho_1+\rho_2}{2}}$ * — A.M

In general, $\rho_{mix} = \frac{\rho_1+\rho_2+\rho_3+---+\rho_n}{n}$

Case-(2) :

If $m_1 = m_2$ then,

$$\rho_{mix} = \frac{m_1+m_2}{V_1+V_2} = \frac{m_1+m_2}{\frac{m_1}{\rho_1}+\frac{m_2}{\rho_2}} = \frac{2}{\frac{1}{\rho_1}+\frac{1}{\rho_2}}$$

$$\boxed{\frac{2}{\rho_{mix}} = \frac{1}{\rho_1}+\frac{1}{\rho_2}}$$ * — H.M

In general for n-liquids

$$\frac{n}{\rho_{mix}} = \frac{1}{\rho_1}+\frac{1}{\rho_2}+---+\frac{1}{\rho_n}$$

Eg: when equal volumes of two metals are mixed together the specific gravity of alloy is 4. when equal masses of the same two metals are mixed to gether, the specific gravity of two alloy

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is now 3. Find gravity of each metal.

Sol:

$$\rho_{mix} = \frac{\rho_1 + \rho_2}{2} = 4 \quad -(1)$$

$$\rho_{mix} = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = 3 \quad -(2)$$

(1) $\Rightarrow$ $\rho_1 + \rho_2 = 8$ $\quad -(3)$

(2) $\Rightarrow$ $2\rho_1\rho_2 = 3(8)$

$$\rho_1\rho_2 = 12$$

$$(\rho_1 - \rho_2)^2 = (\rho_1 + \rho_2)^2 - 4\rho_1\rho_2$$

$$= (8)^2 - 4(12)$$

$$= 64 - 48$$

$$(\rho_1 - \rho_2)^2 = 16$$

$$\rho_1 - \rho_2 = 4 \quad -(4)$$

(3) + (4) $\Rightarrow$ $2\rho_1 = 12$

$$\boxed{\rho_1 = 6}$$

(3) − (4) $\Rightarrow$ $2\rho_2 = 4$

$$\boxed{\rho_2 = 2}$$

★ specific gravity $= \frac{\text{den of body}}{\text{den of water}}$

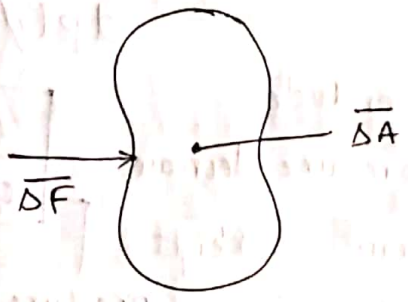
$d_{water} = 1\,gm/cm^3 = 10^3\,kg/m^3$

∴ den of body = (sp. gravity) × $d_{water}$

## Pressure:

$$\boxed{P = \frac{\text{force}}{\text{Area}}}$$

$$\boxed{\Delta\vec{F} = P\,\Delta\vec{A}}$$




* P is a scalar.

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Units :→ dyne/cm² (C.G.S)

N/m² (Pascal) (S.I)

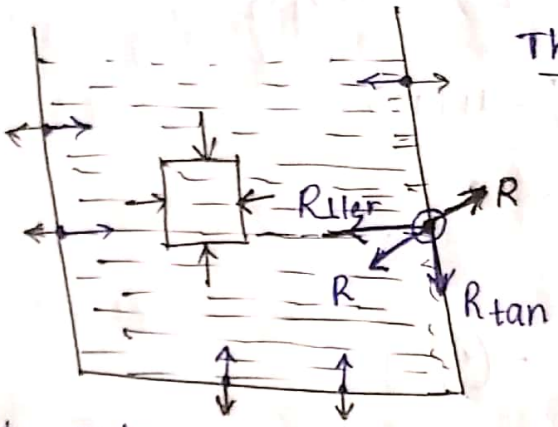
$$1 N/m^2 = \frac{10^5 \text{ dyne}}{(10^2)^2 cm^2} = 10 \text{ dyne}/cm^2$$

* 1 atm = $1.013 \times 10^5$ N/m²

* 1 bar = $10^5$ N/m²

* 1 torr = 1 mm of Hg (height).

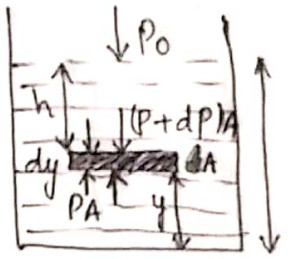
Variation of pressure with depth :




Thrust → force applied by liq on contact surface.

Beaker is at rest on the ground.

Consider a liquid layer.




homogeneous liquid.

$H = h + y \Rightarrow \boxed{h = H - y}$

* since element is rest

$$\vec{F}_{net} = 0$$

$$PA = (P+dp)A + (dm)g$$

$$PA = PA + dPA + (dA)dy \cdot \rho g$$

$$dp(A) = -A(dy)\rho g$$

$$\boxed{\frac{dp}{dy} = -\rho g}$$

* we can conclude that pressure decreases with increase in height.

* graph of P/y

⇒ slope is -ve.

pressure gradient

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$$\Rightarrow \quad dp = -\rho g\, dy$$

$$\int_{p_0}^{p} dp = -\int_{y=y}^{y=H} \rho g\, dy$$

$$p_0 - p = -\rho g (H-y)$$

$$\boxed{p = p_0 + \rho g (H-y)}$$

also can be written as

$$\boxed{p = p_0 + \rho g h} \quad *$$

$h \rightarrow$ depth of a pt from free surface of liq.

* In another way,

$$\boxed{\frac{dp}{dh} = \rho g} \quad *$$

$$* \quad \int_{p_0}^{p} dp = \int_{0}^{h} \rho g\, dh$$

$$\Rightarrow p - p_0 = \rho g h$$

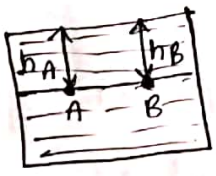
$$\Rightarrow \boxed{p = p_0 + \rho h g}$$

Points:

① If gravity is ignored,

$g = 0 \Rightarrow \boxed{p = p_0}$

②




$h_A = h_B$

$p_A = p_0 + \rho g h_A$

$p_B = p_0 + \rho g h_B$

$\Rightarrow \boxed{p_A = p_B}$ *

hydrostatic paradox.

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③ $P = P_0 + \rho g h$

where $P$ = absolute pressure

$P_0$ = atmospheric pressure

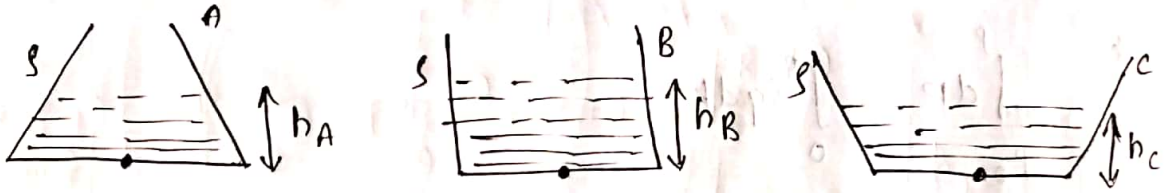
$P - P_0$ = guage pressure

$P - P_0 = \rho g h$ → may be +ve
→ may be -ve
→ may be zero.

④ * Pressure depends on ① $P_0$ ② $\rho$ ③ $g$ ④ $h$.

* pressure doesn't depends on ① amount of fluid
② shape & size of the container.

Eg:




$P_A = P_0 + \rho g h_A$ $P_B = P_0 + \rho g h_B$ $P_C = P_0 + \rho g h_C$

If $h_A = h_B = h_C \Rightarrow$ $\boxed{P_A = P_B = P_C}$

If base area is same $\Rightarrow w_C > w_B > w_A$.

Eg:

$P_A = P_B = P_C = P_D$

$h_A = h_B = h_C = h_D$

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# Pascal's law :

① pressure at a point in the liquid

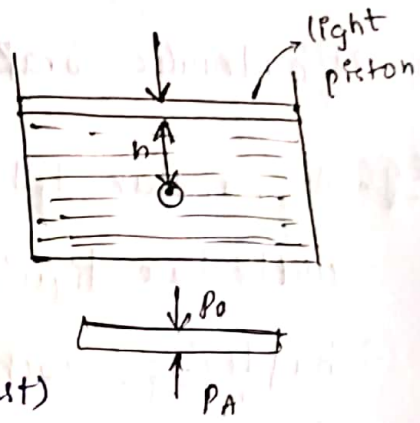
$$P = P_0 + \rho g h$$

$$\Delta P = \Delta (P_0 + \rho g h)$$

$$\Delta P = \Delta P_0 + \Delta (\rho g h)$$

$$\Delta P = \Delta P_0 + \Delta (\rho g h)^{0} \quad (h \text{ is const})$$

* $\boxed{\Delta P = \Delta P_0}$

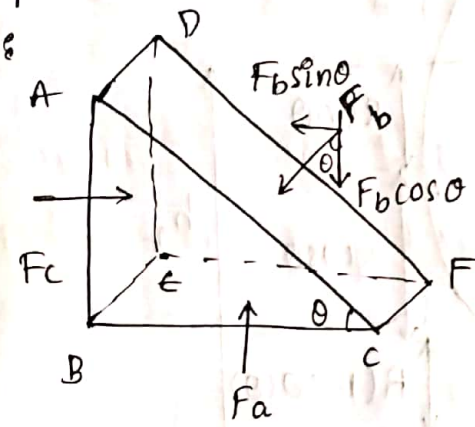



$\boxed{P = P_0}$

* If an external pressure is applied to an enclosed fluid, it is transmitted to every point in the fluid which is undiminished.

* pressure is same at a point in all directions.

Eg:




$AD = BE = CF = \ell$

$AB = AC \sin\theta$

$AC \cos\theta = BC$

* force on face ACDF,

$$F_b = (P_b)(AC)(AD)$$

$$F_b = P_b (AC)(\ell) \quad -①$$

* force on face ABDE,

$$F_c = (P_c)(AB)\ell \quad -②$$

* force on face BCEF,

$$F_a = (P_a)(BC)(\ell) \quad -③$$

now since sample is in equilibrium,

$$F_b \sin\theta = F_c \quad \& \quad F_b \cos\theta = F_a$$

$$P_b (AC) \not{\ell} \sin\theta = P_c (AB) \not{\ell}$$

$$\boxed{P_b = P_c}$$

$$P_b (AC) \not{\ell} \cos\theta = P_a (BC) \not{\ell}$$

$$\boxed{P_b = P_a}$$

clearly $\boxed{P_a = P_b = P_c}$

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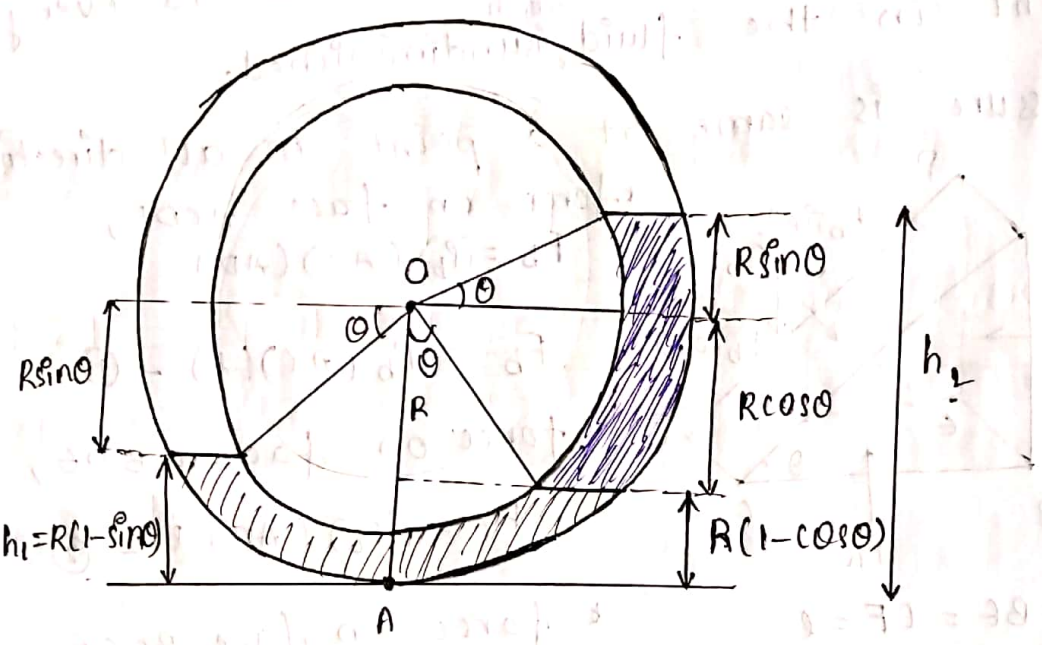
# Applications of Pascal's Law:

1) hydralic lift

2) hydraulic brakes

Eg: A circular tube of uniform cross section is filled with two liquids of densities $\rho_1$ and $\rho_2$ such that half of each liquid occupies a quarter of vol of the tube as shown in figure. If the line joining the free surfaces of the liquid makes an angle $\theta$ with horizontal, then Calculate $\theta$.




Sol:

$h_1 = R - R\sin\theta = R(1-\sin\theta)$

~~$h_2 = R\sin\theta + R\cos\theta + R(1-\cos\theta)$~~

$P_A = P + \rho_1 g h_1 = p + \rho_1 g R(1-\sin\theta) \quad -①$

$P_A = P + \rho_2 g (R\sin\theta + R\cos\theta) + \rho_1 g R(1-\cos\theta) \quad -②$

$① = ② \Rightarrow$ ~~$P + \rho_1 gR(1-\sin\theta) = P + \rho_2 gR(\sin\theta + \cos\theta)$~~

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$$\rho_1 g R(1-\sin\theta) = P + \rho_2 g(R\sin\theta + R\cos\theta) + \rho_1 g R(1-\cos\theta)$$

$$\rho_1(1-\sin\theta) = \rho_2(\sin\theta+\cos\theta) + \rho_1(1-\cos\theta)$$

$$\rho_1(1-\sin\theta-1+\cos\theta) = \rho_2(\sin\theta+\cos\theta)$$

$$\rho_1(\cos\theta-\sin\theta) = \rho_2(\sin\theta+\cos\theta)$$

(÷ with $\cos\theta$ on both sides) $\Rightarrow \rho_1(1-\tan\theta) = \rho_2(\tan\theta+1)$

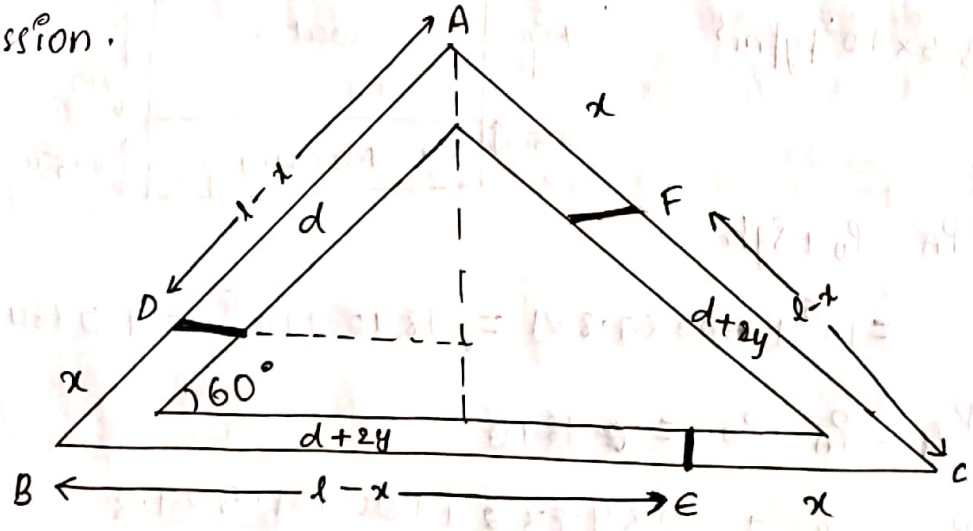
$$\rho_1 - \rho_1\tan\theta = \rho_2\tan\theta + \rho_2$$

$$\rho_1 - \rho_2 = (\rho_1+\rho_2)\tan\theta$$

$$\tan\theta = \frac{\rho_1-\rho_2}{\rho_1+\rho_2}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)}$$

~~Eg~~: A closed tube in the form of an equilateral $\Delta$le of side $\ell$ contains equal volumes of three liquides which do not mix and is placed vertically with it's lowest side horizontal. Find the value of $x$ in the figure, if the densities liquids are in arthmetic progression.




Sol:

$F = PA$ = same $\Rightarrow$ P is same at 'E' from both sides.

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$$P_E = dg(l-x)\sin 60^\circ + (d+2y)gx\sin 60^\circ \quad -①$$

$$P_E = dgx\sin 60^\circ + (d+y)g(l-x)\sin 60^\circ \quad -②$$

① = ②

$$\Rightarrow \frac{dg(l-x)}{g} + \frac{(d+2y)gx}{g} = \frac{dgx}{g} + \frac{(d+y)g(l-x)}{g}$$

$$d(l-x) + (d+2y)x = dx + (d+y)(l-x)$$

$$(d+2y-d)x = (d+y-d)(l-x)$$

$$2y\,x = y(l-x)$$

$$2x = l-x$$

$$3x = l$$

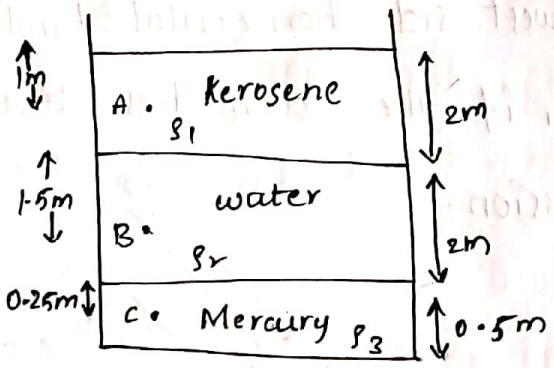
$$\boxed{x = l/3}$$

Eg: Find the absolute pressure and guage pressure at pts A, B, C as shown in figure. Take 1 atm = $10^5$ Pa.

$\rho_1 = 800\ kg/m^3$

$\rho_2 = 10^3\ kg/m^3$

$\rho_3 = 13.6\times 10^3\ kg/m^3$




Soln

① $P_A = P_0 + \rho g h$

$= 10^5 + 800\times 9.8\times 1 = 7840 + 10^5 = 1.0784\times 10^5$

$P_G = P_A - P_0 = 7840$

② $P_B = P_0 + 800\times 9.8\times 2 + 1000\times 1.5\times 9.8$

$= 10^5 + 30380 = 1.3038\times 10^5$

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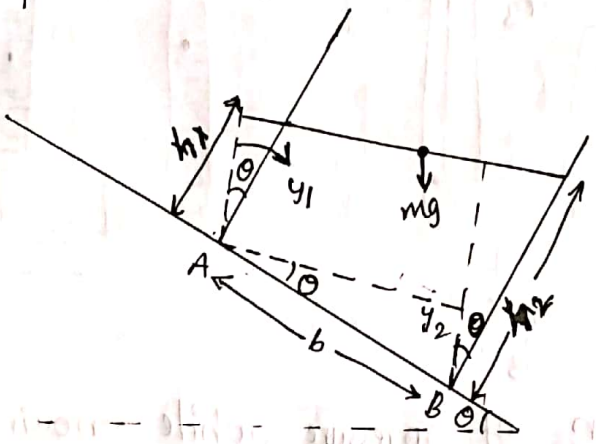
$P_4 = P_B - P_0 = 30380$

③ $P_C = P_0 + 800 \times 9.8 \times 2 + 1000 \times 9.8 \times 2 + 13.6 \times 10^3 \times 9.8 \times 0.25$

$= 10^5 + 68600 = 1.686 \times 10^5$

$P_4 = P_C - P_0 = 68600$

Eg: Find pressure difference b/w A & B.




$\sin\theta = \frac{y_2 - y_1}{b}$

$y_2 - y_1 = b\sin\theta$

$P_A = P_0 + \rho g y_1$ & $P_B = P_0 + \rho g y_2$

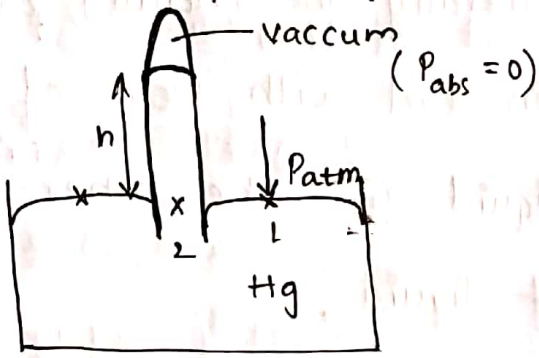
$P_B - P_A = \rho g (y_2 - y_1)$

$= \rho g (b\sin\theta)$

$P_B - P_A = \rho g b \sin\theta$

* "net force" is always ⊥ler to free surface.

Barometer:




$P_1 = P_2$

$P_{atm} = h\rho g$

$P_{atm} = \frac{76}{100} \times 13.6 \times 10^3 \times 9.8$

$= 1.013 \times 10^5$ Pa

to find height of atmosphere:

$P_{atm} = \rho_{air} \times g \times h \Rightarrow h = \frac{P_{atm}}{\rho_{air} \times g} = \frac{1.013 \times 10^5}{1.3 \times 9.8}$

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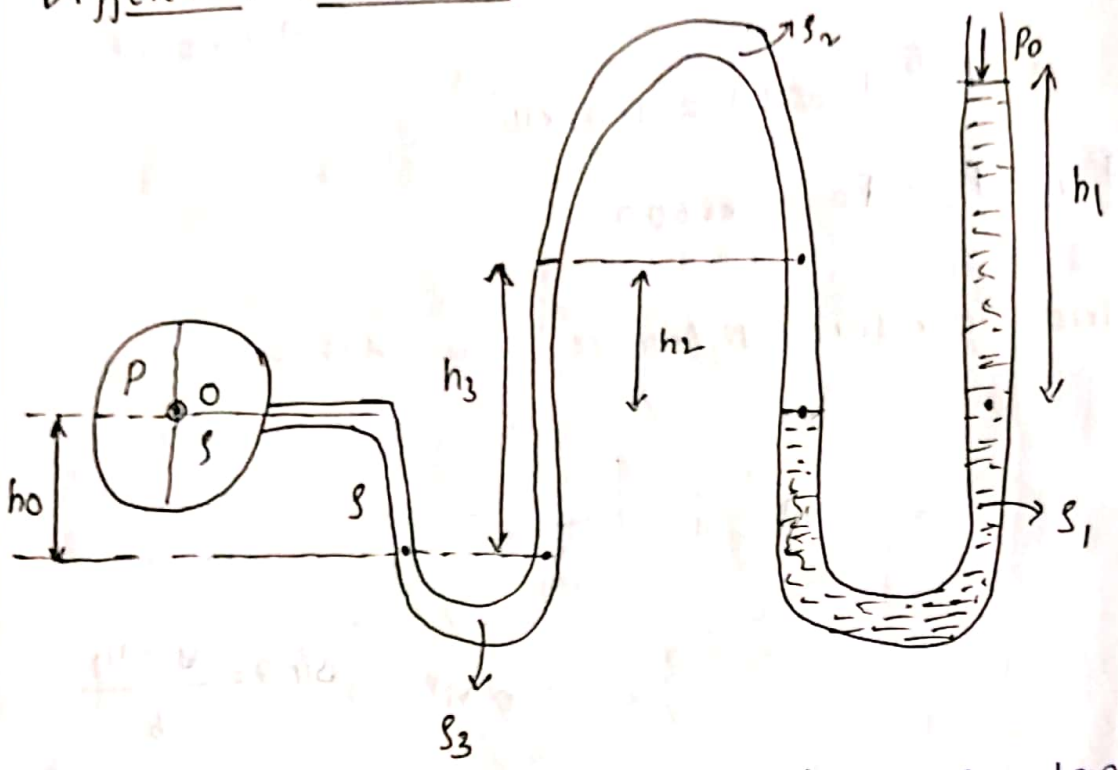
$h \approx 8\,km$

we assume density of air is constant.

Differential manometer:




Taking +ve sign of pressure while moving down.

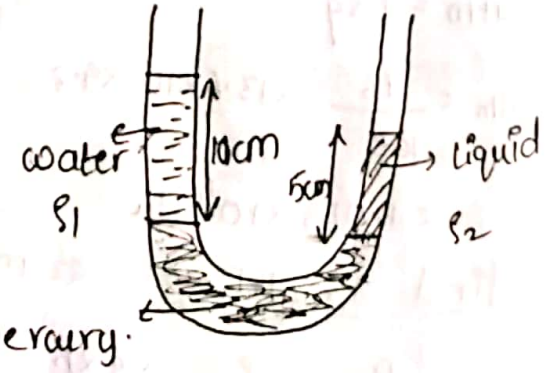
$$P_0 + \rho_1 g h_1 - \rho_2 g h_2 + \rho_3 g h_3 - \rho g h_0 = P$$

$$P - P_0 = \rho_1 g h_1 - \rho_2 g h_2 + \rho_3 g h_3 - \rho g h_0 = \text{guage pressure}$$

Eg: A U-tube contains water and liquid seperated by mercury. The mercury columns in the two arms are in level with 10cm of water in one arm & 5 cm of liquid in the other. what is the specific gravity of liquid?

$\rho_1 = 1\ g/cc$

$\rho_2 = ?$

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$$P_A = P_B$$

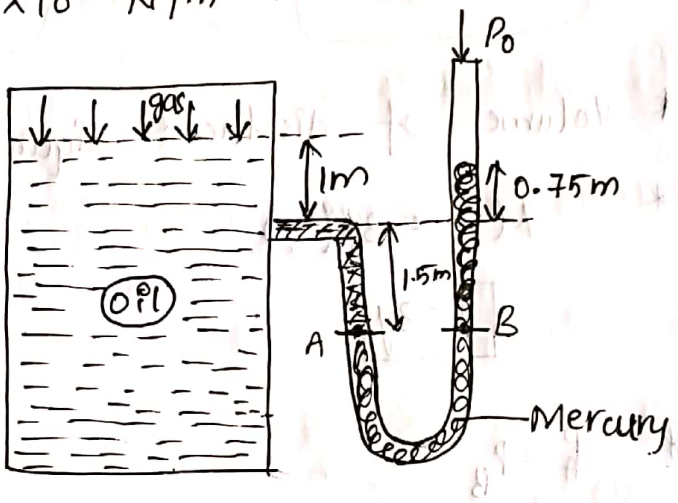
$$P_0 + \rho_1 g h_1 = P_0 + \rho_2 g h_2$$

$$\rho_1 h_1 = \rho_2 h_2$$

$$1 \times 10 = \rho_2 \times 5$$

$$\rho_2 = 2 \text{ g/cm}^3$$

Eg: what is the absolute and gauge pressure of the gas above the liquid surface in the tank shocon in fig. Density of oil = 820 kg/m³, density of mercury = $13.6 \times 10^3$ kg/m³. Given latm pressure = $1.01 \times 10^5$ N/m².




$$P_A = P_B$$

~~$P_g + 820 \times 2.5 \times 9.8$~~

~~$= P_0 + 2.25 \times 13.6 \times 9.8$~~

$$P_g + P_{oil} = P_0 + P_{Mer}$$

$$P_g + 820 \times 2.5 \times 9.8 = 1.01 \times 10^5 + 2.25 \times 13.6 \times 9.8 \times 10^3$$

$$P_g = 1.01 \times 10^5 + 9.8 (13600 \times 2.25 - 820 \times 2.5)$$

$$P_g = 1.01 \times 10^5 + 2.7979 \times 10^5$$

$$P_g = 3.8079 \times 10^5 \text{ Pa}$$

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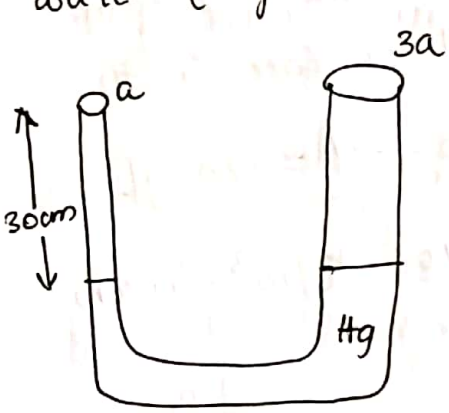
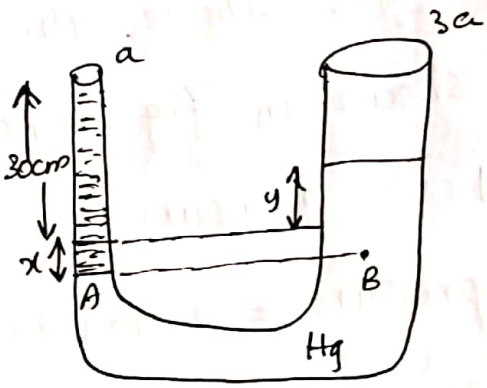
Eg: A U-tube in which the cross-sectional area of the limb on the left is one third of the limb on the right contains mercury (density 13.6 g/cc). The level of the mercury in the narrow limb extends to a distance of 30cm from the upper end of the tube. What will be the rise in the level of mercury in the right limb. If the left limb is filled to the top with water (neglect surface tension effects).




* Volume of displaced liquid is same.

* $a(x) = 3a(y)$

$$\boxed{x = 3y}$$

* $P_A = P_B$

$$P_0 + 1 \times (30+x) \times 9.8 = P_0 + 13.6 \times (x+y) \times 9.8$$

$$30 + x = 13.6 (x+y)$$

$$30 + x = 13.6 \times 4 \times y$$

$$30 + 3y = 54.4y$$

$$51.4y = 30$$

$$\boxed{y = 0.583}$$

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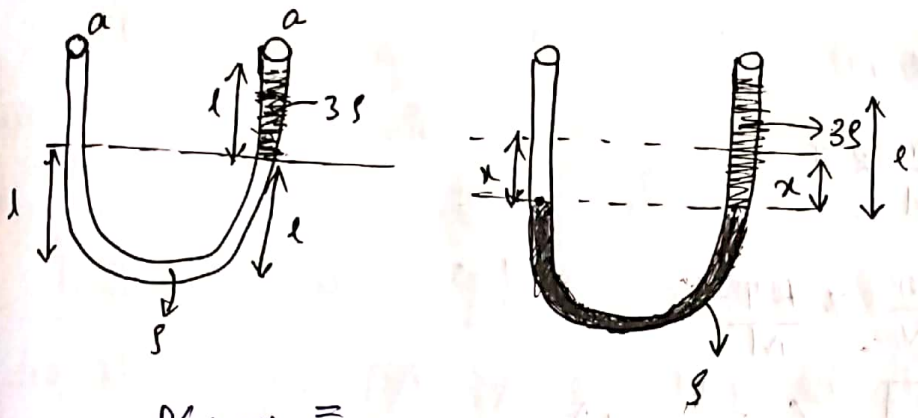


$P_0 + 3\rho g l = P_0 + \rho g(2x)$

$3l = 2x$

$$\boxed{x = 3l/2}$$

* If the fluid is compressible then density is not constant.

For example,

assume, $\rho \propto P \Rightarrow \boxed{\rho = kP}$ (at certain height)

$\therefore \rho_0 = kP_0$ (on earth's atmosphere)

$\Rightarrow \boxed{\rho = \frac{\rho_0}{P_0}(P)}$

we have,

$$\boxed{dp = -\rho g\, dy}$$

$$dp = -\frac{\rho_0}{P_0} P g\, dy$$

$$\int_{P_0}^{P} \frac{dP}{P} = -\frac{\rho_0}{P_0} g \int_0^h dy$$

$$\ln\left(\frac{P}{P_0}\right) = -\frac{\rho_0}{P_0} gh$$

$$\frac{P}{P_0} = e^{-\frac{\rho_0 gh}{P_0}} \Rightarrow \boxed{P = P_0 e^{-\frac{\rho_0 g h}{P_0}}}$$

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WE have,

$$PV = \frac{m}{M} RT$$

$$P = \frac{m}{V} \times \frac{RT}{M}$$

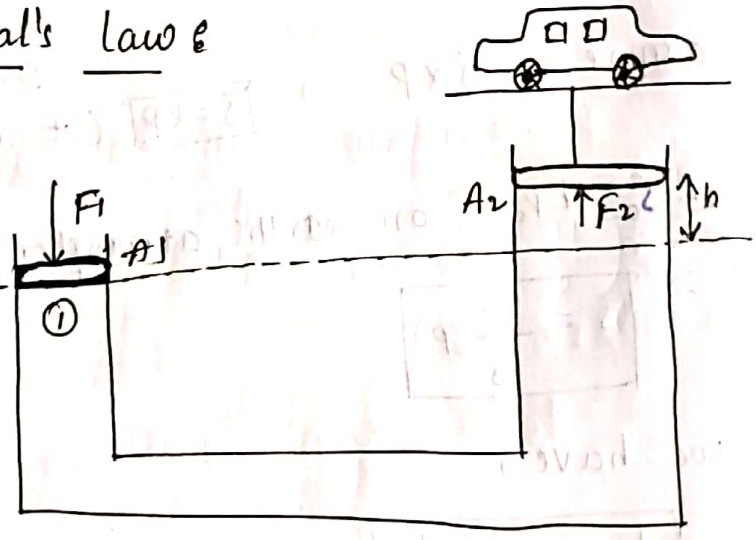
$$\boxed{P \propto \rho} \quad (T = \text{const})$$

i.e, In isothermal condns & molecular mass of composition is same.

$$\boxed{P \propto \rho}$$

## Applications of pascal's law:

### Hydraulic Lift:-




*Increase in pressure,

$$P_1 = \frac{F_1}{A_1} \ (\downarrow)$$

*force on the large piston,

$$F_2 = P_1 A_2 = \frac{F_1}{A_1}(A_2)$$

* In static fluid, $P_1 = P_2$

$$P_0 + \frac{F_1}{A_1} = P_0 + \frac{F_2}{A_2} + \rho g h$$

=) since fluid is incompressible,

$$\boxed{A_1 d_1 = A_2 d_2}$$

* Mechanical Advantage, $= \frac{F_2}{F_1}$

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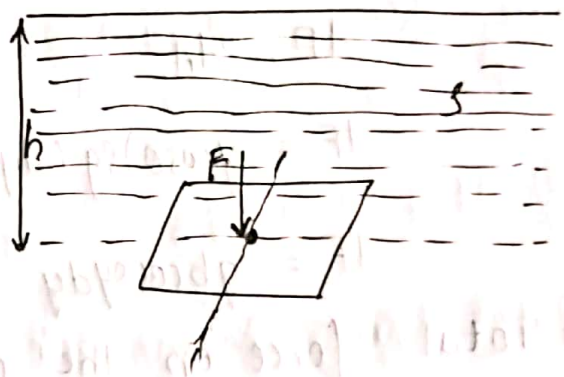
## Hydrostatic force on surfaces :-

when a stafic fluid comes in contact with a surface either plane or curved, a normal force is exerted by the fluid on the surface. This force is known as hydrostatic force. The pt of application of total force is called "centre of pressure"

### case(i) : Hydrostatic force on a plane horizontal surface :

A = area of the surface.

since everypt of surface is at the same depth from free surface.




$\therefore$ pressure $p = hsg$.

* hydrostatic force, $F = PA = (hsg)A$

$$F = vsg$$

$$\boxed{F = mg} = \text{wt of liquid above the liquid surface.}$$

force exerted by liquid on the surface → Thrust

~~Special cases :~~

Total hydrostatic force on inclined / vertical plane surfaces :

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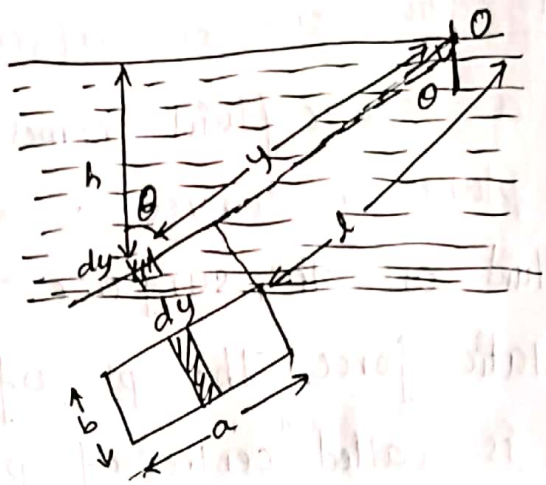
* from figure,

$$\cos\theta = \frac{h}{y}$$

$$h = y\cos\theta \quad -(1)$$




* pressure at the small element,

$$P = hsg = (y\cos\theta)\,sg$$

* force on the small element due to liquid

$$dF = PdA$$

$$dF = (y\cos\theta)sg\,(dy)\cdot b$$

$$dF = sgb\cos\theta y\,dy$$

* Total force on the plane surface.

$$F = \int dF = sgb\cos\theta \int_{y=l}^{y=a+l} y\,dy = sgb\cos\theta \left(\frac{y^2}{2}\right)_{l}^{l+a}$$

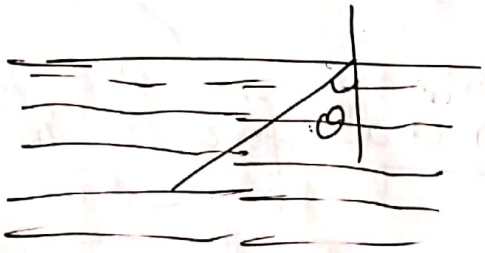
$$F = \frac{1}{2} sgb\cos\theta \left(\frac{(l+a)^2}{1} - \frac{(l)^2}{1}\right)$$

$$F = \frac{1}{2} sgb\cos\theta \left((l+a+l)(l+a-l)\right)$$

$$\boxed{F = \frac{1}{2} sgb\cos\theta\,(a^2+2al)}$$

Special Cases :

(a) If $l = 0$




$$\boxed{F = \frac{sgb\cos\theta\, a^2}{2}}$$

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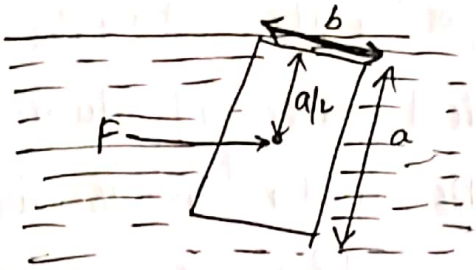
① If $\theta = 0^\circ$ & $l = 0$

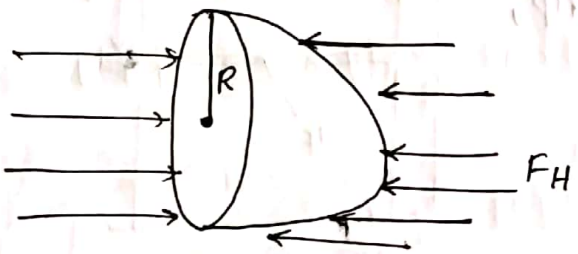


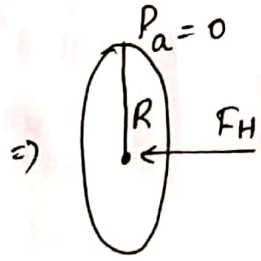

$$F = \frac{\rho g b a^2}{2}$$

$$F = \rho g\left(\frac{a}{2}\right)(ab)$$

$$F_{avg} = \left(\frac{0 + \rho g a}{2}\right)(ab)$$

$$F_{avg} = P_{avg} \times \text{area}$$ (projected area).

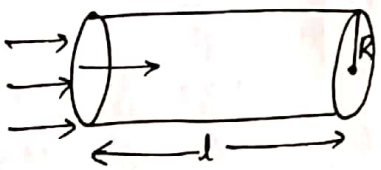



$\Rightarrow$ $P_2 = \rho g(2R)$

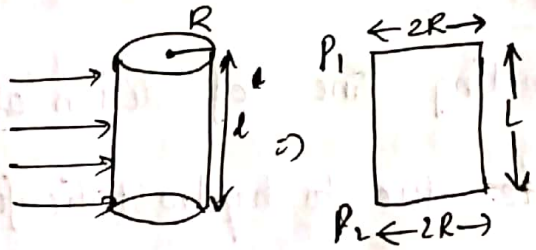
$$F_H = \left(\frac{0 + \rho g(2R)}{2}\right)\pi R^2$$

$$F_H = \rho g \pi R^3$$




$\Rightarrow$


projection area $= \pi R^2$




projection area $= 2RL$

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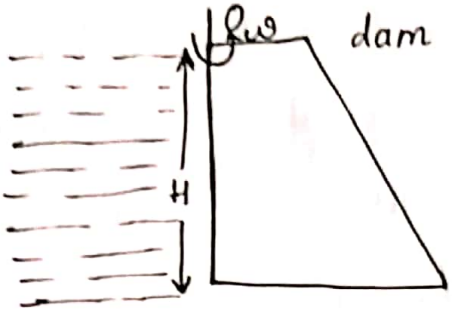
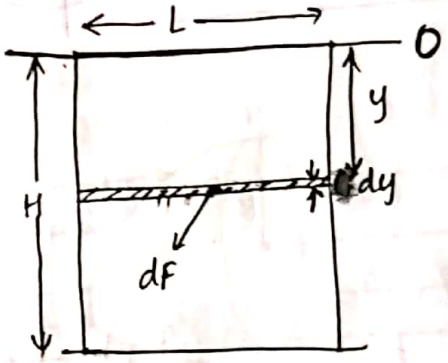
Eg: Suppose water stands at a depth of H behind the vertical face of dam of width L. Find force exerted by the water on the wall & the centre of pressure.




Sol:




* $P = \rho g y$ —①

force on elementary strip,

* $dF = P \cdot dA$

$\Rightarrow dF = (\rho g y) L \cdot dy$

∴ Total force on the gate,

$$F = \int dF = \rho g L \int_{y=0}^{y=H} y \cdot dy$$

$$F = \frac{1}{2} \rho g L \left(\frac{H^2}{1} - 0\right)$$

$$\boxed{F = \frac{1}{2} \rho g L H^2}$$

* $F = P_{avg} \times \text{area} = \left(\frac{0 + \rho g H}{2}\right)(HL)$ *

* Centre of pressure — finding line of action of force

Torque acting on the gate due to hydrostatic force,

* torque on small element about 'O'

$$d\tau = y \cdot dF$$

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$$d\tau = (y)(\rho g y)(L \cdot dy)$$

$$d\tau = \rho g L (y^2 \cdot dy)$$

∴ Total torque, $\tau = \int d\tau$

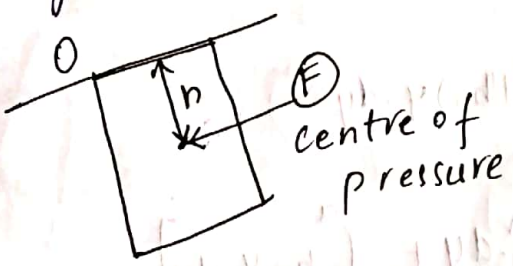
$$\tau = \rho g L \int_0^H y^2 \cdot dy$$

$$\boxed{\tau = \frac{1}{3} \rho g L H^3} \text{ — anti clockwise}$$

Line of action of forces-

It is the pt at which total force is acting on that body.

*




$$\tau_{hydro} = \tau_F$$

$$\frac{\rho g L H^3}{3} = h \times \frac{\rho g L H^2}{2}$$

$$\boxed{h = 2H/3}^*$$

If there is no external force acting on the body,

$$\tau = I\alpha \qquad I = \frac{MH^2}{3}$$

Eg: Figure shows a slar plate submerged in a liquid of density $\rho$. what is the hydrostatic force on one face of the plate?

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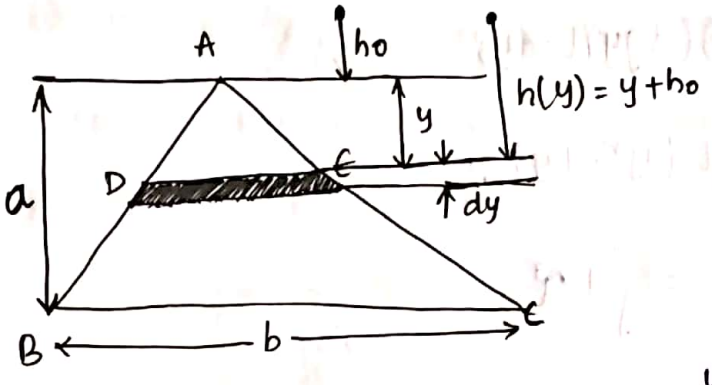



$$\frac{y}{\ell} = \frac{a}{b}$$

$$\Rightarrow \ell = \frac{yb}{a}$$

$$P = \rho g (y + h_0) \quad -①$$

$$dF = P\,dy$$

$$= \rho g (y + h_0)\,\ell \cdot dy$$

$$F = \int_{y=0}^{y=a} dF = \int_0^a \rho g (y + h_0) \frac{yb}{a} dy$$

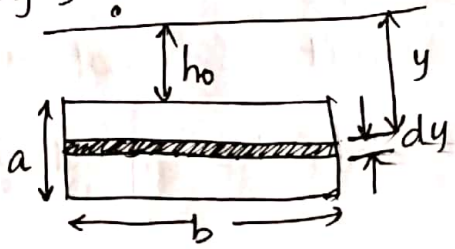
$$= \frac{\rho g b}{a} \int_0^a (y + h_0)\, y \cdot dy$$

$$= \frac{\rho g b}{a} \left[ \int_0^a y^2 \cdot dy + \int_0^a h_0 y \cdot dy \right]$$

$$= \frac{\rho g b}{a} \left[ \frac{a^3}{3} + \frac{h_0 a^2}{2} \right]$$

$$\boxed{F = \frac{\rho g b}{6} (2a^2 + 3h_0 a)}$$

→ During a flood, the water in a basement rises to a height $h_0$ above a window as shown in figure. what is the force on the window due to water of density $\rho$?




$$P = \rho g y$$

$$dF = P\,dA = \rho g y b\, dy$$

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$$F = \int dF = \int_{y=a}^{y=a+h_0} \rho g b y \cdot dy$$

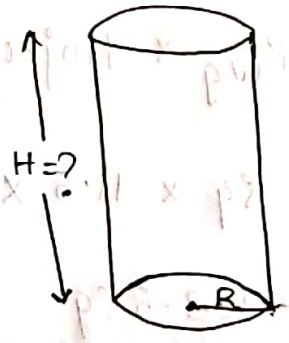
$$F = \rho g b \int_{h_0}^{a+h_0} y \cdot dy$$

$$= \frac{\rho g b}{2} \left((a+h_0)^2 - h_0^2\right)$$

$$F = \frac{\rho g b}{2} \left((a+h_0)(a)\right)$$

$$F = \frac{\rho g b}{2} (a^2 + 2ah_0)$$

To what height should a cylindrical vessel of radius R be filled with a homogeneous liquid to make the force with which the liquid presses on the sides of the vessel equal to the force exerted by the liquid on the bottom of the vessel?




$P_{bottom} = \rho g H$

force on the bottom

$= (\rho g H)\pi R^2$

= wt of liquid above bottom — ①

$F_{side\ wall} = (P_{avg})$ curve surface area

$$= \left(\frac{0 + \rho g H}{2}\right) \times (2\pi R H) \quad — ②$$

① = ② ⇒ $(\rho g H)\pi R^2 = \left(\frac{0+\rho g H}{2}\right)(2\pi R H)$

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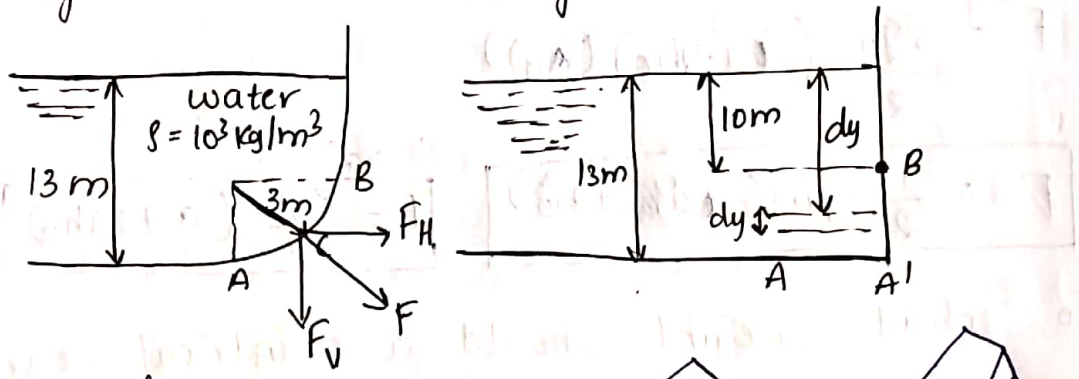
$$\rho g R = \rho g H$$

$$\boxed{H = R}$$

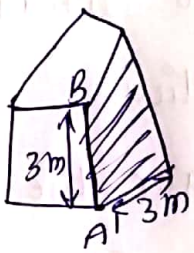
Eg: Compute the horizontal & vatical components of the resultant of the hydrostatic pressure distribution on the gate AB, which is a quater of a cylinder. Assume the gate is 3m wide.




Sol:

$$P_B = \rho g h_1 = \rho g (10) = 10 \rho g$$

$$P_{A'} = \rho g h_2 = \rho g (13) = 13 \rho g$$



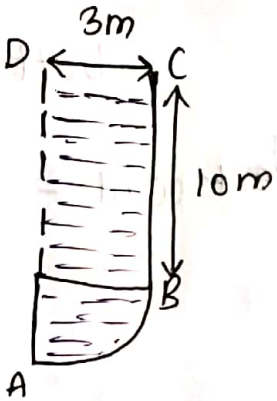

$$P_{avg} = \left(\frac{P_B + P_{A'}}{2}\right) = \rho g \left(\frac{10+13}{2}\right) = \frac{23}{2} \rho g$$

$$\Rightarrow F_H = P_{avg} \times \text{Projection area}$$

$$= \rho g \times 11.5 \times 3 \times 3$$

$$F_H = 103.5 \rho g$$

* To calculate vertical force:




Volume of liquid $= \frac{\pi R^2 \times 3}{4} + 10 \times 3 \times 3$

$$= \frac{3}{4} \times \pi R^2 + 90$$

wt of liquid = volume x density

vertical force ($F_V$)

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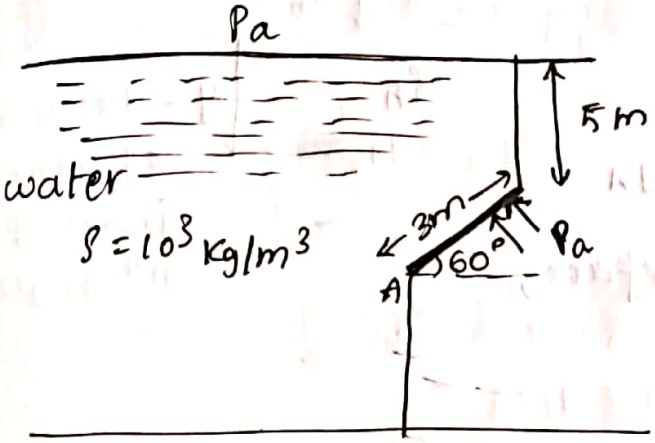
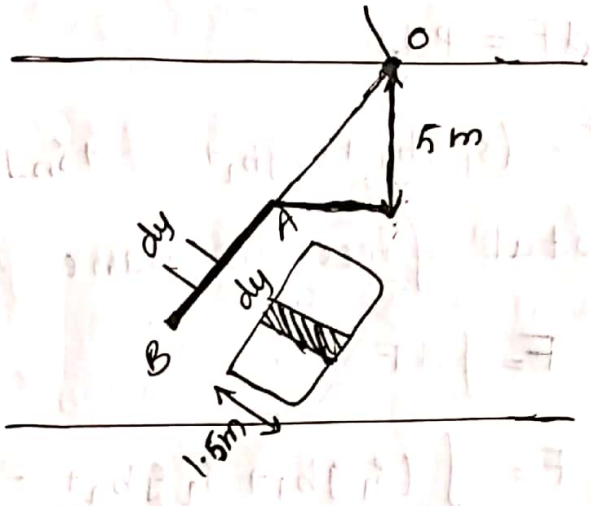
net thrust $F = \sqrt{F_H^2 + F_V^2}$

Eg: Compute the resultant force on the gate AB as a result of hydrostatic pressure in figure. The gate is 1.5m wide. Determine the line of action of this force.

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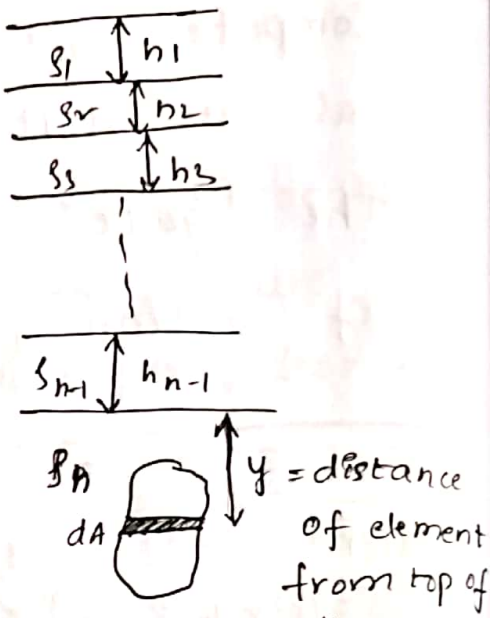
## Hydrostatic force due to multiple layers:




Pressure at the location of small element,

$$P = \rho_1 g h_1 + \rho_2 g h_2 + \cdots + \rho_n g y$$

Force on small element,

$$dF = P\,dA$$

$$dF = (\rho_1 g h_1 + \rho_2 g h_2 + \cdots + \rho_n g y)\,dA$$

∴ Total force on plane surface,

$$F = \int dF$$

$$F = \int (\rho_1 g h_1 + \rho_2 g h_2 + \cdots + \rho_n g y)\,dA$$

$$F = \sum_{i=1}^{n} \rho_i g h_i A + \int \rho_n g y \cdot dA$$

$$F = \sum_{i=1}^{n} \rho_i g h_i A + \rho_n g y_c A$$

$y_c$ → distance of centre of plane surface from top of $n^{th}$ layer.

Short cut –

$$F = (P_{avg}) \times \text{projection area}$$

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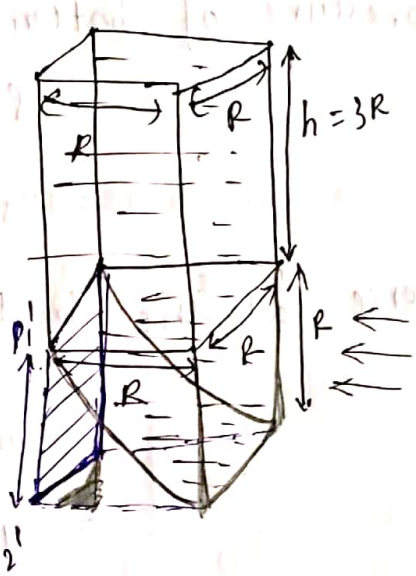
Eg: A body having square crossection at the top & curved surface at the bottom completely filled with liquid of density $\rho$ as shown. Find

① Thrust on right vertical wall.

② Thrust on curved surface.




① Force on right vertical surface

$= P_{avg} \times$ Projected area

$= \left(\frac{P_1 + P_2}{2}\right)(R \times 1 \times R)$

$= (4\rho gR)(2R^2)$

$\boxed{F = 8\rho gR^3}$

② Thrust on curved surface,

i) vertical thrust ($F_V$) = wt of liquid about the curved surface

$= \left(\frac{\pi R^2 \times R}{4} + R \times R \times 3R\right)\rho g$

$= \frac{(12+\pi^2)}{4} R^3 \rho g$

ii) horizontal thrust ($F_H$) = $P_{avg} \times$ Projected area

$= \left(\frac{P_1' + P_2'}{2}\right) \times R \times R$

$= \left(\frac{3R\rho g + 4R\rho g}{2}\right) \times R^2$

$= \frac{7R^3 \rho g}{2}$

Net thrust $= \sqrt{F_V^2 + F_H^2} = \frac{R^3 \rho g}{2} \sqrt{\frac{(12+\pi)^2}{4} + 49}$

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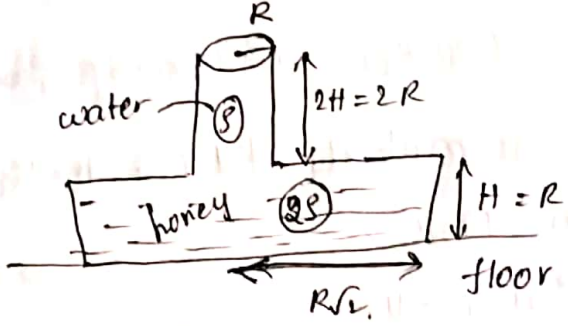
Eg:




*before mixing,

pressure of bottom of vessel,

$= \rho g(2H) + 2\rho gH$

$= 4\rho gH$

Force at the bottom of vessel,

$= P_{bottom} \times$ base area

$= 4\rho gH \times \pi (R\sqrt{2})^2$

$= 4\pi\rho gH(2R^2)$

$= 8\pi\rho gR^2H$

*After mixing,

pressure of bottom $= \rho_{mix} \times g(3H) = \frac{9\rho gH}{2}$.

$\rho_{mix} = \frac{m_1+m_2}{V_1+V_2} = \frac{V_1\rho_1+V_2\rho_2}{V_1+V_2} = \frac{\pi R^2(2H)\rho + \pi(R\sqrt{2})^2 H(2\rho)}{\pi R^2(2H)+\pi(R\sqrt{2})^2 H}$

$= \frac{2\rho+6\rho}{4} = \frac{6\rho}{4} = \frac{3\rho}{2}$.

Force at bottom,

$= P_{bottom} \times$ area of base

$= \frac{9\rho gH}{2} \times \pi (R\sqrt{2})^2 = 9\pi\rho gHR^2 = 9\pi\rho gR^3$.

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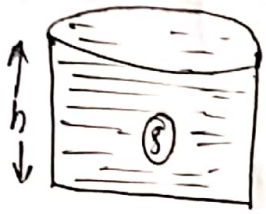
here normal rxn exerted by the floor on the bottom of beaker is not changing.

Imp points:

- In some situations, force at bottom of vessel is not equal to wt of liquid in that vessel.

Ex:




Here pressure of bottom of vessel due to liquid = $h s g$

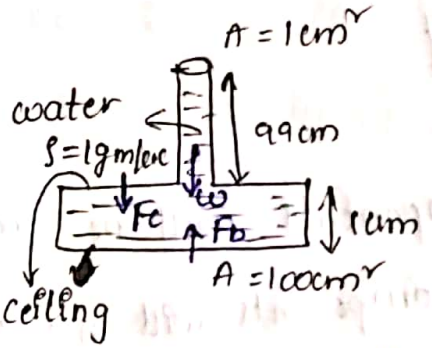
force at the base of vessel

$$\boxed{F_b = h s g A}$$

$\therefore$ $F_b = sgv = mg = w$

$$\boxed{F_b = w}$$

Ex6




pressure due to liquid at the base of vessel,

$$P = hsg = \frac{(99+1)}{100} \times 10^3 \times 9.8 = 9.8 \times 10^3 \text{ Pa}$$

Force at the bottom of vessel,

$$F_b = P \times \text{base area} = 9.8 \times 10^3 \times 100 \times 10^{-4} = 98 \text{ N}.$$

weight of liquid in the vessel,

$$w = (V_1 + V_2)\, sg = (100 \times 10^{-6} + 99 \times 10^{-6}) \times 10^3 \times sg$$

$$w = 1.96 \text{ N}.$$

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clearly $F_b > w$

bcoz, ceiling exert force on liquid downwards.

• In equilibrium, * $F_{net} = 0$ (on liquid)

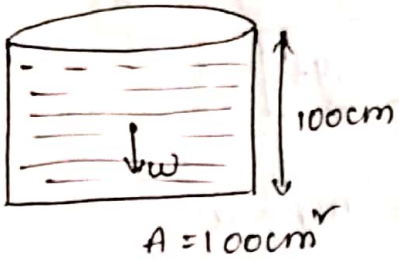
$$w + F_c = F_b$$

$$F_c = F_b - w$$

$$= 98 - 1.96$$

$$\boxed{F_c = 96.04\ N}$$

*




$$w = V s g$$

$$= 100 \times 100 \times 10^{-6} \times 10^3 \times 9.8$$

$$\boxed{w = 98N}$$

Ex:

horizontal components will be cancelled but vertical components will not.

bcoz P is same on same horizontal line by P increase by going from top to bottom (so force also increases)

$$P = hsg$$

$$F_b = PA$$

In equilibrium,

$$F_b + F_v = w$$

$$\boxed{F_v = w - F_b}$$

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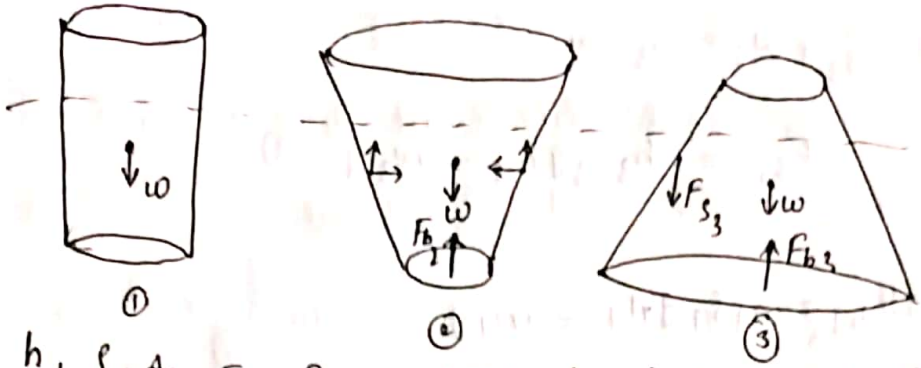
Ex:8




① ② ③

let h, $\rho$, A, $F_s$, $F_b$ represent height of liquid, density of liquid, area of base, force on liquid due to side walls, force on liquid due to base respectively. Then choose the correct options.

1) The vessels have same base area & contain equal mass of liquid with equal heights of the liquid column. then

$\rho_3 > \rho_1 > \rho_2$.

2) If the vessels contain equal masses of same liquid with same base area & If $F_{s_2} = F_{s_3}$ then the heights of liquid columns must be related as

$h_3 - h_2 = 2h_1$

3) If the vessels contain equal masses of same liquid to equal heights and if $F_{s_1} = F_{s_3}$ then base areas of the containers must be related as $A_3 + A_2 = 2A_1$

Sol:

Fig ① $\Rightarrow F_b = W$

$\Rightarrow h_1 \rho_1 g A_1 = m_1 g$

$h_1 \rho_1 A_1 = m_1$ — ①

Fig ② $\Rightarrow F_{b_2} + F_{s_2} = W_2$

$F_{s_2} = m_2 g - h_2 \rho_1 A_2 g$ — ②

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Fig ③ ⇒ $F_{S_3} + w_3 = F_{b_3}$

$$F_{S_3} = h_3 S_3 g A_3 - m_3 g \quad -③$$

opt 1:

$$F_{b_1} = h_1 S_1 g A_1 = m_1 g$$

$$F_{b_2} = h_2 S_2 g A_2 = m_2 g - F_{S_2}$$

$$F_{b_3} = h_3 S_3 g A_3 = m_3 g + F_{S_2}$$

$$\left.\begin{array}{l} m_1 = m_2 = m_3 \\ h_1 = h_2 = h_3 \\ A_1 = A_2 = A_3 \end{array}\right\}$$

$h_3 S_3 g A_3 > h_1 S_1 g A_1 > h_2 S_2 g A_2$

$$\boxed{S_3 > S_1 > S_2} \checkmark$$

Opt-2:

$$F_{S_2} = F_{S_3}$$

$$m_2 g - h_2 S_2 A_2 g = h_3 S_3 A_3 g - m_3 g$$

$$m_2 - h_2 S_2 A_2 = h_3 S_3 A_3 - m_3$$

$$h_1 S_1 A_1 - h_2 S_2 A_2 = h_3 S_3 A_3 - h_1 S_1 A_1$$

$$\boxed{2h_1 = h_2 + h_3} \checkmark$$

$m_1 = m_2 = m_3$

$A_1 = A_2 = A_3$

$S_1 = S_2 = S_3$

Opt-3:

$$F_{S_2} = F_{S_3}$$

$$m_2 - h_2 S_2 A_2 = h_3 S_3 A_3 - m_3$$

$$h_1 S_1 A_1 - h_2 S_2 A_2 = h_3 S_3 A_3 - h_1 S_1 A_1$$

$$\boxed{2A_1 = A_2 + A_3} \checkmark$$

$m_1 = m_2 = m_3$

$h_1 = h_2 = h_3$

$S_1 = S_2 = S_3$

Eg: A conical flask with long neck as shown is filled completely with liquid of density $S$. Find

i) force at the bottom of the vessel.

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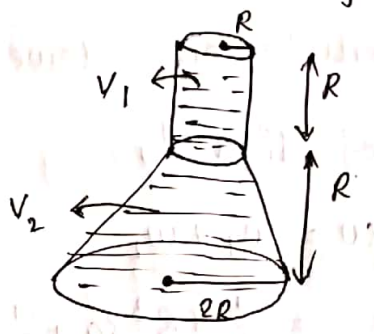
ii) wt of the liquid in the vessel.

iii) Thrust force on liquid due to side walls of vessel.

i) pressure at bottom $= hsg$

$= (2R)sg$




force at bottom of vessel

$F_b = P \times$ base area

$= (2Rsg) \times \pi(2R)^2$

$$\boxed{F_b = 8\pi R^3 sg}$$

ii) wt of liquid,

$w = (V_1 + V_2)sg$

$V_2 = \frac{1}{3}\pi(2R)^2(2R) - \frac{1}{3}\pi R^2 \times R$

$= \frac{8}{3}\pi R^3 - \frac{1}{3}\pi R^3$

$= \frac{(8-1)\pi R^3}{3} = \frac{7\pi R^3}{3}$

$V_1 = \pi R^3$

$w = \left(\frac{7\pi R^3}{3} + \pi R^3\right) sg$

$$\boxed{w = \frac{10}{3}\pi R^3 sg}$$

iii) $w + F_v = F_b$

$F_v = F_b - w = \frac{24\pi R^3 sg - 10\pi R^3 sg}{3}$

$$\boxed{F_v = \frac{14}{3}\pi R^3 sg}$$

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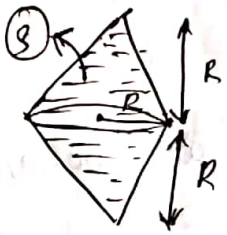
Find

① Thrust on upper side wall due to liquid.

② Thrust on lower side wall due to liquid.




wt of liquid,

$$w = w_1 + w_2$$

$$= \frac{1}{3}\pi R^3 \rho g + \frac{1}{3}\pi R^3 \rho g$$

$$\boxed{w = \frac{2}{3}\pi R^3 \rho g}$$

$$F_b = (h\rho g)\pi R^2$$

$$\boxed{F_b = \pi R^3 \rho g}$$

$$F_1 + w_1 = F_b$$

$$F_1 = F_b - w_1$$

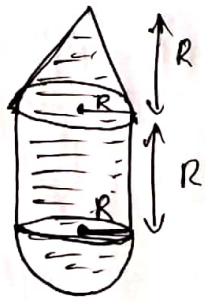
$$= \pi R^3 \rho g - \frac{2}{3}\pi R^3 \rho g$$

$$F_1 + w = F_2$$

$$F_2 = \frac{2}{3}\pi R^3 \rho g + \frac{2}{3}\pi R^3 \rho g$$

$$\boxed{F_2 = \frac{4}{3}\pi R^3 \rho g}$$

Eg:




Sol: wt of liquid,

$$w = w_1 + w_2 + w_3$$

$$= \frac{1}{3}\pi R^3 \rho g + \pi R^3 \rho g + \frac{2}{3}\pi R^3 \rho g$$

$$\boxed{w = 2\pi R^3 \rho g}$$

If $\rho$ is the density of liq in the container the ratio of vertical thrust on the spherical surface to the conical surface due to liquid?

$$F_b = \pi R^3 \rho g$$

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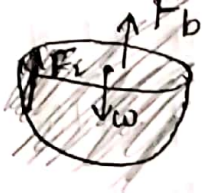


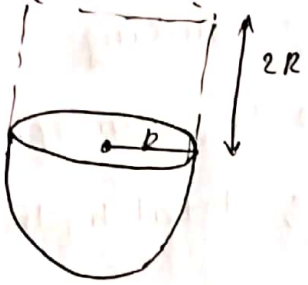


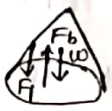


$F_1 + \omega = F_b$

$F_1 = \pi R^3 \rho g - \omega_1$

$= \pi R^3 \rho g - \frac{1}{3}\pi R^3 \rho g$

$= \frac{2}{3}\pi R^3 \rho g$

$F_2 = \frac{2}{3}\pi R^3 \rho g + \pi R^2 (2R) \rho g$

$= \frac{2}{3}\pi R^3 \rho g + 2\pi R^3 \rho g$

$F_2 = \frac{8\pi R^3 \rho g}{3}$

$F_2 : F_1 = 4 : 1.$

Eg: A solid hemisphere a radius r is just made to sink in a liquid of density $\rho$. Find

① vertical thrust on the curved surface.

② vertical thrust on plane surface.

③ Side thrust on the hemisphere.

Volume of liquid above liquid surface

$= \pi R^3 - \frac{1}{3}\pi R^3$

$= \frac{1}{3}\pi R^3$

wt of liquid along the curved surface

$= \frac{1}{3}\pi R^3 \rho g$ = vertical component (thrust) on curved surface.

vertical thrust on base = $\pi \cdot R^3 \rho g$

side thrust = zero.

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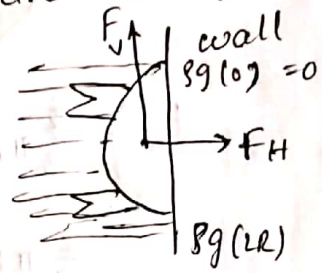
Eg: A tortoise is just sinking in water of density $\rho$. It is assumed to be hemisphere of radius R.

Find vertical thrust & total hydrostatic force.




$$F_V = \frac{2}{3}\pi R^3 \rho g$$

$$F_H = \pi R^3 \rho g$$

$$F = \sqrt{F_H^2 + F_V^2}$$

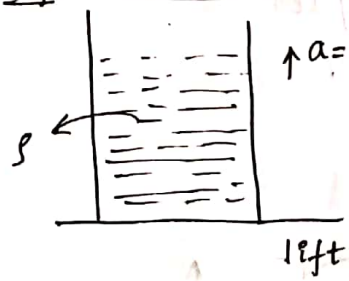
$$= \pi R^3 \rho g \sqrt{1 + \frac{4}{9}}$$

$$= \frac{\sqrt{13}}{3}\pi R^3 \rho g$$

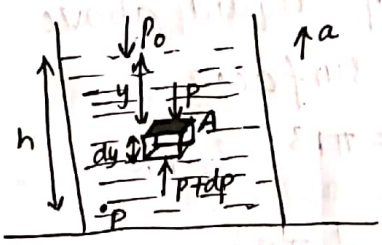
Variation of pressure in accelerated frame

* lift accelerates upwards

* Vertical direction *




Net force on the sample,




$$(P+dp)A - PA - dmg = dma$$

$$dpA = dm(g+a)$$

$$dpA = (A \cdot dy)\rho(g+a)$$

$$\boxed{\frac{dp}{dy} = \rho(g+a)}$$ — pressure gradient

const

now $$\int_{P_0}^{P} dp = \int_0^h \rho(g+a)\,dy$$

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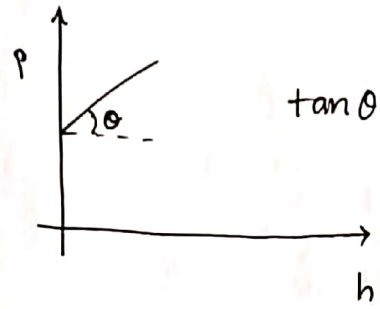
$P - P_0 = \rho h(g+a)$

$$\boxed{P = P_0 + \rho(g+a)h}$$

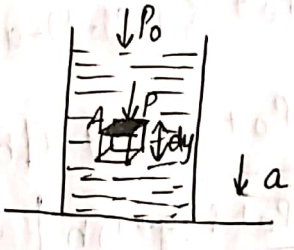



$\tan\theta = \rho(g+a)$

* Lift accelerates downwards :

net force on the sample,

$PA - (P+dp)A + dm\,g = dm\,a$




$-dpA = dm(a-g)$

$dpA = (A \cdot dy)\,\rho(g-a)$

$$\boxed{\frac{dp}{dy} = \rho(g-a)}$$ — pressure gradient.

now $\int_{P_0}^{P} dp = \int_0^h \rho(g-a)\,dy$

$P - P_0 = \rho h(g-a)$

$$\boxed{P = P_0 + \rho(g-a)h}$$

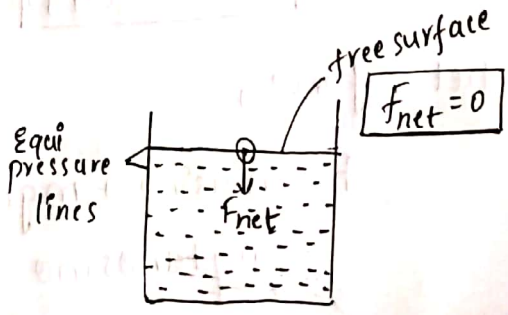
* If lift falls freely,

$a = g \Rightarrow \boxed{P = P_0}$

Equi pressure lines :

In general,

$$\boxed{P = P_0 + \rho\,|\bar{g} - \bar{a}|\,h}$$

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$\downarrow \bar{a} = -a\uparrow$

& $\bar{g} = g(-\uparrow)$

$P = P_0 + \rho \left| g(-\uparrow) - a(-\uparrow) \right| h$

$= P_0 + \rho \left| -g + a \right| h$

$\boxed{P = P_0 + \rho(g-a)h}$

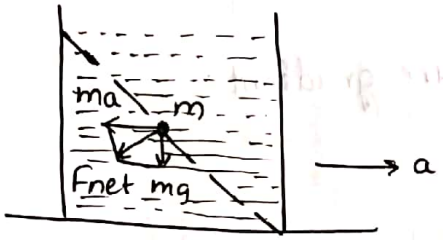
& $\uparrow \bar{a} = a\uparrow$

$P = P_0 + \rho \left| g(-\uparrow) - a(\uparrow) \right| h$

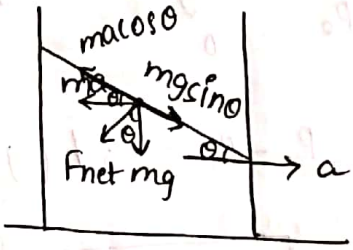
$= P_0 + \rho \left| -g - a \right| h$

$\boxed{P = P_0 + \rho(g+a)h}$

* acceleration in Horizontal direction *




Since net force along the free surface = 0,




$\Rightarrow ma\cos\theta = mg\sin\theta$

$\frac{a}{g} = \tan\theta$

$\boxed{\theta = \tan^{-1}(a/g)}$

net force,

$F = ma\sin\theta + mg\cos\theta$

$= mg\tan\theta\sin\theta + mg\cos\theta$

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$$= mg\left(\frac{\sin^2\theta}{\cos\theta} + \cos\theta\right)$$

$$= mg\left(\frac{\sin^2\theta + \cos^2\theta}{\cos\theta}\right)$$

$$\boxed{F = mg\sec\theta}$$

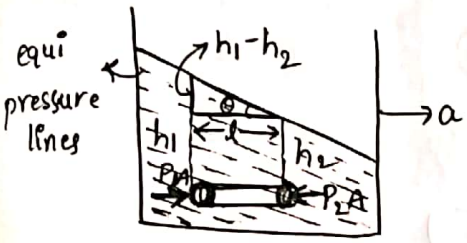
$$F = mg\sqrt{1+\tan^2\theta}$$

$$= mg\sqrt{1+\frac{a^2}{g^2}}$$

$$\boxed{F = m\sqrt{a^2+g^2}}$$

$$F = m g_{eff} \Rightarrow \boxed{g_{eff} = \sqrt{a^2+g^2}}^{*}$$

*




Net force on the sample in horizontal direction,

$$P_1A - P_2A = (dm)a$$

$$(P_1 - P_2)A = A \cdot \ell\, \rho\, a$$

$$\boxed{P_1 - P_2 = \rho \ell a}^{*} \quad \text{—— (1)}$$

$$\rho g(h_1 - h_2) = \rho \ell a$$

$$\frac{h_1 - h_2}{\ell} = \frac{a}{g}$$

$$\boxed{\tan\theta = \frac{a}{g}}^{*}$$

we have,

$$P_1 = P_0 + \rho g h_1 \;,\; P_2 = P_0 + \rho g h_2$$

$$P_1 - P_2 = \rho g(h_1 - h_2)$$

since $h_1 > h_2 \Rightarrow \boxed{P_1 > P_2}$

we have,

$$\boxed{\frac{dP}{dx} = -\rho a}^{*}$$

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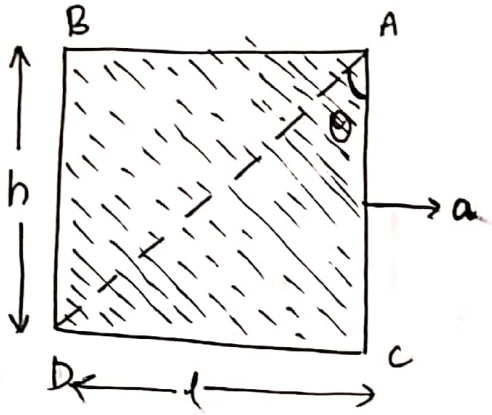
i.e, pressure decreases linearly in the direction of acceleration of liquid (horizontal).

Eqn




$$P_A = P_0 \quad -(1)$$

$$P_B = P_A + \rho l a$$

$$P_D = P_B + \rho g h$$

$$P_D = P_A + \rho l (la + gh) \quad -(2)$$

$$P_C = P_A + \rho g h \quad -(3)$$

$$P_D = P_C + \rho l a$$

$$P_D = P_A + \rho (la + gh) \quad -(4)$$

we can write,

$$P_D = P_A + \rho g_{eff} (\sqrt{l^2 + h^2})$$

$$P_D = P_A + \rho (\sqrt{g^2 + a^2})(\sqrt{l^2 + h^2})$$

$$P_D = P_A + \rho g h \left(\sqrt{1 + \frac{a^2}{g^2}}\right)\left(\sqrt{1 + \frac{l^2}{h^2}}\right)$$

$$= P_A + \rho g h (1 + \tan^2\theta)$$

$$= P_A + \rho g h \left(1 + \frac{a}{g} \times \frac{l}{h}\right)$$

$$\boxed{P_D = P_A + \rho g h + \rho a l}$$

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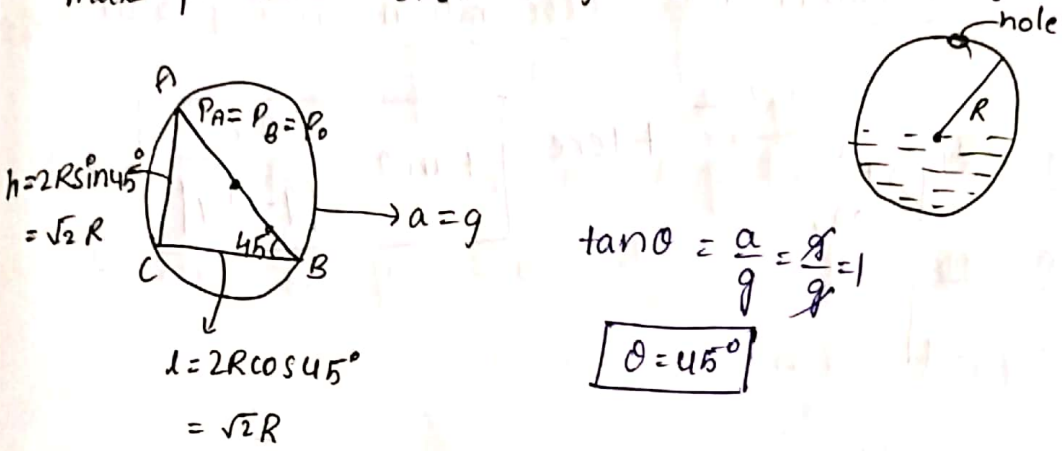
Eg: A cylinder of radius 'R' is half filled with liquid of density ρ. There is a small hole at the top. Cylinder is given a horizontal acceleration $a = g$. Find Max pressure exerted by liquid on the cylinder.




$$\tan\theta = \frac{a}{g} = \frac{g}{g} = 1$$

$$\boxed{\theta = 45°}$$

Max pressure is at pt 'C'.

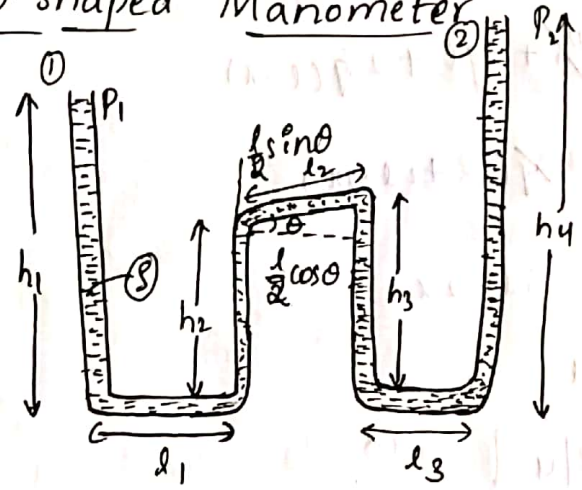
$$P_{max} = P_0 + \rho g h$$

$$= P_0 + \rho g \sqrt{2} R$$

Or $P_{max} = P_B + \rho l a$

$$\boxed{P_{max} = P_0 + \rho g \sqrt{2} R}$$

Eg: U-shaped Manometer




If the table is at rest,

$$\boxed{P_1 + \rho g h_1 - \rho g h_2 - \rho g l_2 \sin\theta + \rho g h_3 - \rho g h_4 = P_2}$$

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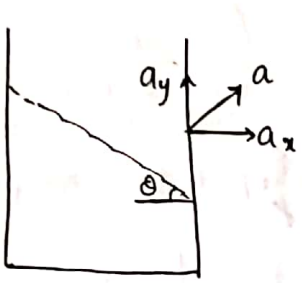
If the table accelerates towards right with 'a':

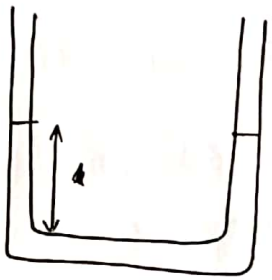
$$P_1 + \rho g h_1 - \rho a l_1 - \rho g h_2 - \rho l_2 \sin\theta a - \rho l_2 \cos\theta a + \rho g h_3 - \rho l_3 a - \rho g h_4 = P_2.$$

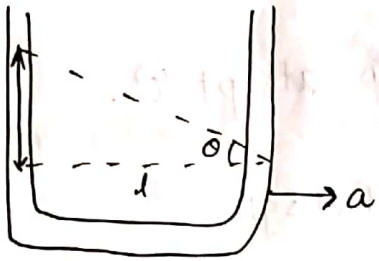
* If the container has acceleration in x-y plane:




Here $\boxed{\tan\theta = \frac{a_x}{|g - a_y|}}$

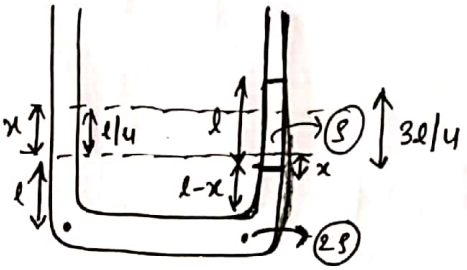
*




$$\tan\theta = \frac{h}{l} = \frac{a}{g}$$

*




$$P_0 + 2\rho g(l+x) = P_0 + \rho g l + 2\rho g(l-x)$$

$$2\rho g(l+x) = \rho g(l + 2l - 2x)$$

$$2l + 2x = 3l - 2x$$

$$4x = l$$

$$\boxed{x = l/4}$$

Eg: Two liquids of densities $\rho$ & $2\rho$ are taken in U-tube with equal volumes. If tube is given acceleration a towards right, the level difference in the limbs is zero.

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Find ① height 'h'

② If the system is stopped, find length of liquid of density $2\rho$ in horizontal part at eqm.

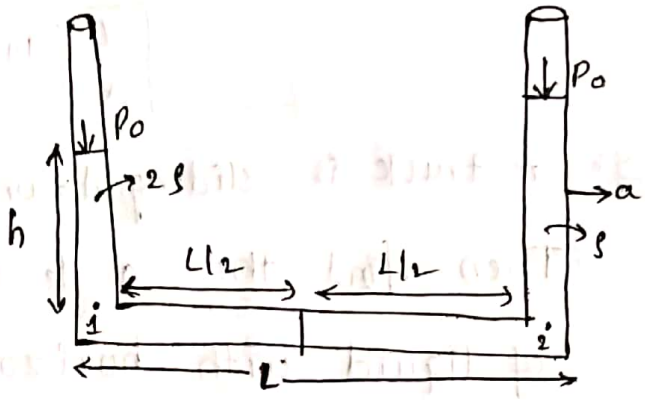
Sol⁶

① clearly,

$P_1 > P_2$

$P_0 + \rho_1 gh > P_0 + \rho_2 gh$

$\rho_1 > \rho_2$.




now, $P_0 + 2\rho gh - \rho\left(\frac{L}{2}\right)a - \rho\left(\frac{L}{2}\right)a - \rho gh = P_0$

$\rho gh = \frac{3\rho La}{2}$

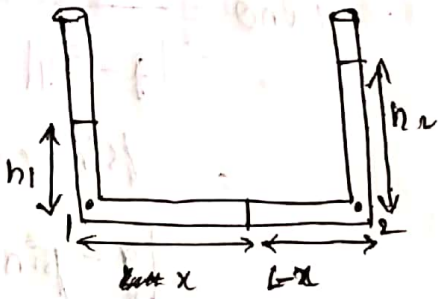
$$h = \frac{3La}{2g} \quad *$$

②

we have,

$h + \frac{L}{2} = h_1 + x$

$h_1 = h + \frac{L}{2} - x$




again,

$h + \frac{L}{2} = h_2 + L - x$

$\therefore h_2 = h - \frac{L}{2} + x$

now,

$P_1 = P_2$

$P_0 + h_1(2\rho)g = P_0 + h_2(\rho)g$

$2h_1 = h_2$

$2h + L - 2x = h - \frac{L}{2} + x$

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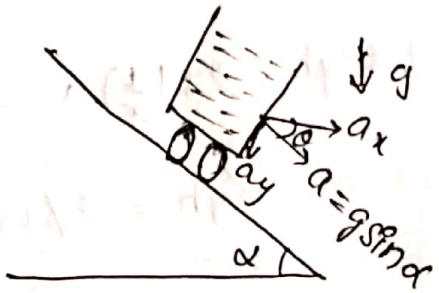
$$h + \frac{3L}{2} = 3x$$

$$\frac{3L}{2} + \frac{3L}{2}\left(\frac{a}{g}\right) = 3x$$

$$\boxed{\frac{L}{2}\left(1 + \frac{a}{g}\right) = x}$$

Eg: A truck is sliding down on smooth inclined plane. Then find the angle made by the free surface of liquid with horizontal.




Here $a = g\sin\alpha$

$$a_x = a\cos\alpha$$
$$= g\sin\alpha\cos\alpha$$

$$a_y = g\sin\alpha = g\sin^2\alpha$$

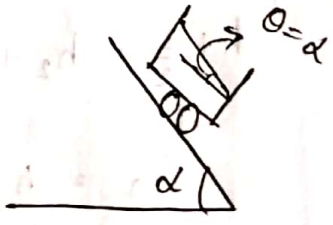
we have,

$$\tan\theta = \frac{a_x}{|g - a_y|}$$

$$= \frac{g\sin\alpha\cos\alpha}{|g - g\sin^2\alpha|}$$

$$= \frac{\sin\alpha\cos\alpha}{\cos^2\alpha}$$

$$\boxed{\tan\theta = \tan\alpha}$$




i.e, free surface is parallel to inclined plane

Eg: Let $P_1$ & $P_2$ are the pressures at 1 & 2 & $\beta = \frac{P_1 - P_2}{\rho g d}$

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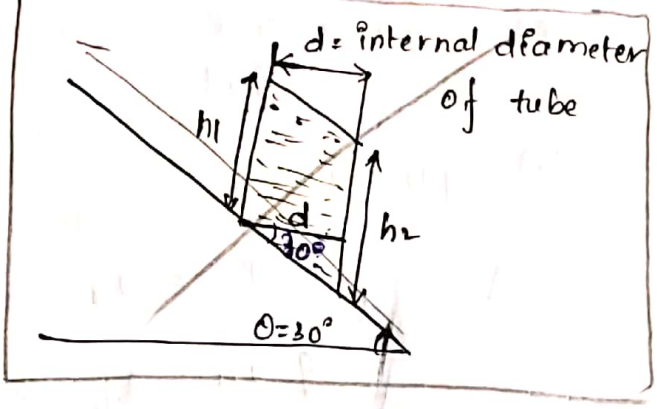


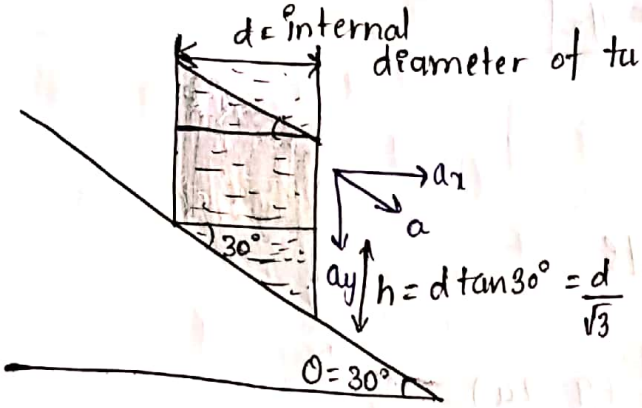


$a_x = a\cos 30° = \frac{\sqrt{3}}{2}a$ , $a_y = a\sin 30° = \frac{a}{2}$

$P_1 - \rho(g - a_y)h = P_3$

$P_1 - P_3 = \rho\left(g - \frac{a}{2}\right)\left(\frac{d}{\sqrt{3}}\right)$

& $P_2 - \rho d\frac{\sqrt{3}}{2}a = P_3$

$P_2 - P_3 = \frac{\rho d a\sqrt{3}}{2}$

finally, $P_1 - P_2 = P_1 - P_3 + P_3 - P_2$

$= (P_1 - P_3) - (P_2 - P_3)$

$= \frac{\rho d}{\sqrt{3}}\left(g - \frac{a}{2}\right) - \left(\frac{\rho d a\sqrt{3}}{2}\right)$

$= \frac{\rho d}{\sqrt{3}}\left(g - \frac{a}{2} - \frac{3a}{2}\right)$

$P_1 - P_2 = \frac{\rho d}{\sqrt{3}}(g - 2a)$

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here, $\beta = \frac{P_1 - P_2}{\rho g d}$

① for $\beta = 0$.

$P_1 - P_2 = 0$

$g - 2a = 0$

$\boxed{a = g/2}$

② $\beta = \sqrt{3}$

$P_1 - P_2 = \sqrt{3}\rho g d$

& $P_1 - P_2 = \frac{\rho d}{\sqrt{3}}(g - 2a)$

$\sqrt{3}\rho g d = \frac{\rho d}{\sqrt{3}}(g - 2a)$

$3g = g - 2a$

$2g = -2a$

$\boxed{a = -g}$

③ $\beta > 0$

$P_1 - P_2 > 0$

$g - 2a > 0$

$g > 2a$

$\boxed{a < g/2}$

④ $\beta < 0$

$g - 2a < 0$

$2a > g$

$\boxed{a > g/2}$

Eg: Open top cubical box of side length 'A' is filled with liquid of density $\rho$. Now box is given a horizontal accelention 'a' forwards. Then find

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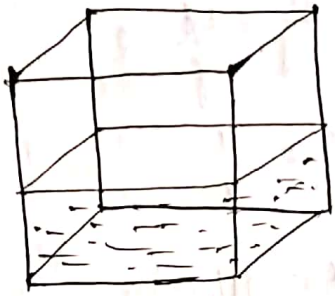
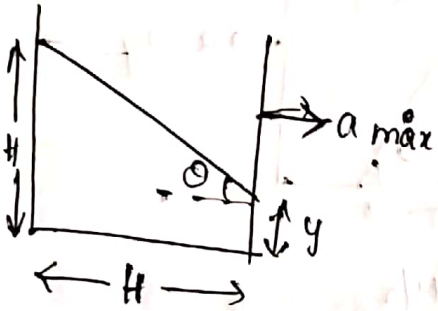
① Max acceleration of box so that liquid will not spill out.






we have, $\tan\theta = \frac{a_{max}}{g}$ —①

$\tan\theta = \frac{H-y}{H}$ —②

from eqn ① & ②

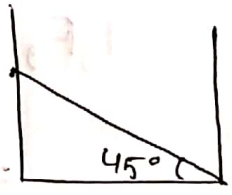
$a_{max} = g\left(1-\frac{y}{H}\right)$ —③

* since liquid is not spilling out, volume of liquid is const.

$H \times H \times \frac{H}{2} = \frac{1}{2}(H+y) \times H \times H$

$H = H+y$

$\Rightarrow \boxed{y=0}$




from eq ③ $\boxed{a_{max} = g}$

eqn ② $\tan\theta = 1$

$\boxed{\theta = 45^\circ}$

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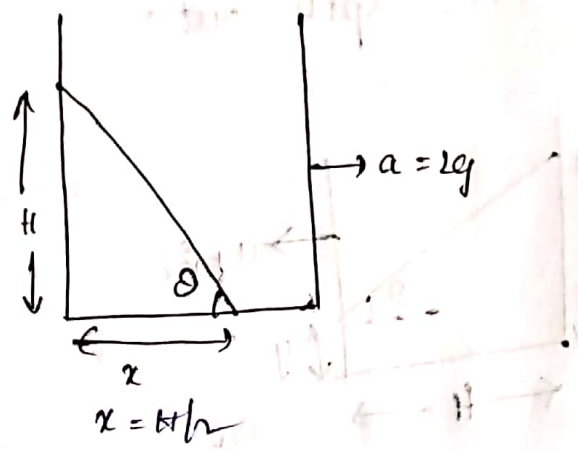
* If $a = 2g$ $(> a_{max})$

$$\tan\theta = \frac{a}{g} = \frac{2g}{g} = 2$$

$$\tan\theta = \frac{H}{x}$$

$$\boxed{x = H/2}$$




* Final volume of liquid,

$$V_f = \frac{1}{2} \times \left(\frac{H}{2}\right) \times H \times H = \frac{H^3}{4}.$$

* mass of remaining liquid,

$$m_f = \frac{\rho H^3}{4}$$

* Thrust force on rear wall:

before giving acceleration,

$$F_1 = \left(\frac{0 + \rho g H}{2}\right)\left(H \times \frac{H}{2}\right)$$

$$= \frac{\rho g H^3}{8}$$

After giving acceleration,

$$F_2 = \left(\frac{0 + \rho g H}{2}\right)(H \times H)$$

$$= \frac{\rho g H^3}{2}$$

* Increase in thrust force on rear wall,

$$\Delta F = F_2 - F_1 = \frac{\rho g H^3}{2} - \frac{\rho g H^3}{8} = \frac{3\rho g H^3}{8}$$

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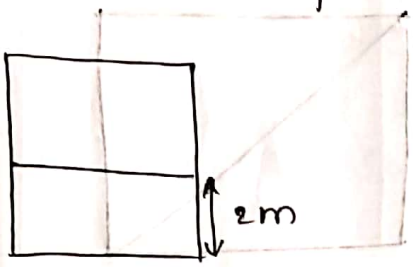
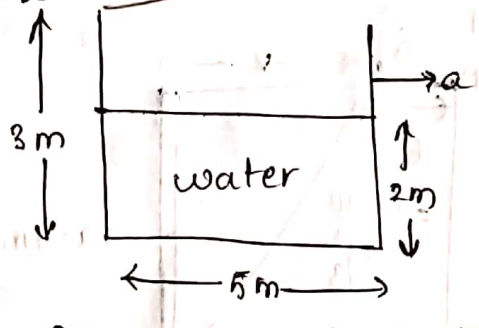
Eg: ① Find max acceleration that can be given without spilling water. 4 m/s²

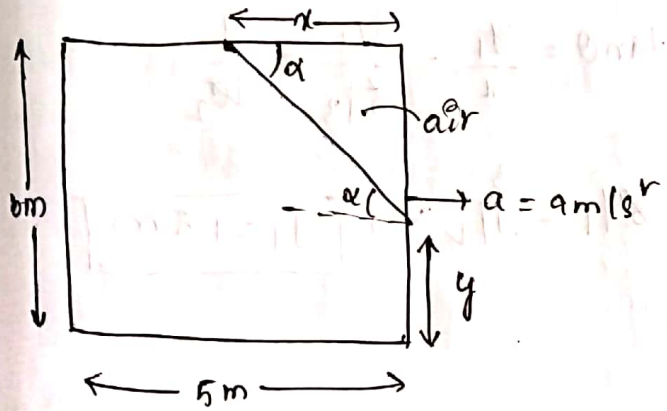
② If acceleration is increased by 20%, calculate the % of water spills over. 10%

③ If initially the tank is closed at the top & is accelerated horizontally by 9 m/s². Find the guage pressure at the bottom of the front end rear wall of the tank. 36.89.




Open rectangular tank of dimensions 5×4×3 m.




Here volume of air is const.

$$5 \times 1 \times x = \frac{1}{2}(3-y)(x) \times 4$$

$$10 = (3-y)(x) \quad -①$$

$$\tan\alpha = \left(\frac{3-y}{x}\right) = \frac{a}{g} = \frac{9}{10} = 0.9$$

$$3-y = 0.9x \quad -②$$

put ② in ①

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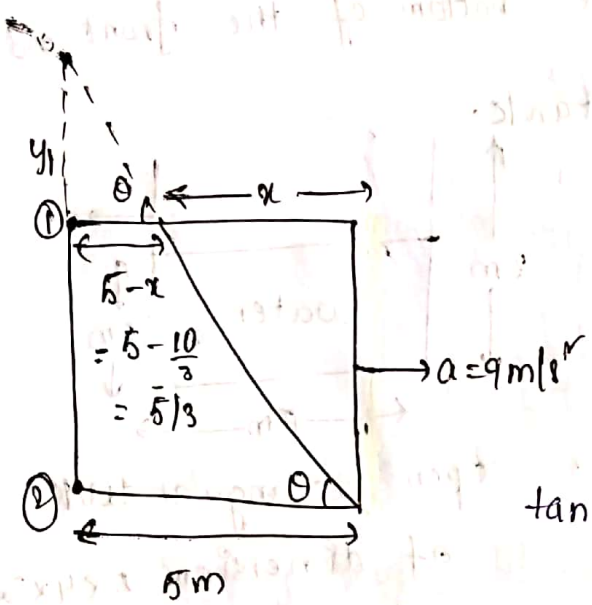
$$0.9x^2 = 10$$

$$x^2 = \frac{10 \times 10}{9}$$

$$\boxed{x = 10/3}$$

$$3 - y = 0.9x = \frac{0.3}{0.9} \times \frac{10}{3}$$

$$3 - y = 3 \Rightarrow \boxed{y = 0}$$




$$\tan\theta = \frac{y_1}{x} = \frac{y_1}{5/3} = \frac{3}{10}$$

$$3y_1 = 9/2 \Rightarrow \boxed{y_1 = 1.5\,m}$$

$$P_1 = y_1 \rho g$$

$$= 1.5 \rho g$$

$$P_2 = 1.5\,\rho g + 3\rho g = 1.5\,\rho g + 3\rho g$$

$$\boxed{P_2 = 4.5\,\rho g}$$

* Thrust on rear wall,

$$= P_{avg} \times \text{Projection area}$$

$$= \frac{P_1 + P_2}{2} \times 3 \times y_1^2$$

$$= 6\rho g \times 6 = \underline{36\rho g}.$$

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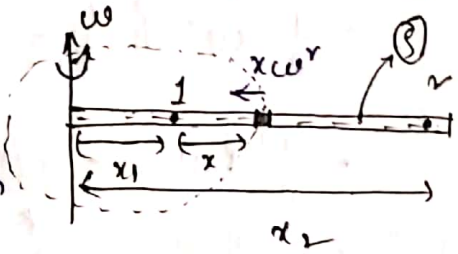
## Pressure difference in Rotating Frame :

In a fluid,

① pressure is minimum at AOR.




② pressure decreases as we move towards AOR.

* pressure difference accross the ends of small element,

$$dp = \rho (dx) a$$

$$dp = \rho (dx)(x\omega^2)$$

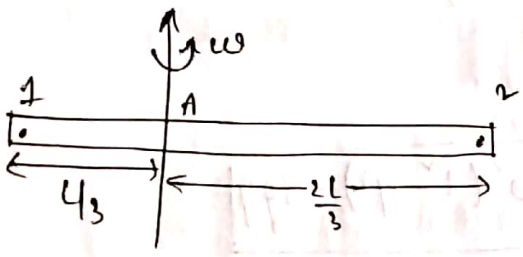
$$dp = \rho \omega^2 x \cdot dx$$

$$\int_{P_1}^{P_2} dp = \rho \omega^2 \int_{x_1}^{x_2} x \cdot dx$$

$$\boxed{P_2 - P_1 = \frac{1}{2} \rho \omega^2 (x_2^2 - x_1^2)} \quad *$$

clearly, $P_2 > P_1$.

Eg :




Find pressure diff b/w 1 & 2.

$$P_1 - P_A = \frac{1}{2} \rho \omega^2 \left( \left(\frac{l}{3}\right)^2 - 0^2 \right)$$

& $$P_2 - P_A = \frac{1}{2} \rho \omega^2 \left( \left(\frac{2l}{3}\right)^2 - 0^2 \right)$$

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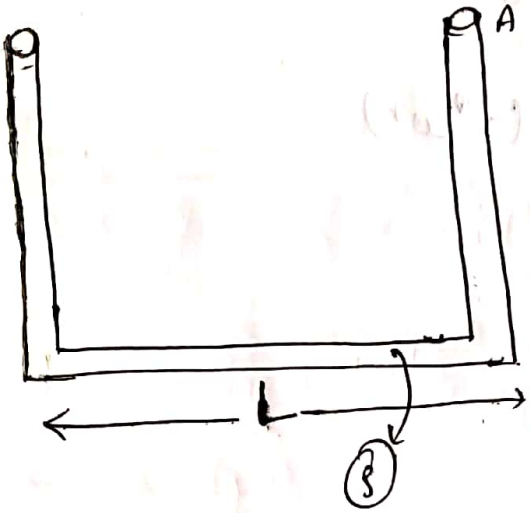
$$P_1 - P_2 = \frac{1}{2}\rho\omega^2\left(\frac{l^2}{9} - \frac{4l^2}{9}\right)$$

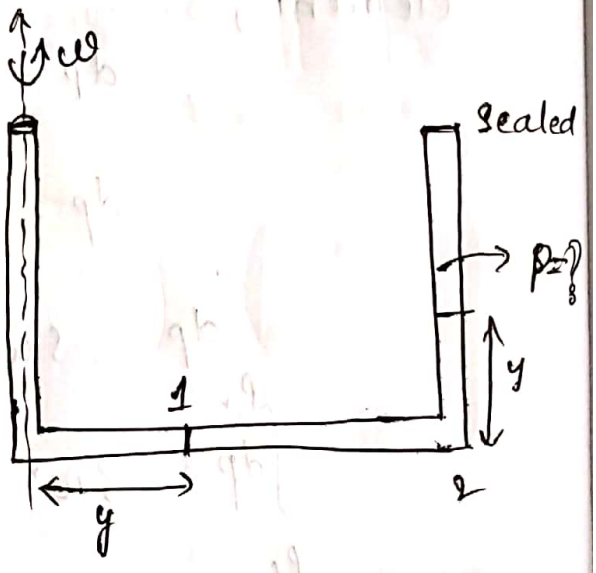
$$= \frac{1}{2}\rho\omega^2\left(-\frac{3l^2}{9}\right)$$

$$= -\frac{1}{2}\rho\omega^2\frac{l^2}{3}$$

$$\boxed{P_2 - P_1 = \frac{1}{6}\rho\omega^2 l^2}$$

*




$$P_2 - P_1 = \frac{1}{2}\rho\omega^2(l^2 - y^2)$$

$$P_2 = P_1 + \frac{1}{2}\rho\omega^2(l^2 - y^2)$$

now, $P_2 = \rho g y = P$

$$\boxed{P = P_1 + \frac{1}{2}\rho\omega^2(l^2 - y^2) = \rho g y}$$

$$P_1 = P_{atm}.$$

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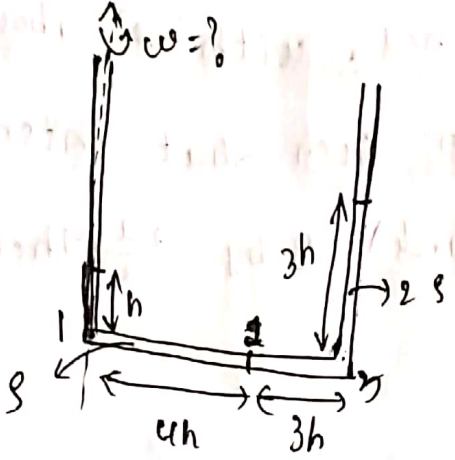
Eg:




$P_2 - P_1 = \frac{1}{2} s\omega^2((7h)^2 - 0^2)$

$\therefore P_2 - P_1 = \frac{1}{2} s((4h)^2 - 0^2)\omega^2$

$P_3 - P_2 = \frac{1}{2}(2s)((7h)^2 - (4h)^2)\omega^2$

$P_3 - P_1 = \left[\frac{1}{2}(2s)(49h^2 - 16h^2) + \frac{1}{2}s(16h^2)\right]\omega^2$

$= \omega^2 s[(33h^2) + 8h^2]$

$P_3 - P_1 = s\omega^2(41h^2)$

$P_0 + 3hsg(2) - P_0 - hsg = s\omega^2(41h^2)$

$6g - g = \omega^2(41h^2)$

$$\boxed{\omega = \sqrt{\frac{5g}{41h}}}$$

Eg: The interface of two liquids of density $s$ & $2s$ respectively lies at the pt A in a U-tube at rest. height of liquid column above A is $\frac{8}{3}a$ where $AB = 2a$. The crossectional area of tube is $s$.

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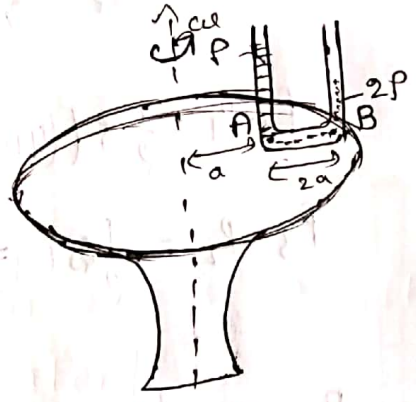
Now the table is spinned with 'ω' about a vertical axis as shown such that interfaces of liquids shifts towards B by $\frac{2a}{3}$. Then at this instant. Find

① $P_B - P_A$ in $\rho g a$

② The force exerted by liq 1 on liq 2 is at the interface: $\frac{5}{4}\rho a g \times a l 4$

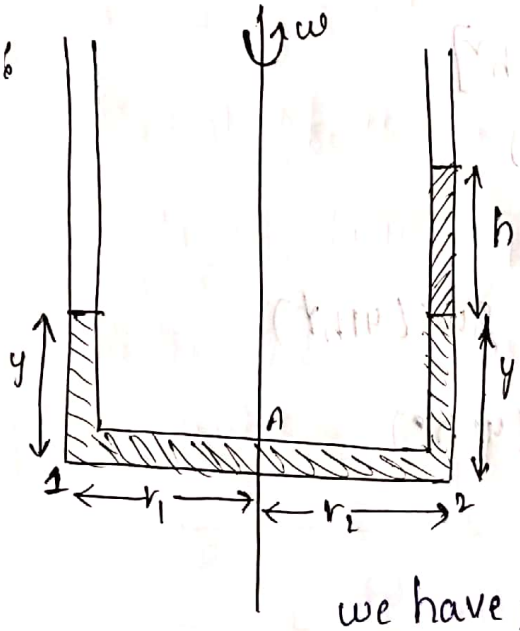
③ $\omega = \sqrt{\frac{9g}{32a}}$




$$P_0 + \frac{4}{3}a\rho g = P_0 + \rho g h_1$$

$$\boxed{h_1 = 4a/3}$$

Eg:




Find diff of levels of liquid in the U-tube.

Sol: $P_1 - P_A = \frac{1}{2}\rho\omega^2(r_1^2 - 0^2)$

$P_2 - P_A = \frac{1}{2}\rho\omega^2(r_2^2 - 0^2)$

we have, $P_1 = P_0 + \rho g y$

$P_2 = P_0 + \rho g h + \rho g y$

$\rho g h = \frac{1}{2}\rho\omega^2(r_2^2 - r_1^2)$

$$\boxed{h = \frac{\omega^2(r_2^2 - r_1^2)}{2g}}$$

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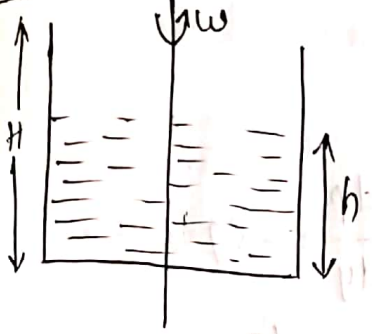
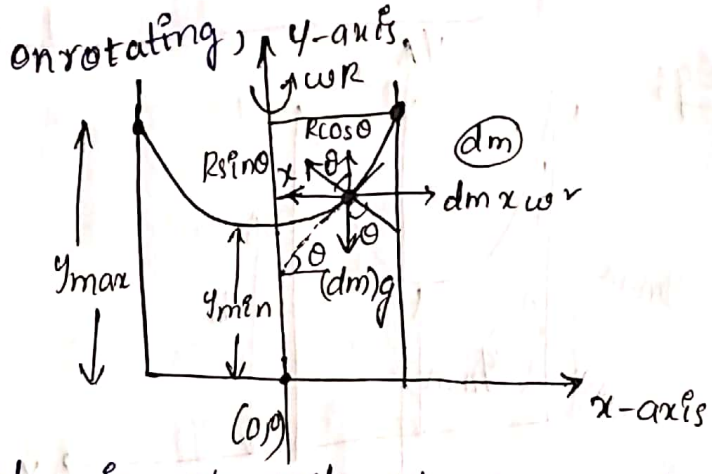
# Pressure in a fluid of rotating vessels




since fluid particle is at rest w.r.to vessel,

$$R\cos\theta = (dm)g$$

$$R\sin\theta = (dm)\,r\omega^2$$

$$\boxed{\tan\theta = \frac{x\omega^2}{g}}$$

we have, slope of tangent, $\tan\theta = \frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{x\omega^2}{g}$$

$$dy = x \cdot dx \frac{\omega^2}{g}$$

$$y = \frac{\omega^2 x^2}{2g} + c \quad -①$$

$\therefore$ At $x = 0$, we have $y = y_{min}$, $\therefore y_{min} = c$

$$\boxed{y = \frac{x^2\omega^2}{2g} + y_{min}} \Rightarrow \text{eqn of parabola}$$

when $x = R$, then $y = y_{max}$

$$y_{max} = \frac{R^2\omega^2}{2g} + y_{min}$$

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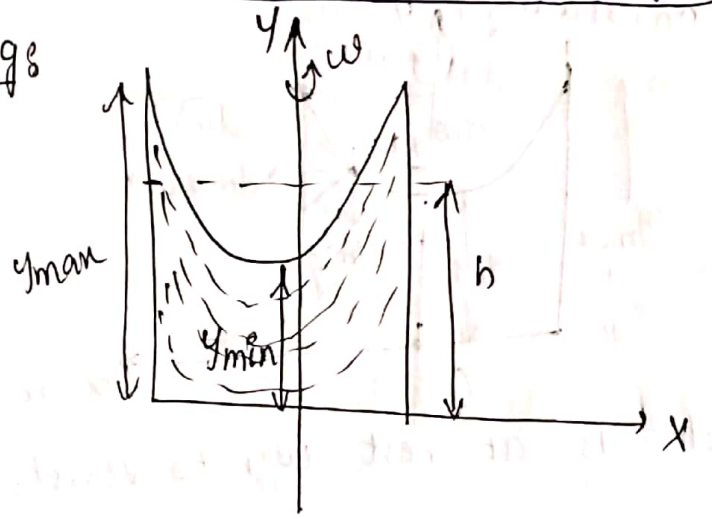
$$y_{max} - y_{min} = \frac{R^2\omega^2}{2g}$$

Eqs




$$\frac{dp}{dy} = -\rho g$$

$$\frac{dp}{dx} = \rho a_x$$

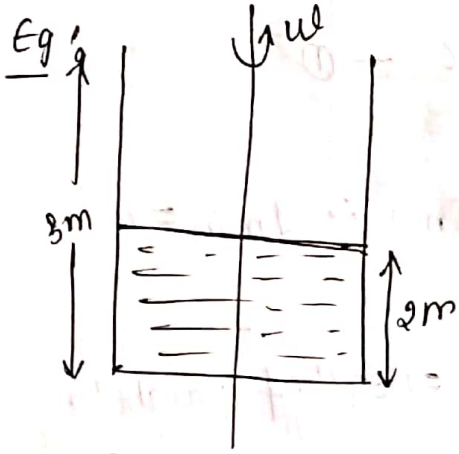
Imp points:

① $y_{max} - h = h - y_{min}$ [when there is no spill of water]

$$h = \frac{y_{max} + y_{min}}{2}$$

② The vol of the paraboloid (due to rotation of parabola) is always equal to half of the vol of cylinder circumscribing that parabola.

Eg:




Find

① Max angular velocity of rotation so that no liquid can spill over

② the angular velocity ($\omega$) so that base is just visible.

③ the angular velocity ($\omega$) so that half of base area is just visible.

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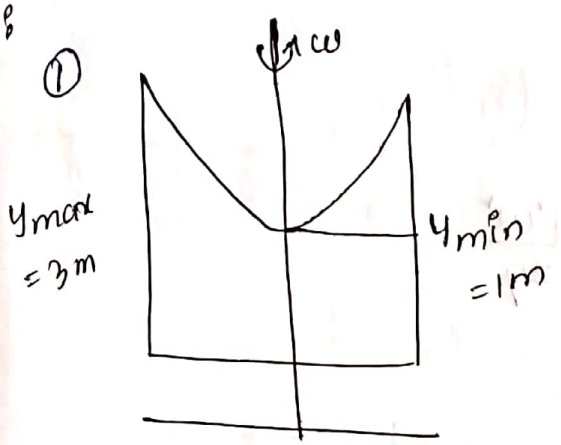
④ Angular velocity ($\omega$) for which half of the initial vol of liquid left in the vessel.

⑤ If final volume is 80% of the intial volume then find ($\omega$).

Sol:

①



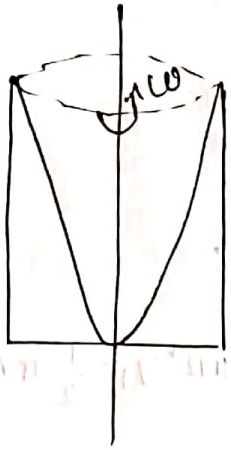

$$y_{max} - y_{min} = \frac{R^2\omega^2}{20}$$

$$40 = \omega^2$$

$$\omega = \sqrt{40}$$

$$\boxed{\omega = 2\sqrt{10}\ rad/s}$$

②




here
$y_{min} = 0$
$y_{max} = H = 3m$

$$3 - 0 = \frac{R^2\omega^2}{20}$$

$$\omega = \sqrt{60} = 2\sqrt{15}\ rad/s$$

here vol of liquid left in the vessel.

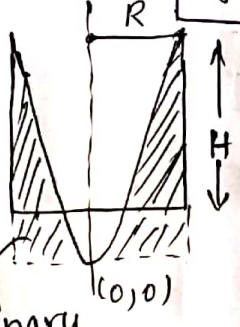
$$V_f = \frac{1}{2}\pi R^2(3) = \frac{3\pi}{2}$$

& intial vol of liq $= \pi(1)(2) = 2\pi$

$$\frac{V_f}{V_i} = \frac{3\pi/2}{2\pi} = \frac{3}{4} = 0.75$$

$$\boxed{V_f = 75\%\ V_i}$$

③




If origin is taken at vertex

$$y = \frac{x^2\omega^2}{2g}$$

$$x^2 = \frac{2gy}{\omega^2} \Rightarrow \boxed{x^2 = 4ay}$$

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Given,

$r^2 = 4ay_1$

$R^2 = 4a(H+y_1)$

$\frac{1}{2} = \frac{y_1}{H+y_1}$

$\boxed{H = y_1 = 3m}$

Given,

$\pi r^2 = \frac{1}{2}\pi R^2$

$r^2 = \frac{R^2}{2}$

now, $y_{max} \div y_{min} = \frac{R^2\omega^2}{2g}$

$6 - 0 = \frac{\omega^2}{20}$

$\omega = \sqrt{120}$

$\boxed{\omega = 2\sqrt{30}\ rad/s}$

vol of liquid left in the vessel :

$V_f = \frac{1}{2}\pi R^2(y_1+H) - \left[\pi R^2 y_1 - \frac{1}{2}\pi r^2 y_1\right]$

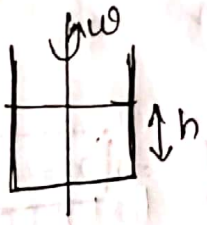
$= \frac{1}{2}\pi(1)(6) - \pi(1)(3) + \frac{1}{2}\pi\left(\frac{1}{2}\right)(3)$

$\boxed{V_f = \frac{3\pi}{4}}$

now, $\frac{V_f}{V_i} = \frac{\frac{3\pi}{4}}{2\pi} = \frac{3}{8}$

$\boxed{V_f = \frac{3}{8}V_i}$ less than $\frac{V_i}{2}$

(4)




①

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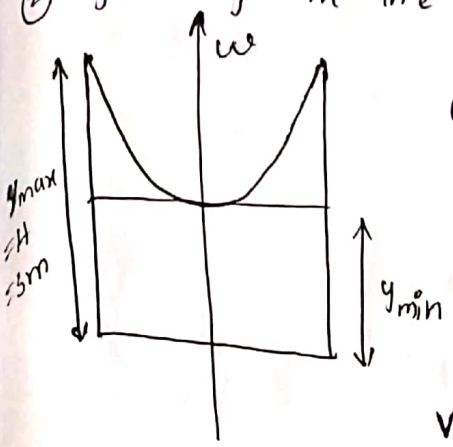
Find 'ω' when base is just visible.

② find $V_f$ in the vessel. now, if $V_f$ is less than $\frac{V_i}{2}$




ⓐ now,

$$H - y_{min} = \frac{R^2\omega^2}{2g} \quad -①$$

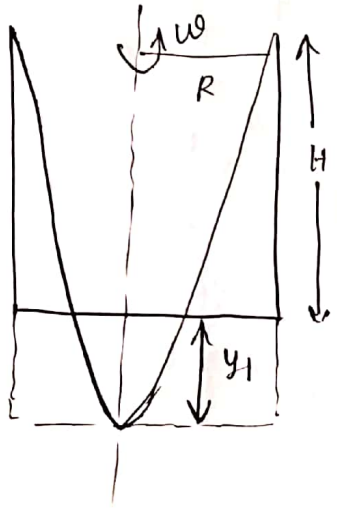
& $V_f = \frac{1}{2} V_i$

$$V_f = \pi R^2 y_{min} + \frac{1}{2}\pi R^2 (H - y_{min})$$

from this we get $y_{min}$.

From eq① we get 'ω'.

ⓑ If $V_f$ is more than $\frac{V_i}{2}$.




& $(H+y_1) = \frac{R^2\omega^2}{2g} \quad -①$

& $V_f = \frac{1}{2} V_i$

$$V_f = \frac{1}{2}\pi R^2 (y_1 + H) - \left[\pi R^2 y_1 - \frac{1}{2}\pi R^2 y_1\right]$$

from this we get $y_1$

from eq① we get 'ω'.

* $y_{min} + H = \frac{R^2\omega^2}{2g}$ , $y_{min} = \frac{r^2\omega^2}{2g}$

$$\frac{y_{min}}{H + y_{min}} = \frac{r^2}{R^2}$$

$V_f = \pi$

$$\pi = \frac{1}{2}\pi R^2 (3 + y_1) - \pi R^2 y_1 + \frac{1}{2}\pi R^2 \left(\frac{y}{3+y}\right) y_1$$

$$1 = \frac{3}{2} + \frac{y_1}{2} - y_1 + \frac{y_1^2}{(3+y)^2}$$

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$$y_1 - 1 = \frac{y_1^2}{3+y_1}$$

$$3y_1 + y_1^2 - 3 - y = y_1^2$$

$$2y_1 = 3$$

$$\boxed{y_1 = 3/2}$$

$$\omega = \sqrt{90} \text{ rad/s}$$

* $$v = \frac{8\phi}{10\phi}(2\pi) = \frac{8\pi}{5}$$

$$\pi - y = \frac{\omega^2}{20}$$

$$\frac{8\pi}{5} = \pi(1)(y) + \frac{1}{2}\pi(1)(3-y)$$

$$\frac{8}{5} = y + \frac{3}{2} - \frac{y}{2}$$

$$-\frac{y}{2} = \frac{3}{2} - \frac{8}{5}$$

$$\pm\frac{y}{2} = \pm\frac{1}{10}$$

$$\boxed{y = \frac{1}{5}}$$

$$\frac{14}{5} = \frac{\omega^2}{20}$$

$$\boxed{\omega = 2\sqrt{14} \text{ rad/s}}$$

→ The fig shows a semi-cylindrical massless gate of unit length ⊥ler to the plane of the page & is pivoted at the point 'O' holding a stationary

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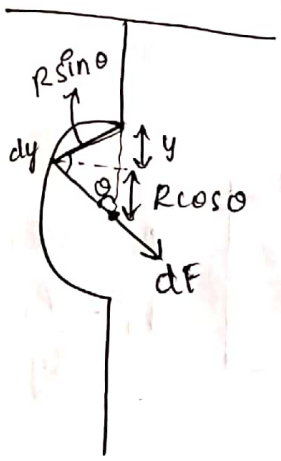
liquid of density $\rho$. A horizontal force F is applied at lowest position to keep it stationary.

F is ____




$\tau = F \times 2R$

$dF = \rho g (R+y) \times dA$

$dA = (\text{width})\, dy = dy = R d\theta$

$dF = \rho g (R+y) \cdot R d\theta$

$d\tau = dF \cdot r_{\perp}$

$$\int d\tau = \int_0^{\pi} \rho g (R+y) \cdot R d\theta \times R\sin\theta$$

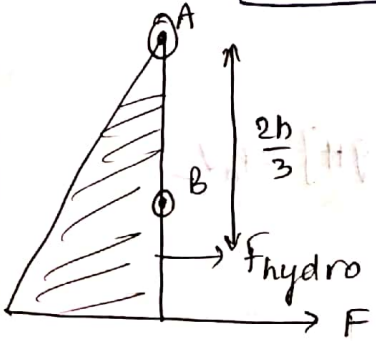
$y = R(1-\cos\theta)$

$$\tau = \int_0^{\pi} \rho g (R + R(1-\cos\theta)) R^2 \sin\theta \cdot d\theta$$

$\tau = 4 \rho g R^3$

$4 \rho g R^3 = F \times 2R$

$$\boxed{F = 2 \rho g R^2}$$

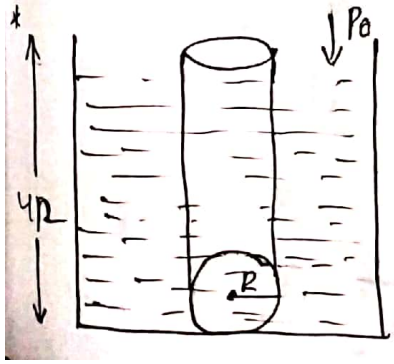



$\tau_A$

$F_{hydro} \left(\frac{2h}{3}\right) = F(h)$

$\tau_B$

$F_{hydro} \left(\frac{2h}{3} - \frac{h}{2}\right) = F\left(\frac{h}{2}\right)$




$F_{thrust}$ on upper surface on the sphere is

$F_{th} = F_{due\ to\ \omega} + F_{due\ to\ P_o}$

$= \left( (\pi R^2)(3R) - \frac{2}{3}\pi R^3 \right) \rho g + P_o \times \pi R^2$

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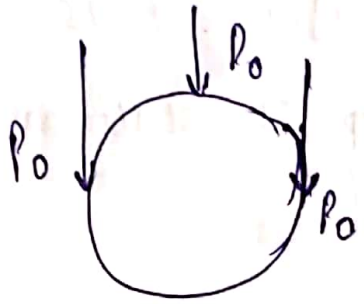
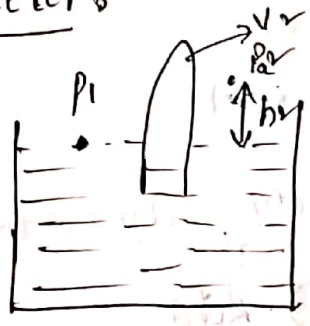



$F = P_0 \times \pi R^2$

## Barometer:

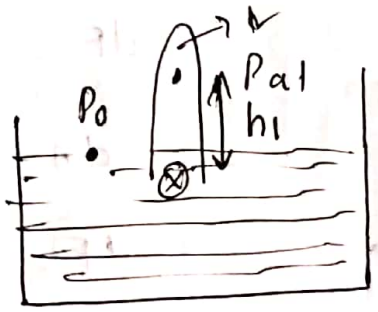



$P_1 = \rho g h_2$

$P_1 < P_0$

$\rho g h_2 < \rho g h_1$

$h_2 < h_1$




$P_0 - \rho g h_1 = P_{a1}$

$P_1 - \rho g h_1 = P_{a2}$

$P_1 + \rho_{air} g H = P_0$

$\rho g h_2 + \rho_{air} g H = \rho g h_1$

temp $\longrightarrow$ const

$P_1 V_1 = P_2 V_2$

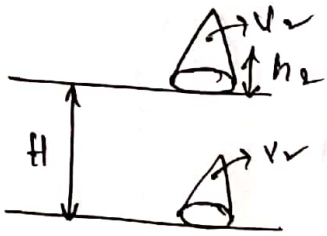
$[P_0 - \rho g h_1] V_1 = [P_1 - \rho g h_2]$

$P_1 V_1 = V_2$

$V_1 [P_0 + \rho_{air} g H] = P_0 V_2$

$P_0 = \rho g h_1$

$h_1 = \frac{P_0}{\rho g}$

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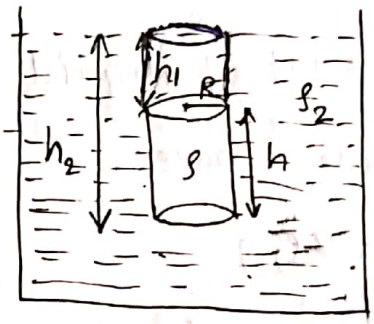
# Force of Buoyancy:

* It is the force acting on immersed body due to pressure difference of surrounding fluid.
* It always acts in upward direction ⊥er to free surface.

eg:

Force on the top face of cylinder due to liquid.




$F_1$ = wt of liq in the cylinder above the top surface.

$$F_1 = \pi R^2 h_1 \rho_2 g \ (\downarrow)$$

(or) $F_1$ = pressure at top surface × Area (for horizontal plane surface)

$$= h_1 \rho_2 g \pi R^2 \ (\downarrow)$$

In general,

$$F_1 = h_1 \rho_2 g \pi R^2 + P_0 \pi R^2$$

$P_0 \pi R^2$ ← projection area.

Force on bottom of cylinder due to liquid,

$F_2$ = wt of liq in the cylinder above the base

$$= \pi R^2 h_2 \rho_2 g + P_0 \pi R^2 \ (\uparrow)$$

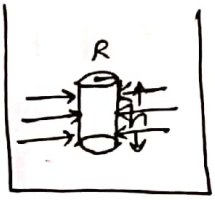
(or) $F_2$ = pressure at bottom × base area

$$= h_2 \rho_2 g \pi R^2 + P_0 \pi R^2 \ (\uparrow)$$

* Buoyant force,

$$F_2 - F_1 = (h_2 - h_1) \rho_2 g \pi R^2$$

$$F_B = h(\pi R^2) \rho_2 g$$




A = area of face $= \pi R^2$

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$$\boxed{F_B = V\rho_l g}$$

where,

V = vol of displaced liquid (vol of body immersed in liq)

$\rho_l$ = density of liquid.

g = acceleration due to gravity.

$\Rightarrow F_B$ = wt of liquid displaced by body.

Again we have,

$$F_B = V\rho_l g$$

$$= V\rho_l g \left(\frac{\rho_b}{\rho_b}\right)$$

$$\boxed{F_B = mg\left(\frac{\rho_l}{\rho_b}\right)}$$ (completely submerged body)

mg = True wt or wt of body in air.

now,

wt of body in liq ($w_{app}$),

$$w_{app} = mg - F_b$$

$$\boxed{w_{app} = mg\left(1 - \frac{\rho_l}{\rho_b}\right)}$$

If $\rho_l = \rho_b$ then $\boxed{w_{app} = 0}$ *

* Buoyancy force acts as an interfnal force b/w the liquid & body.

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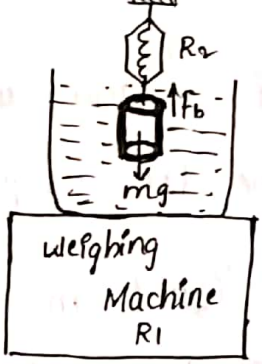
Eg:

Reading $R_1$

= wt of (beaker + liq)

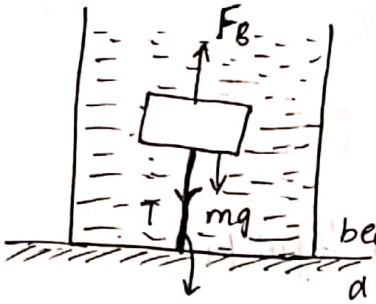
$+F_B$




Reading $R_2$

$= mg - F_B$

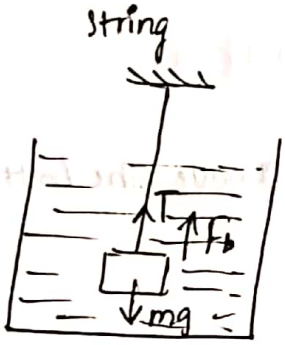
Eg:




$(\rho_b < \rho_l)$

$$T = F_B - mg$$

Eg:

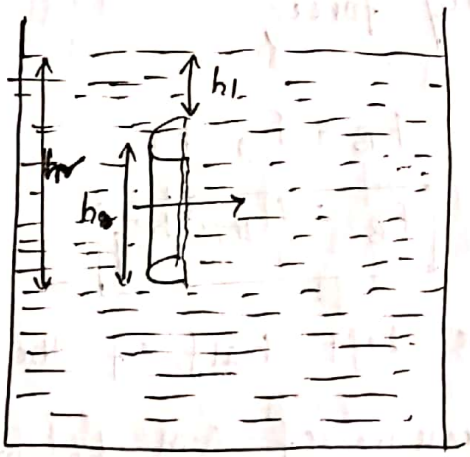


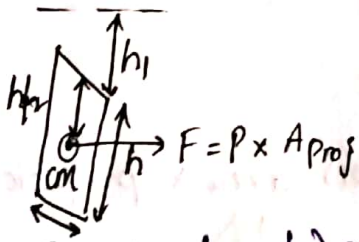

$T + F_b = mg$

$$T = mg - F_b$$

## Force on left half of the cylinder:

F = pressure at CM of projected area × projection area




$F = P \times A_{proj}$

$$F = \left(h_1 + \frac{h}{2}\right) \rho_l g (2Rh)$$

* In eqm,

$$F_B = mg$$

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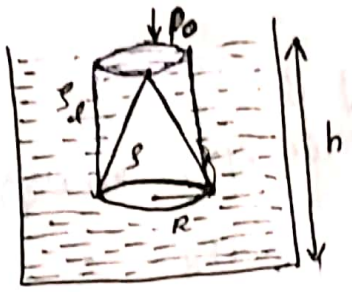
Eg:

Force on curved surface of cone due to liquid,




$F_1$ = wt of liquid in the cylinder above the curved surfaced.

$$F_1 = \left(\pi R^2 h - \frac{1}{3}\pi R^2 h\right)\rho_l g + P_0(\pi R^2)$$

$$= \frac{2}{3}\pi R^2 h + P_0 \pi R^2$$

* force on the bottom of cone due to liquid,

$$F = \frac{2}{3}\pi R^2 h \rho_l g$$

$F_2$ = wt of liquid in the cylinder above the bottom,

$$\boxed{F_2 = \pi R^2 h \rho_l g}$$

(or) $F_2$ = pressure of bottom x Area of base.

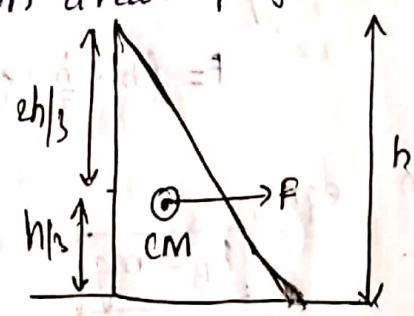
* buoyancy force,

$$F_B = F_2 - F_1$$

$$\boxed{F_B = \frac{1}{3}\pi R^2 h \rho g}$$

force on one left half of the cone,

F = pressure of C.M of projection area x projection area.




~~$F = \frac{1}{3}\rho_l g(\frac{1}{2}\pi R^2)$~~

$$F = \frac{2h}{3}\rho_l g\left(\frac{1}{2}\right)(h)(2R) = \frac{2h^2 R \rho g}{3}$$

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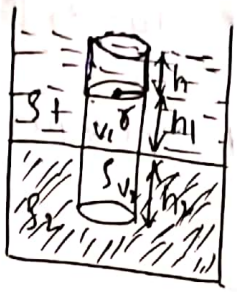
Eg:




Immiscible liquids

$\rho_1 < \rho < \rho_2$.

* force due to liquid on the top surface,

$F_1$ = wt of liq in cylinder above the top surface.

$= \pi R^2 h \rho_1 g + P_0 \pi R^2 \ (\downarrow)$

or $F_1 = (\rho_1 g h)\pi R^2 + P_0 (\pi R^2) \ (\downarrow)$

* force due to liquid on the bottom of cylinder,

$F_2 = \pi R^2 (h+h_1) \rho_1 g + \pi R^2 h_2 \rho_2 g + P_0 \pi R^2 \ (\uparrow)$

(or)

$F_2 = [(h+h_1)\rho_1 g + h_2 \rho_2 g]\pi R^2 + P_0 \pi R^2 \ (\uparrow)$

now, $\boxed{F_B = F_2 - F_1}$

$F_B = \pi R^2 h_1 \rho_1 g + \pi R^2 h_2 \rho_2 g$

$\boxed{F_B = V_1 \rho_1 g + V_2 \rho_2 g}$

* In eqm,

$F_B = mg$

$V_1 \rho_1 + V_2 \rho_2 = (V_1 + V_2)\rho$

$\rho = \dfrac{V_1 \rho_1 + V_2 \rho_2}{V_1 + V_2}$ $\boxed{\rho = \dfrac{V_1 \rho_1 + V_2 \rho_2}{V_1 + V_2}}$

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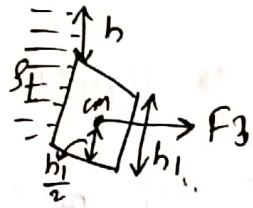
force on left half of the cylinder due to liquid,

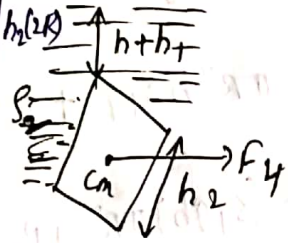
F = pressure of cm of projected area × projection area

here,

$F_3 = \rho_1 g(h + h_1/2) h_1 (2R)$ [due to $\rho_1$]

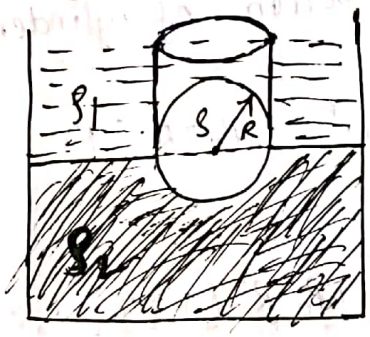



Eq $F_4 = [\rho_1 g(h+h_1) + \rho_2 g(\frac{h_2}{2})] h_2 (2R)$ [due to $\rho_2$]




∴ net force, $F_5 = F_3 + F_4 + P_0 (2R)(h_1+h_2)$.

Eq:




Clearly, $\rho_1 < \rho < \rho_2$.

* force on top curved surface of sphere due to liquid.

$F_1$ = wt of liq in the cylinder above the curved surface.

$= (\pi R^2 h - \frac{2}{3}\pi R^3)\rho_1 g + P_0 \pi R^2$

* force on lower half of curved surface

$F_2 = \pi R^2 h \rho_1 g + \frac{2}{3}\pi R^3 \rho_1 g + P_0 \pi R^2$ (↑)

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now,

buoyancy force $F_b = F_2 - F_1$

$$F_B = \frac{2}{3}\pi R^3 \rho_1 g + \frac{2}{3}\pi R^3 \rho_2 g = V_1 \rho_1 g + V_2 \rho_2 g$$

*In eqm,

$$F_b = mg$$

$$V_1 \rho_1 + V_2 \rho_2 = (V_1 + V_2)\rho$$

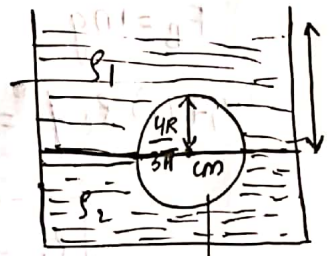
$$\boxed{\rho = \frac{V_1 \rho_1 + V_2 \rho_2}{V_1 + V_2}}$$

force on left half of the sphere,

F = pressure of CM of projected area x area of projection

$$F_3 = \left(h - \frac{4R}{3\pi}\right)\rho_1 g \left(\frac{\pi R^2}{2}\right)$$

[due to liquid $\rho_1$]




Semicircular disc

& $F_4 = \left(h\rho_1 g + \frac{4R}{3\pi}\rho_2 g\right)\left(\frac{\pi R^2}{2}\right)$

∴ Net force,

$$F_B = F_3 + F_4 + P_0(R\pi^2)$$

Eg: Two objects of equal volumes & densities $\rho_1 = 400\ kg/m^3$ & $\rho_2 = 600\ kg/m^3$ have identical flat portions of area $S = 100\ cm^2$ on their surfaces. These flat portions are glued to eachothes. Thus composite body

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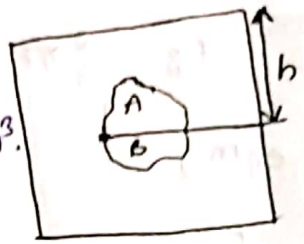
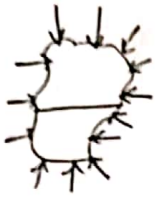
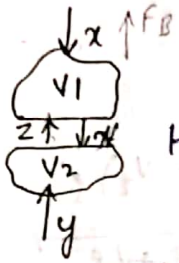
is formed. If glue can with stand a force F = 500N. at what height 'h' in the liq can the common portion be in eqm.




Sol$^n$

$V_1 = V_2 = 1m^3$, $\rho_1 = 400 kg/m^3$, $\rho_2 = 600 kg/m^3$.




Here $z = h \rho_l g S$

we have,

$$z - x = V_1 \rho_l g \quad -①$$

& $$y - z = V_2 \rho_l g \quad -②$$

$\therefore$ eqn ① + eqn ②

$$F_B = y - x = (V_2 + V_1) \rho_l g \quad -③$$

In eqm,

$$F_B = mg$$

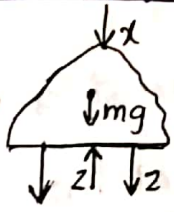
$$(V_1 + V_2) \rho_l g = (V_1 \rho_1 + V_2 \rho_2) g$$

$$\boxed{\rho_l = \frac{V_1 \rho_1 + V_2 \rho_2}{V_1 + V_2}}$$

$$\rho_l = \frac{\rho_1 + \rho_2}{2} = \frac{10^3}{2}$$

$$\boxed{\rho_l = 500 kg/m^3}$$

Again,




here $z - x = V_1 \rho_l g$

& $z = h \rho_l g_1 S$

F = real force

In eqm, $z - x = F + m_1 g + z$

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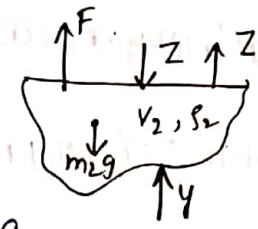
$V_1 \rho_l g + = F + h \rho_l g s + V_1 \rho_1 g$

$(1)(500)(10) = 500 + h \times 500 \times 10 \times 10^{-2} + 1 \times 1000$

$500 = h \times 500 \times 10 \times 10^{-2}$

$\boxed{h = 10m}$




$F + (y-z) + z = m_2 g$

$F + V_2 \rho_l g + h \rho_l g s = V_2 \rho_2 g$

$500 + 1 \times 500 \times 10 + h \times 500 \times 10 \times 10^{-2} = 1 \times 600 \times 10$

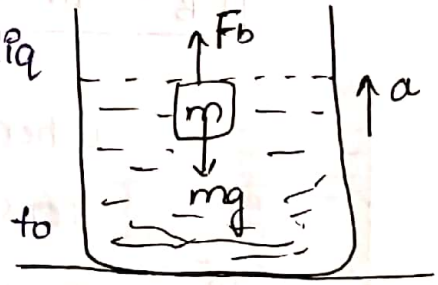
$1 + h \times 0.1 = 2$

$\boxed{h = 10m}$

## Buoyancy Force in accelerating frames

(1)

* block of mass 'm' kept in a liq of density '$d_l$'.
* when acceleration 'a' is given to container as shown,




$F_B - mg = ma$

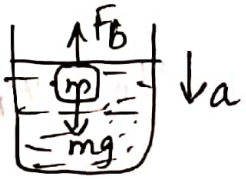
$F_B = m(g+a)$

$\boxed{F_B = V_{im} \rho_l (g+a)}$

Here body sinks in the liquid

$\therefore d_b = d_l$.

(2)




$mg - F_b = ma$

$F_b = m(g-a)$

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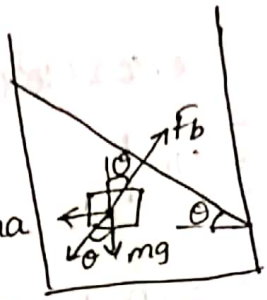
$$\boxed{F_B = V_{im} d_l (g-a)}$$

③ w.r.to container block is at rest.

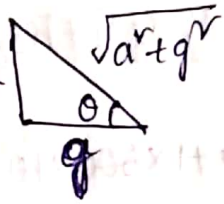



$\rightarrow a = g\tan\theta$

$F_B = ma\sin\theta + mg\cos\theta$

$F_B = mg\tan\theta\sin\theta + mg\cos\theta$

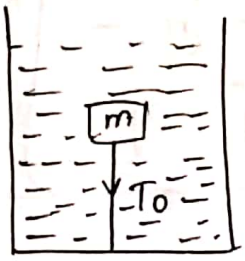
$= \frac{mg}{\cos\theta} = \frac{mg}{g}\sqrt{a^2+g^2}$




$F_B = m\sqrt{a^2+g^2}$

$$\boxed{F_B = V_{im} d_l \sqrt{g^2+a^2}}$$

In general, $\boxed{F_B = V_{im} d_l g_{eff}}$

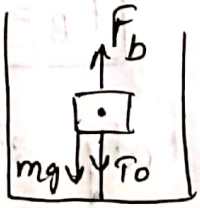
Eg: 


when the beaker is at rest, tension in the string = $T_0$

If the beaker moves upwards with acceleration 'a'. Find tension in the string.

Clearly, $\boxed{d_b < d_l}$

when beaker is at rest,




$F_B = mg + T_0$

$T_0 = F_B - mg$

$T_0 = V_{im} d_l g - mg$

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$$= \left(\frac{m}{d_B}\right) d_l\, g - mg$$

$$T_0 = mg\left(\frac{d_l}{d_B} - 1\right) \quad -①$$

when beaker is moving up with acceleration 'a'.

$$F_B - mg - T = ma$$

$$T = F_b - m(g+a)$$

$$T = v_{im}\, d_l\, (g+a) - m(g+a)$$

$$= \frac{m}{d_B} d_l (g+a) - m(g+a)$$

$$T = m(g+a)\left(\frac{d_l}{d_B} - 1\right)$$

$$= \frac{T_0}{g}\left(\frac{d_l}{d_B} - 1\right)(g+a)$$

$$\boxed{T = T_0\left(1 + \frac{a}{g}\right)} \quad *$$

* If the string is cut, then $\boxed{T = 0}$

∴ acceleration of block,

$$a' = \frac{F_b - mg}{m} = \frac{m\left(\frac{d_l}{d_B}\right)(g+a) - mg}{m}$$

$$a' = \frac{d_l}{d_B}(g+a) - g$$

* acceleration of block w.r.to container,

$$a_{rel} = a' - a$$

we have, $s_{rel} = u_{rel}^{0} t + \frac{1}{2} a_{rel} t^2$

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time taken by the block to come out of the liquid,

$$t = \sqrt{\frac{2s_{rel}}{a_{rel}}}$$

* velocity of block w.r.to container vehicle coming out of surface,

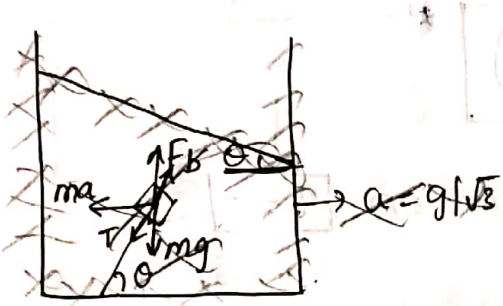
$$v_{rel}^2 - u_{rel}^2 = 2a_{rel} \cdot s_{rel} \quad (u \to 0)$$

$$v_{rel} = \sqrt{2 a_{rel} \cdot s_{rel}}$$

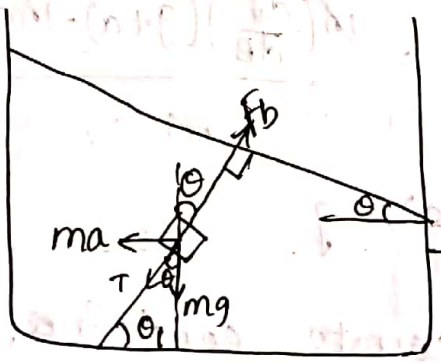
$$v_{rel} = v_{block} - v_{container}$$

$$\text{[} v_c = at \text{]}$$

Eg:




Eg:




$d_b = \frac{1}{2} d_l$ · Find

① tension in the string

② angle $\theta_1$

$$\tan\theta = \frac{a}{g} = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

we have, $\theta + \theta_1 = 90^\circ$

$$\theta_1 = 60^\circ$$

~~ueber~~

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w.r.to container, block is at rest.

$$F_B = T + ma\sin\theta + mg\cos\theta$$

$$\frac{m}{d_B} \times d_\ell \sqrt{g^2+a^2} = T + \frac{mg}{\sqrt{3}}\sin 30^\circ + mg\cos 30^\circ$$

$$\frac{4m}{3}\sqrt{g^2+a^2} = T + \frac{mg}{2\sqrt{3}} + \frac{mg\sqrt{3}}{2}$$

$$4m\sqrt{g^2+a^2} = 2mg\left(\sqrt{3}+\frac{1}{\sqrt{3}}\right) = 2T$$

$$4m\sqrt{\frac{4g^2}{3}} - mg\left(\frac{4}{\sqrt{3}}\right) = 2T$$

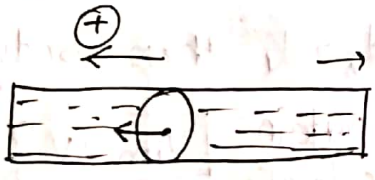
$$\frac{4m \times 2g}{\sqrt{3}} - mg\frac{4}{\sqrt{3}} = 2T$$

$$\boxed{T = \frac{2mg}{\sqrt{3}}}$$

* when string is cut, $T = 0$

$$a_{rel} = \frac{F_B - (ma\sin\theta + mg\cos\theta)}{m} = \frac{2g}{\sqrt{3}}$$

Eg: A ball of relative density $\frac{1}{4}$ is taken in a tube enclosed with water. Tube is given forward acceleration $a = g$. Find immediate acceleration of the ball w.r.to tube.




Sol: $ma - F_B = ma'$

$$\rho d_B a - \rho d_w a = \rho d_B a'$$

$$\Rightarrow a' = a\left(1 - \frac{d_w}{d_B}\right) = a(1-4)$$

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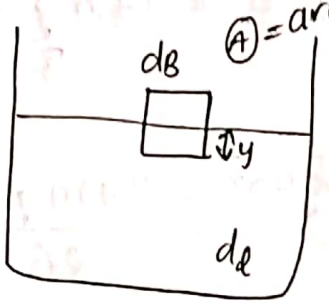
$a' = -3a = -3g$

$\boxed{a' = 3g}$, forward

## Laws of floatation :

In case of floating,




$V_{im} + V_{out} = V$

$F_B = mg$

$V_{im}\, \rho_l \times g = mg$

$V_{im}\, \rho_l = V\, d_B$

* fraction of vol of the body submerged in liquid

$$\boxed{\frac{V_{im}}{V} = \frac{d_B}{d_l} = f_{sub}}$$

* fraction of vol of body out side the liquid.

$$f_{out} = \frac{V_{out}}{V} = \frac{V - V_{in}}{V} = 1 - \frac{V_{in}}{V}$$

$$\boxed{f_{out} = 1 - \frac{d_B}{d_l}}$$

Points:

1) If $d_B < d_l$ then $\boxed{V_{im} < V}$

2) If $d_B = d_l$ then $\boxed{V_{im} = V}$

3) If $d_B > d_l$ then body completely sinks to bottom with certain acceleration. $\boxed{V_{im} > V}$

→ rest

$$a = \frac{F_{net}}{m} = \frac{w - F_B}{m} = \frac{V d_B g - V d_l g}{V d_B}$$

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$$\boxed{a = g\left(1 - \frac{d_1}{d_B}\right)}$$

* block hits the bottom with velocity,

$$\boxed{v = \sqrt{2as}}$$

* time taken by the block to hit the bottom,

$$s = \frac{1}{2} a t^2$$

$$\boxed{t = \sqrt{\frac{2s}{a}}}$$

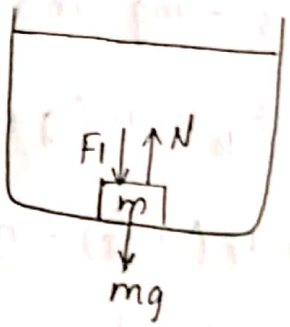
* Now, if there is no liquid blw block & base of beaker,

force exerted by beaker on block is,

$$\boxed{N = mg + F_1}$$

$$F_1 = N - mg$$

$$\boxed{F_1 = P_1 A} + mg$$

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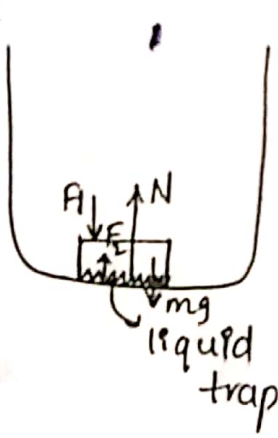



In this case,

$$N + F_2 = F_1 + mg$$

$$N = mg - (F_2 - F_1)$$

$$\boxed{N = mg - F_B}$$

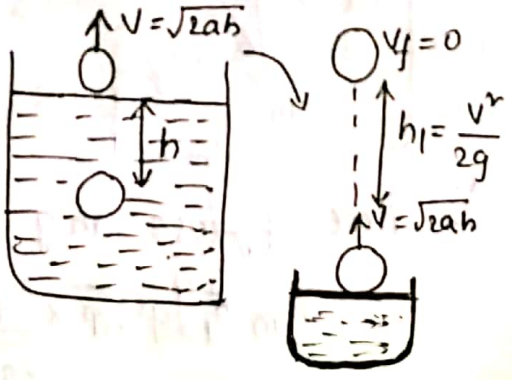
Eg:

$$d_{Ball} = \frac{1}{2} d_l$$

acceleration of ball,

$$a = \frac{F_B - mg}{m}$$

$$= V d_l g - V d_B g \,/\, V d_B$$

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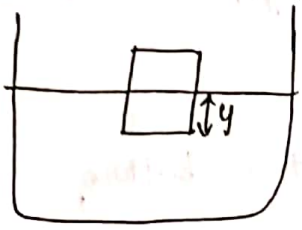
$$\boxed{a = \left(\frac{d_l}{d_B} - 1\right) g} \quad (\uparrow)$$

** Important point:



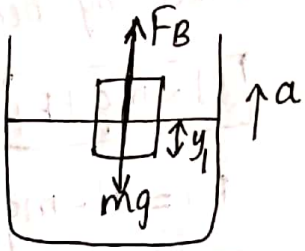

when beaker is at rest,

$$F_B = mg$$

$$V_{im} \rho_l g = mg$$

$$A y d_l = m \quad -①$$

① when beaker is moving with acceleration a,




$$F_B - mg = ma$$

$$F_B = m(g+a)$$

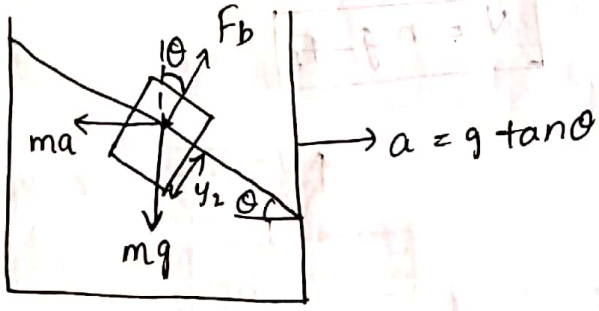
$$A y_1 d_l (g+a) = m(g+a)$$

$$A y_1 d_l = m \quad -②$$

from eqn ①, ②

$$\boxed{y = y_1}$$

②




here $F_B \cos\theta = mg$

$$V_{im} d_l \sqrt{g^2+a^2} \times \frac{g}{\sqrt{g^2+a^2}} = mg$$

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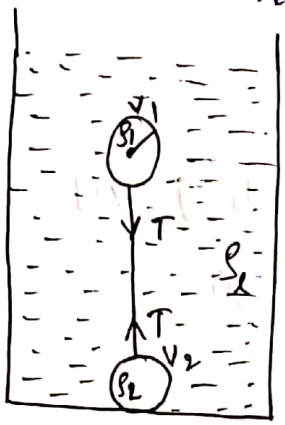
$$A y_2 d\ell = m \quad -③$$

from eqn ① & ③,

$$\boxed{y = y_2}$$

i.e, length of the submerged part is always remains same even beaker is moving with acceleration.

Eg:




$[\rho_2 > \rho_1]$. Find tension in the string.

Sol: for upper body:

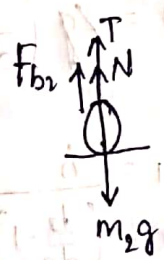
$$F_{B_1} = m_1 g + T$$

$$V_1 \rho_2 g = V_1 \rho_1 g + T \quad -①$$

for lower body:

$$F_{B_2} + T + N = m_2 g$$

$$V_2 \rho_1 g + T + N = V_2 \rho_2 g \quad -②$$




Solving ① & ② we get T & N.

* If $F_{B_1} + F_{B_2} > (m_1 + m_2) g$

here 'T' is internal force,

$$a = \frac{(F_{b_1} + F_{b_2}) - (m_1 + m_2) g}{m_1 + m_2}$$

$$a = \frac{(V_1 \rho_1 + V_2 \rho_1) g}{m_1 + m_2} - g$$

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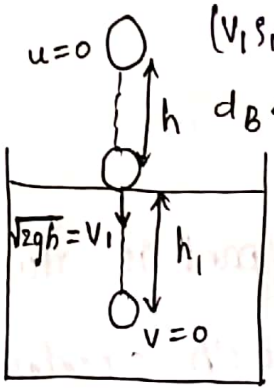
$$a = \frac{(V_1+V_2)\rho_1 g}{(V_1\rho_1+V_2\rho_2)} - g$$




Eg: Find maximum depth up to which body sinks in liquid.

M-I :

In liquid,

$$a = \frac{F_{net}}{m} = \frac{F_B - w}{m} = \frac{\left(\frac{m}{d_B}\cdot d_l\right)g - mg}{m}$$

$$\boxed{a = \left(\frac{d_l}{d_B} - 1\right)g} \quad (\uparrow)$$

we have,

$$v^2 - a^2 = 2gh_1$$

$$0^2 + \left(\sqrt{2gh}\right)^2 = +2g\left(\frac{d_l}{d_B} - 1\right)h_1$$

$$2gh = 2g\left(\frac{d_l}{d_B} - 1\right)h_1$$

$$h_1 = \frac{h}{\left(\frac{d_l}{d_B} - 1\right)}$$

$$\boxed{h_1 = \frac{d_B\, h}{d_l - d_B}}$$

M-II :

W-E Theorem :-

$$W_g + W_{F_B} = \Delta K^0$$

$$mg(h+h_1) + (-F_B h_1) = 0$$

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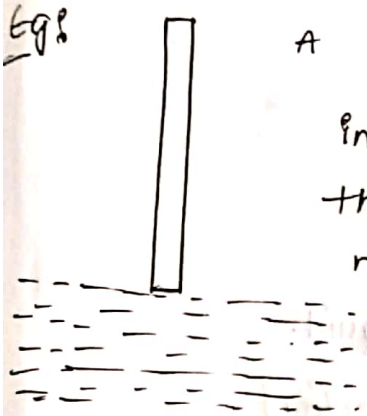
$$\rho g(h+h_1) = \frac{m}{d_B} \times d_1 \times g h_1$$

$$h + h_1 = \frac{d_l}{d_B} \times h_1$$

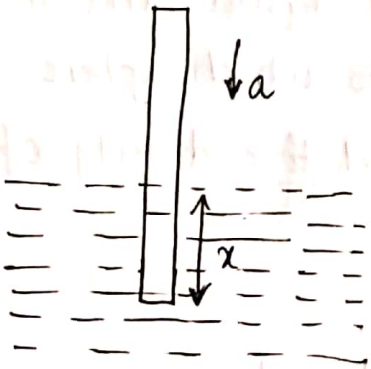
$$h = \left(\frac{d_l}{d_B} - 1\right) h_1$$

$$\boxed{h_1 = \frac{d_B h}{d_l - d_B}}$$

Eg: 

A uniform rod of length $l = 4m$ is released into deep lake by touching one end to the liquid surface as shown. Find max displacement of rod. [given $\frac{\sigma}{\rho} = \frac{1}{2}$]

At any intermediate state,




$$F_{net} = ma$$

$$mg - F_B = ma$$

$$Alg - Ax\rho g = Al\sigma a$$

$$a = g\left(1 - \frac{x\rho}{l\sigma}\right)$$

$$a = g\left(1 - \frac{2x}{l}\right)$$

$$\int_0^{v=0} v\,dv = \int_0^{x_{max}} g\left(1 - \frac{2x}{l}\right) dx$$

$$\frac{v^2}{2}\Big|_0^0 = g\left[x_m - \frac{2x_m^2}{2l}\right] \Rightarrow x_m = \frac{x_m^2}{l}$$

$$\boxed{x_{max} = l = 4m}$$

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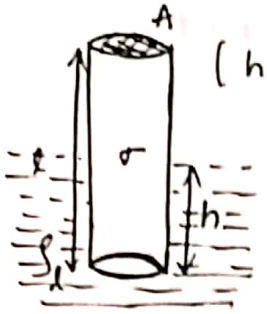
Eg: (here area of crossection is uniform)




body is floating partially in a liquid,

$$F_B = W$$

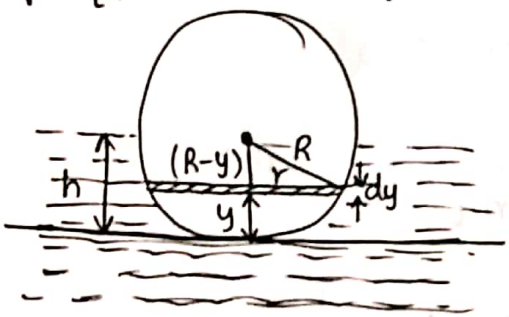
$$V_{im}\, \rho_L g = mg$$

$$Ah\, \rho_L g = Al\sigma g$$

$$\boxed{h = \frac{l\sigma}{\rho_L}}$$

depth upto which body sinks in liquid.

Eg: [for non-uniform Area of cross-section]




A sphere of radius R=2feet is flowing in liquid. Find the depth upto which sphere sinks if relative density of sphere = 0.4.

* Vol of small elementary disc,

$$dV = \pi r^2 dy$$

where,

$$R^2 = (R-y)^2 + r^2$$

$$r^2 = R^2 - (R-y)^2$$

$$= 2Ry - y^2$$

$$dv = \pi (2Ry - y^2) \cdot dy$$

∴ Total immersed volume,

$$V_{im} = \int dv = \pi \int_0^h (2Ry - y^2) \cdot dy$$

$$= \pi \left[ Ry^2 - \frac{y^3}{3} \right]_0^h = \pi \left( Rh^2 - \frac{h^3}{3} \right) = \frac{\pi}{3}(3Rh^2 - h^3)$$

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$$V_{im} = \frac{\pi}{3} h^2 (3R - h)$$

For a floating body,

$$F_b = mg$$

$$V_{im} d_l g = \frac{4}{3}\pi R^3 . d_b g$$

$$\frac{\pi}{3} h^2 (3R-h) \cdot d_l = \frac{4}{3}\pi R^3 . d_b$$

$$h^2 (3R-h) = 0.4 R^3 \times 4$$

$$3Rh^2 - h^3 - 0.4R^3 \times 4 = 0$$

$$h^3 - 3Rh^2 + 0.4R^3 \times 4 = 0$$

$$h^3 - 6h^2 + 3.2 \times 4 = 0$$

$$10h^3 - 60h^2 + 32 \times 4 = 0$$

$$5h^3 - 30h^2 + 16 \times 4 = 0$$

$$5h^3 - 30h^2 + 64 = 0$$

$$h \approx 1.75 \text{ feet}$$

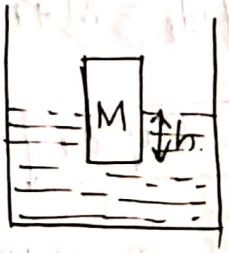
Eg:

Case (i):

$$F_b = W$$

$$Ah s_l g = Mg$$

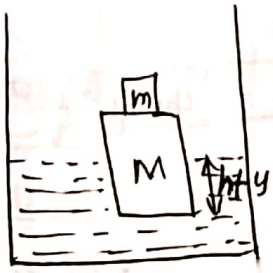
$$M = Ah s_l \quad -(1)$$




case (ii):

$$F_b' = w'$$

$$A(h+y) s_l g = (M+m) g$$

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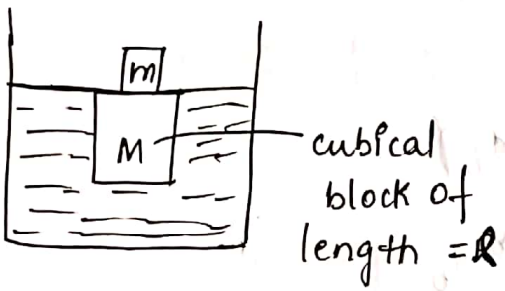
$A\cancel{h}\rho g + Ay\rho g = \cancel{M} + m$

$m = Ay\rho g$

$\boxed{m \propto y}$

Eg: 

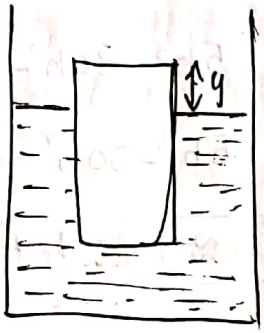

$F_B = W$

$V d_\ell g = (M+m) g$

$\ell^3 d_\ell g = (M+m) g$

$\ell^3 d_\ell = M+m \quad \text{—①}$

on remaining mass,



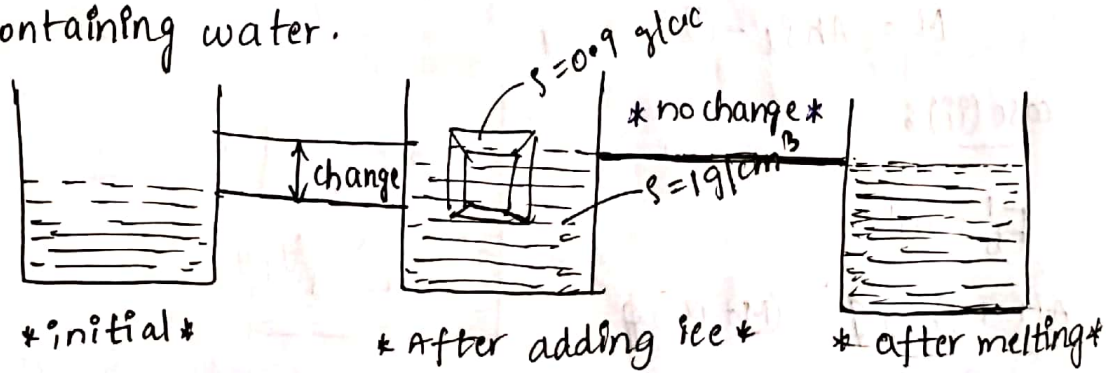

$\ell^2 (\ell - y)\, d_\ell g = Mg$

$\ell^2 (\ell - y)\, d_\ell = M$

$\cancel{\ell^3 d_\ell} - \ell^2 y \cdot d_\ell = \cancel{M+m} - m$

$\boxed{m = \ell^2 y \cdot d_\ell}$

Eg: Ice of mass m-grams is taken in a beaker containing water.

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In floating case,

$$F_B = mg$$

$$V_{disp}\, \rho_w\, g = m g$$

$$\boxed{V_{dis} = \frac{m}{\rho_w}}$$

After melting the ice,

$$m_{ice} = m_{water}$$

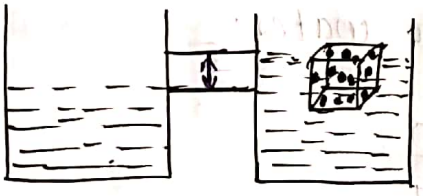
here volume of water formed due to melting of ice,

$$V_f = \frac{m_{ice}}{\rho_w} \quad -(2)$$

from eqn (1) & (2)

water level does not change on melting of ice.

Eg: Ice containing Impurities:



In floating case,

$$F_B = W$$

$$V_{disp}\, \rho_w\, g = (M+m)g$$

$$V_{dip} = \frac{M}{\rho_w} + \frac{m}{\rho_w} \quad -(1)$$

After melting the ice,

Vol of water formed,

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$$V_f = V_1 + V_2$$

where, $V_1 = \frac{M}{\rho_w}$ & $V_2 = \frac{m}{\rho_w}$

$$\therefore V_F = \frac{M}{\rho_w} + \frac{m}{\rho_w} \quad -(2)$$

~~$V_{disp} = V_{formed}$ i.e level of liq will not change on melting of ice.~~

(a) If $\rho_{imp} < \rho_w$

∴ impurities float on water surface

$$\therefore F_B' = w'$$

$$V_2 \rho_w = m$$

$$V_2 = \frac{m}{\rho_w}$$

∴ $V_{disp} = V_{formed}$, i.e level of liq will not change on melting of ice.

(b) If $\rho_{imp} > \rho_w$

∴ impurities sink into water.

$$\therefore V_2 = V_{imp} = \frac{m}{\rho_1}$$

$$\therefore V_f = \frac{M}{\rho_w} + \frac{m}{\rho_1} \quad -(3)$$

from (1) & (3), $V_F < V_{Disp}$

∴ level of water 'falls'.

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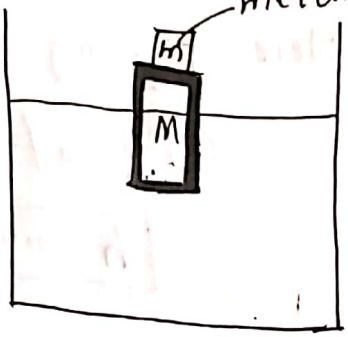
Eg: metal block of mass 200gm and Sp.G=2




If block is dropped into water. Find change in level of liquid.

Base area $= 10cm^2$.

Sol:

① initially,

$$F_B = W$$

$$V_{dis}\rho_w g = (M+m)g$$

$$V_{dis} = \frac{M+m}{\rho_w} \quad -①$$

After metal block drops into water,

$$V_F = V_1 + V_2$$

$$= \frac{M}{\rho_w} + \frac{m}{\rho_b}$$

Clearly $V_F < V_{Disp}$

∴ decrease in level of liquid

$$\Delta h = \frac{\Delta V}{A} = \frac{V_{Disp} - V_F}{A}$$

$$= \frac{\frac{m}{\rho_w} - \frac{m}{\rho_b}}{A}$$

$$= \frac{20\phi - \frac{200}{2}}{10\phi}$$

$$\boxed{\Delta h = 10cm}$$

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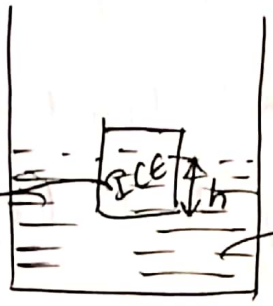
Eg:




Cubical block of side length 1cm

water $\rho_w = 1 g/cc$

Base area $= 10cm^2$

Here $F_B = mg$

$(1)^2 h \rho_w g = 1^3 \rho_{ice} g$

$\boxed{h = 0.9cm}$

* on melting the ice,

level of water remain same.

i.e, $1cm^3$ of ice converts into $0.9cm^3$ of water.

* decreasing in P.E of water,

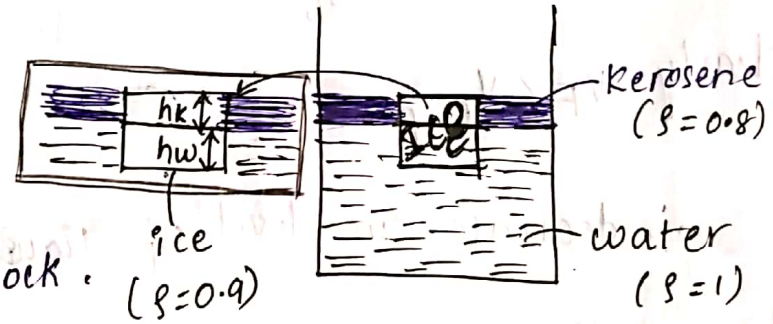
$$\Delta U = mg(0.5 - 0.45)$$

$$\Delta U = 1^3 \times 0.9 \times 10 \times 0.05 \times 100$$

$$\boxed{\Delta U = 0.45\ J} \text{ (decrease)}$$

Now,

when kerosene is poured, just to sink the ice block.




∴ here,

$F_b' = W$

$$h_w \rho_w g + h_K \rho_K g = h \rho_{ice} g$$

$$h_w \rho_w + h_K \rho_K = h \rho_{ice}$$

$$h_w \times 1 + h_K \times 0.8 = h(0.9)$$

$$h_w + (h - h_w) \times 0.8 = 0.9$$

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$$h_w - 0.8h_w = 0.1$$

$$0.2h_w = 0.1$$

$$\boxed{h_w = 0.5\text{ cm}}$$

* now, on melting the ice, kerosene level fall.

fall in level of kerosene $= \Delta h_k$

$$\Delta h_k = \frac{0.5 \times 1 \times 1}{A}$$

$$= \frac{0.5}{10}$$

$$\boxed{\Delta h_k = 0.05\text{ cm}}$$

* rise in water level,

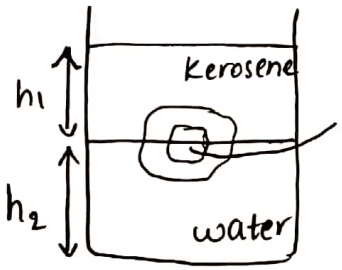
$$\Delta h_w = \frac{0.9 - 0.5 \times 1 \times 1}{10} = 0.04\text{ cm}$$

$$\boxed{\Delta h_w = 0.04\text{ cm}}$$

∴ net fall in liquid level $= 0.05 - 0.04$

$= 0.01\text{ cm}$.

Eg:




On melting the ice,

$h_1$ decreases, $h_2$ increases

* $h_1 + h_2 \rightarrow$ decreases *

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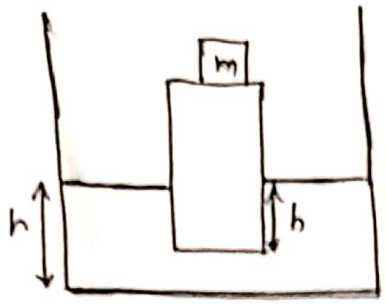
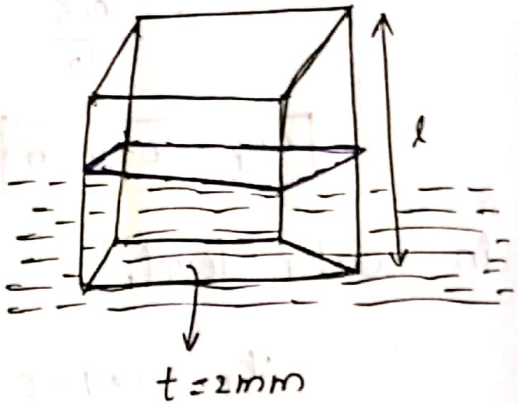
Ex:




On dropping the block into water both 'h' & 'd' decreases.

Ex: Cubical box made of iron of density 8 gm/cm³. Find dimensions of box so that half of the volume of box submerged in water.




$F_B = w$

$l^2 \times \frac{l}{2} \rho_w g = 6 l^2 t \rho_{iron} g$

$\frac{l}{2} \times 1 = 6 \times 2 \times 8 \times 0.1$

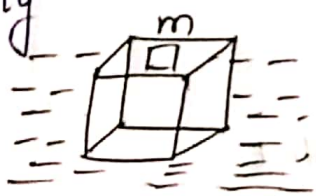
$l = 6 \times 4 \times 8 \times 0.1$

$= 24 \times 8 \times 0.1$

$\boxed{l = 19.2 \text{ cm}}$

● Just to submerge the box,

①-possibility




$F_B = w$

$l^3 \rho_w g = 6 l^2 t \rho_{iron} g + m_1 g \quad -①$

from this we get $m_1$:

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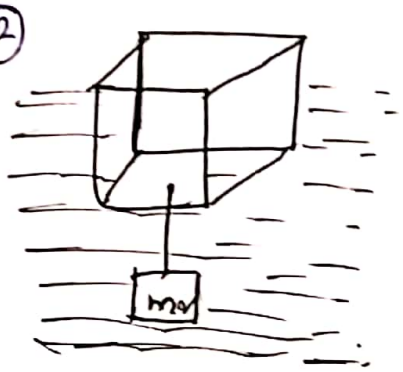
(2)

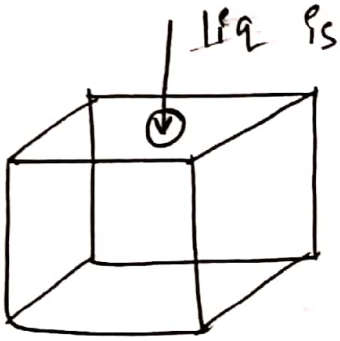



$F_B = w$

$l^3 \rho_w g + V_{block} \rho_w g = 6l^2 t \rho_{non} g + m_2 g$

from this we get $m_2$

(3)



Liq is poured through hole.

$F_B = w$

$l^3 \rho_w g = 6l^2 t \rho_1 g + m_{liq} g$

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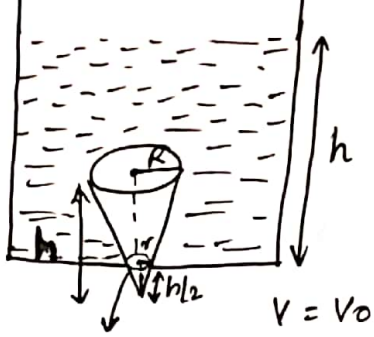
Eg:




Find the Buoyancy force acting on the cone.

V = Volume of cone

protudes out of the hole

Sol:

$$V = \frac{1}{3}\pi R^2 h \quad -(1)$$

from fig $r = \frac{R}{2}$

∴ Volume of cone outside the beaker,

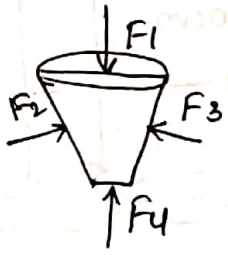
$$V_0 = \frac{1}{3}\pi r^2 \left(\frac{h}{2}\right) = \frac{1}{3}\pi \left(\frac{R}{2}\right)^2 \left(\frac{h}{2}\right)$$

$$\boxed{V_0 = V/8}$$

∴ Volume of cone inside the liquid,

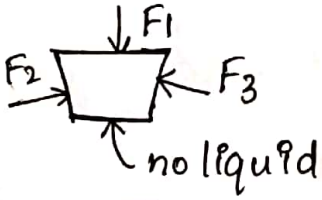
$$V_{in} = V - \frac{V}{8} = \frac{7V}{8}.$$

We have,




$$\therefore \bar{F}_1 + \bar{F}_2 + \bar{F}_3 + \bar{F}_4 = \bar{F}_B \quad -(1)$$

here,




$$\bar{F}_B' = \bar{F}_1 + \bar{F}_2 + \bar{F}_3 \quad -(2)$$

$$\bar{F}_4 = 0$$

From eqns (1) & (2),

$$\bar{F}_B' = \bar{F}_B - \bar{F}_4$$

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now, $F_B = \frac{7V}{8} \rho g \ (\uparrow)$

$F_4 = h \rho g \times \pi r^2$

$= h \rho g \times \pi \frac{R^2}{4}$

$F_4 = \frac{3V\rho g}{4} \ (\uparrow)$

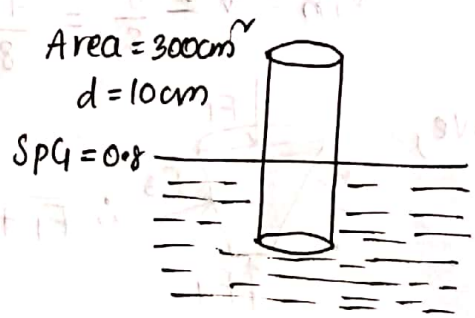
$\therefore \ \overline{F_B} = \frac{7V}{8} \rho g - \frac{3V}{4} \rho g$

$$\boxed{\overline{F_B} = \frac{V}{8} \rho g}$$

Eg: A cylinder of area $300cm^2$ and ~~length~~ 10cm made up of a material of specific gravity 0.8 is floated in water with it's axis vertical. It is pushed downwards slowly so as to just immerse. Calculate the work done by the external force to push the cylinder completely in water.

Take $g = 10m/s^2$

* Here level of liquid remains same.




For floating,

$F_b = W$

$A y \rho_w g = A l \rho_b g$

$y = l \left(\frac{\rho_b}{\rho_w}\right)$

$y = 10 (0.8) = 8cm$

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* on immersing $mg \longrightarrow$ const

$F_B \longrightarrow$ increases

$$F_{ext} = F_B - mg$$

$$\int F_{ext} \cdot dy = \int F_B \cdot dy - \int mg \cdot dy$$

$$W_{ext} = \int_{y=8}^{y=10} A\ell\rho_w g \cdot dy - mg \int_{y=8}^{y=10} dy$$

$$W_{ext} = A\ell\rho_w g \left[\frac{y^2}{2}\right]_8^{10} - mg[y]_8^{10}$$

$$= 300\times10^{-4}\times10^{3}\times10\times\frac{36\times10^{-4}}{2} - A\ell\rho g[10-8]$$

$$= [3\times10^{-1}\times10^{4}\times10^{-2}\times18 - 300\times10^{-4}\times10^{3}\times10\times2\times10^{-2}] \quad (800)$$

$$= [54 - 60]\times$$

$$= [-6]\times$$

$$\Rightarrow 300\times10^{-4}\times10^{3}\times10\times\frac{18\times10^{-4}}{2} - 300\times10^{-4}\times10\times10^{-2}\times\frac{2}{2}\times800\times10\times2\times10^{-2}$$

$$\Rightarrow 54\times10^{-2} - 48\times10^{-2}$$

$$\Rightarrow 10^{-2}[54-48]$$

$$\Rightarrow 6\times10^{-2} = 0.06$$

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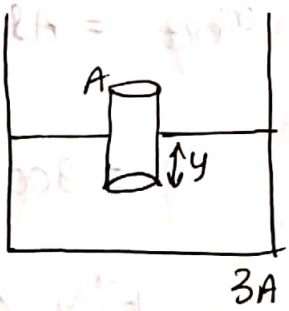
Eg: A liquid of density 2ρ is filled in a cylindrical vessel whose area is 3A. a wooden cylinder of height $h$ & area $A$ has density $\rho$ is floating in the liquid at eqm with it's axis vertical. Find the ~~increased~~ Min work with which the external agent has to do to pull the cylinder completely out of the liquid.




Initially,

$$F_B = W$$

$$Ay(2\rho)g = AH(\rho)(g)$$

$$\boxed{y = \frac{H}{2}}$$

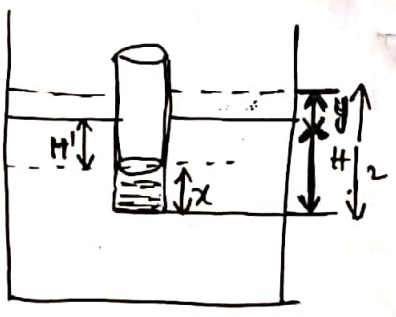
when it is pulling out slowly,

$mg \longrightarrow$ const

$F_b \longrightarrow$ decreases

$$\boxed{F_{net} = mg - F_b}$$

& $$\int F_{ext} \cdot dx = \int mg \cdot dx - \int F_b \cdot dx \quad -①$$




here volume of displaced liquid is const.

$$(3A - A)\, y = A(x)$$

$$\boxed{2y = x}$$

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from fig,

$$H' + x + y = \frac{H}{2}$$

$$\Rightarrow \boxed{H' = \frac{H}{2} - \frac{3x}{2}}$$

when rod completely pulls out,

$$H' = 0$$

$$\frac{H}{2} = \frac{3x}{2} \Rightarrow \boxed{x = \frac{H}{3}}$$

limits $\longrightarrow$ $x = 0$ to $x = H/3$

① $\Rightarrow$ $\int F_{ext} \cdot dx = \int mg \cdot dx - \int F_b \cdot dx$

$$W_{ext} = \int_0^{H/3} mg \cdot dx - \int_0^{H/3} A H' s_l g \, dx$$

($dx$: displacement of CM)

$$W_{ext} = mg\left(\frac{H}{3}\right) - A s_l g \int_0^{H/3} \left(\frac{H}{2} - \frac{3x}{2}\right) \cdot dx$$

$$= mg\frac{H}{3} - A(2s)g\left[\frac{H}{2} \times \frac{H}{3} - \frac{3}{2} \times \frac{H^2}{2 \times 2}\right]_0^{H/3}$$

$$= \frac{mgH}{3} - 2Asg\left[\frac{H^2}{6} - \frac{3H^2}{12}\right]$$

$$= \frac{AHsgH}{3} - Asg\left[\frac{H^2}{6}\right]$$

$$= Asg H^2\left(\frac{1}{3} - \frac{1}{6}\right)$$

$$\boxed{W_{ext} = \frac{1}{6} Asg H^2}$$

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## Archimedi's Principle:

when a body is partially or completely submerged in a liquid then the buoyancy force acting on body is equal to loss of wt of body in water.

## Applications:

① To determine specific gravity of solid:-

Let $w_1$ = wt of solid in air (true wt)

$w_2$ = wt of solid in water (apparent wt)

loss in wt of solid = $w_1 - w_2$

$$F_B = w_1 - w_2$$

where $F_B = mg\left(\frac{dw}{d_B}\right) = w_1\left(\frac{dw}{d_B}\right)$

$\therefore w_1 \cdot \frac{dw}{d_B} = w_1 - w_2$

$$\boxed{w_2 = w_1\left(1 - \frac{dw}{d_B}\right)}$$

Specific gravity of solid,

$$\frac{d_B}{d_w} = \frac{w_1 - w_2}{w_1} \qquad \boxed{\frac{d_B}{dw} = \frac{w_1}{w_1 - w_2}}$$

$$\boxed{\frac{d_B}{d_w} = 1 - \frac{w_2}{w_1}}$$

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② Specific gravity of liquid :-

$w_1$ = wt of solid in air

$w_3$ = wt of solid in liq

$w_2$ = wt of solid in water

∴ loss in wt of solid in liquid,

$$F_B = w_1 - w_3$$

$$w_1\left(\frac{d_l}{d_s}\right) = w_1 - w_3$$

$$\frac{d_l}{d_s} = \frac{w_1 - w_3}{w_1}$$

$$\boxed{\frac{d_s}{d_l} = \frac{w_1}{w_1 - w_3}}$$

now specific gravity of liquid,

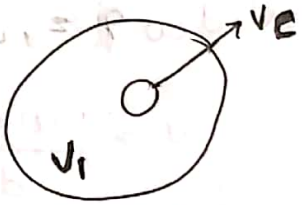
$$\frac{d_l}{dw} = \frac{\frac{d_s}{dw}}{\frac{d_s}{d_l}} = \frac{\frac{w_1}{w_1 - w_2}}{\frac{w_1}{w_1 - w_3}} = \frac{w_1 - w_3}{w_1 - w_2}$$

$$\boxed{\frac{d_l}{dw} = \frac{\text{loss of wt of solid in liq}}{\text{loss of wt of solid in water}}}$$

③ To find vol of cavities in a body :-

* Vol of metal,

$$V_1 = \frac{M}{\rho}$$




M = mass of metal

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$\sigma$ = density of metal.

∴ Total vol of body, $\boxed{V = V_1 + V_2}$

* $w_1$ = wt of body in air

$w_2$ = wt of body in water

now, $F_b = w_1 - w_2$

$V\rho g = w_1 - w_2$ , $\rho$ — density of water

$$V = \frac{w_1 - w_2}{\rho g} \quad -②$$

finally, $V_c = V - V_1$

$$= \left(\frac{w_1 - w_2}{\rho g}\right) - \left(\frac{M}{\sigma}\right)$$

④ To find the purities of ornaments :-

let the alloy consits of two metals.

Total mass,

$$m = m_1 + m_2 \quad -①$$

let $w_1$ = wt in air

$w_2$ = wt in water

now,

$F_b = w_1 - w_2$

$V \cdot d_w g = w_1 - w_2$

$$V = \frac{w_1 - w_2}{d_w g}$$

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$$V_1 + V_2 = \frac{w_1 - w_2}{d_w g}$$

$$\frac{m_1}{d_1} + \frac{m_2}{d_2} = \frac{w_1 - w_2}{d_w g} \quad -(2)$$

from (1) & (2),

we get $m_1$ & $m_2$.

Eg: A piece of gold sample weighs 36gms in air & 34 gm in water. If the sample has some copper mixed in it. Then calculate the amount of copper in the sample if spg of gold is 19.3 & copper is 8.9. Now assume that the same mass of sample is made of pure gold but with some air cavities inside it and it weighs the same in water. Then calculate volume of cavities in the sample.

Ans: i) 2.225gms

ii) 0.135

Sol:

$w_1 = 36g$

$m_1 + m_2 = 36$

$m_c + m_g = 36$

$m_g = 36 - m_c$

$$\frac{m_c}{d_c} + \frac{m_g}{d_g} = \frac{w_1 - w_2}{s_w g}$$

$$\frac{m_c}{8.9} + \frac{36 - m_c}{19.3} = 2$$

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$$19.3 m_c + 36 \times 8.9 - 8.9 m_c = 2 \times 8.9 \times 19.3$$

$$10.4\, m_c = (38.6 - 36) 8.9$$

$$10.4\, m_c = 2.6 \times 8.9$$

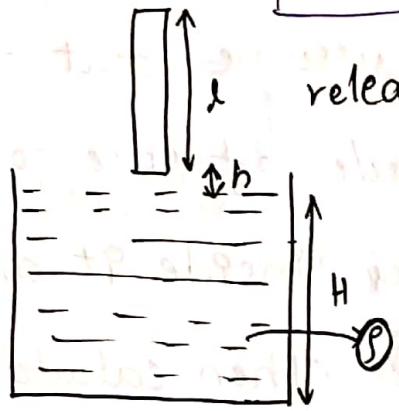
$$\boxed{m_c = 2.225 \text{ gms}}$$

If only gold is present with some cavities in it. then,

$$V_c = \frac{2}{1} - \frac{36}{19.3}$$

$$= 2 - 1.865$$

$$\boxed{V_c = 0.135}$$

Eg: 


released from rest.

Find velocity of rod when reaches to the bottom of the tank?

$$\sqrt{2gh + \frac{gl}{2} + gH}$$

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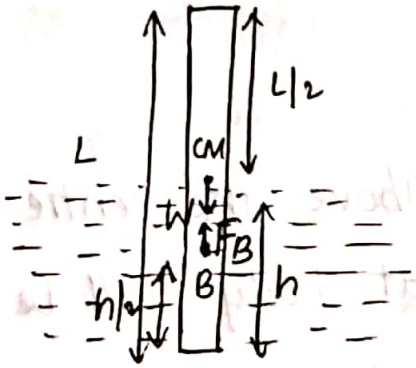
## Equilibrium :

* Centre of buoyancy – centre of mass of immersed volume of body.

* centre of gravity – geometric centre of body.

Equilibrium $\begin{cases} F_{net} = 0 \\ \tau_{net} = 0 \end{cases}$

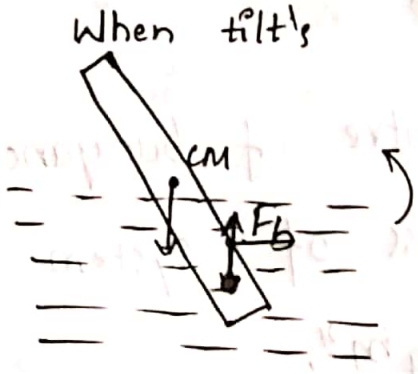
①




$\boxed{W = F_B} \Rightarrow F_{net} = 0$

Torque about c.M of body $\boxed{\tau_{net} = 0}$

When tilt's




Torque about c.M,

$\boxed{\tau_c \neq 0}$

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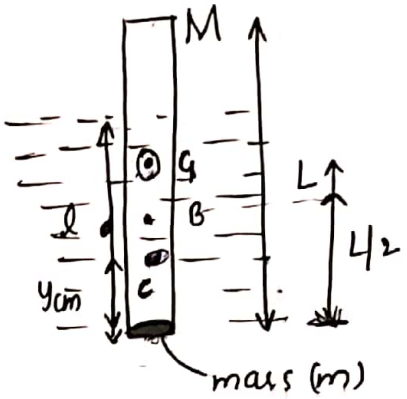


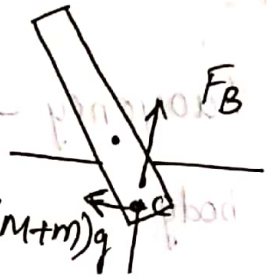

CM of system,

$$y_{cm} = \frac{M(L/2)}{M+m} = \frac{ML}{2(M+m)}$$

In floating,

$$F_B = (M+m)g$$

$$A l \rho_l g = AL\rho_b g + mg$$




Torque about C.M of system

$$\tau_{cm} \neq 0 \text{ (clockwise)}$$

Imp points

* If centre of buoyancy lies above the centre of mass of system then that body will be in "static eqm".

For static eqm,

$$\frac{l}{2} \geq y_{cm}$$

* To find min mass,

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$$\frac{l}{2} = y_{cm}$$

$$\frac{l}{x} = \frac{ML}{x(M+m)}$$

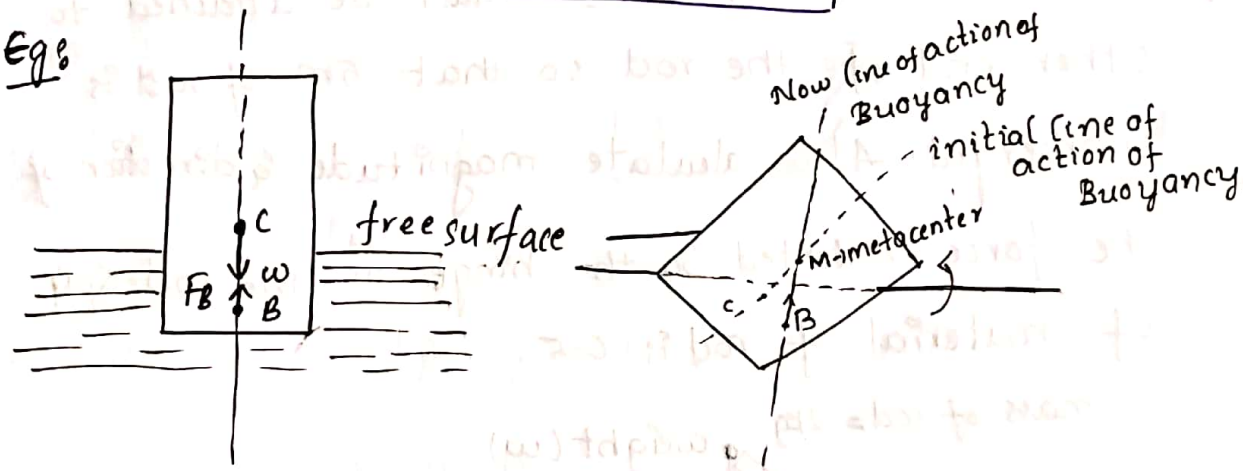
$$(M+m)l = ML$$

$$Ml + ml = ML$$

$$ml = M(L-l)$$

$$\boxed{m = M\left(\frac{L}{l} - 1\right)}$$

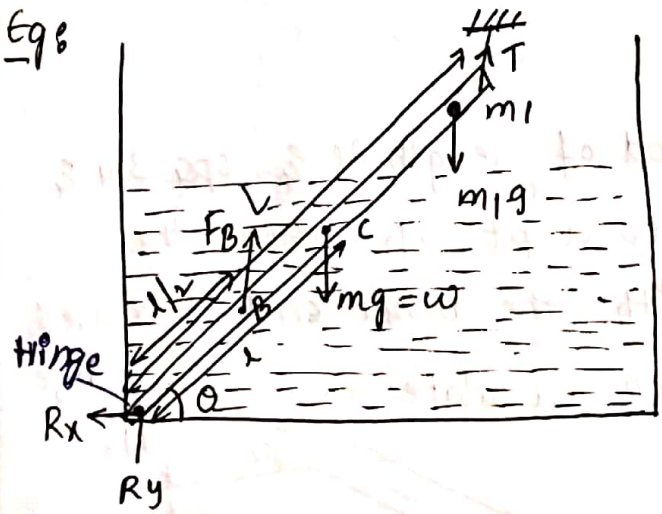
Eg:




Imp pt:

* If meta centre lies above the centre of gravity of body then body is in static eqm.

Eg:

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$F_B = A l S_l g$ — ①

rod, $w = AL S_b g$ — ②

$\Rightarrow \Sigma F_x = 0 \Rightarrow R_x = 0$ — ③

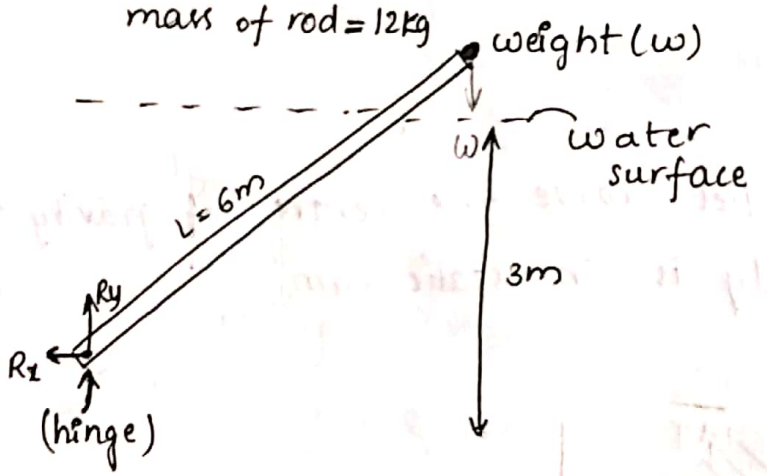
$\Sigma F_y = 0 \Rightarrow T + F_B = R + w + m_1 g$ — ④

& $\tau_{hinge} = 0$

$$F_B \frac{l}{2}\cos\theta(\curvearrowright) + w * \frac{L}{2}\cos\theta(\curvearrowright) + m_1 g \times \cos\theta(\curvearrowright) + T L \cos\theta(\curvearrowleft) = 0$$

Let $(\curvearrowleft) = +ve$.

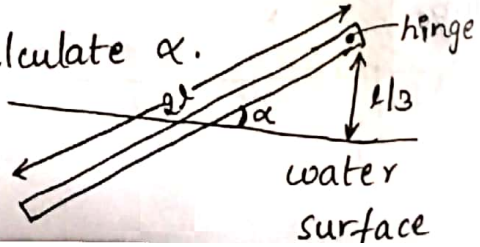
Eg: 1) Calculate the wt which must be attached to other end of the rod so that 5m of rod is submerged. Also calculate magnitude & direction of the force exerted by the hinge on the rod. Sp.G of material of rod is 0.5.




Ans: 1) $w = 2.33$ Kgwt

2) $R_y = -5.67$ Kgwt

2) A thin uniform rod of length $2l$ & sp.g $3/4$ & hinged at one end to a pt $l/2$ above the surface of water with the other end immersed. If the rod is in eqm. calculate $\alpha$.

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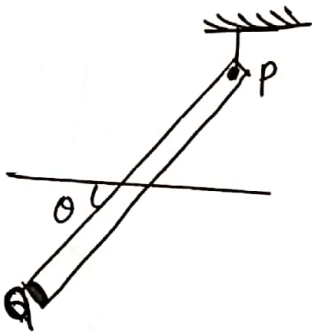
3) A uniform rod PQ 4m long & weighing 12kg is supported at end P with a 6kg lead weight attached at Q. The rod floated with one half of it's length submerged as shown in figure. Assuming lead mass has negligable volume. Calculate i) Tension in the string – 20N

ii) Total volume of rod – $3.2 \times 10^{-2} m^3$

iii) $\theta$.

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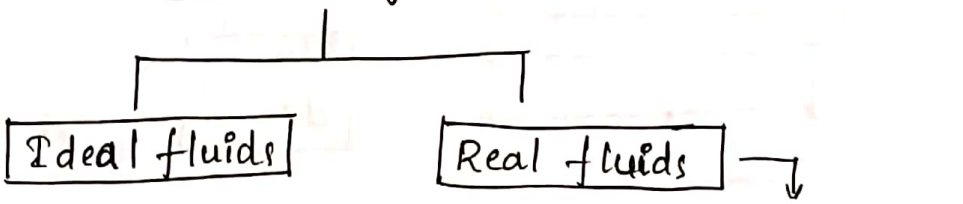
# Fluid Dynamics




characterstics:

1) Steady & laminar
2) Non-Viscous
3) Incompressible
4) Irrotational

## *Stream Lines:

* Stream lines is a curve or a line along which the particle is moving.
* the tangent drawn at any pt on the stream line gives the direction of velocity of particle.
* No two stream lines can intersect each other.
* Tube of flow:
* It is the imaginary tube containing stream lines.

### 1) Steady flow:

* Velocity of fluid particles crossing at a pt in flow is same both in magnitude & direction & does not change with time.

$$\text{i.e, } \frac{dv}{dt} = 0$$

* Here stream lines are smooth.

### 2) Non-Viscous:

* Here there is no velocity gradient in path of fluid flow.

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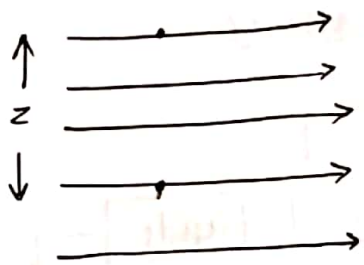



$$\frac{dv}{dz} = 0$$

3) Incompressible :

* density of fluid is same at all pts.

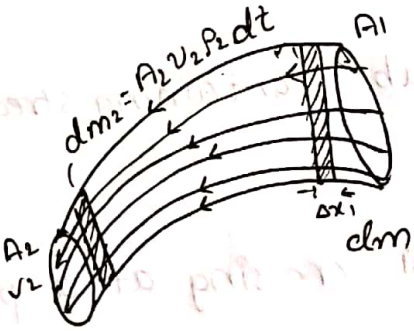
4) Irrotational :

* Floating objects can be carried away by the fluid with out rotation, then the flow is irrotational

*** Equation of Continuity :-

Statement: when incompressible, irrotational, non-viscous steady fluids flowing then the product of area of crossection & speed of the fluid flow at any cross-section is constant.

proof :




$dm_1 = V_1 \rho_1 = A_1 \Delta x_1 \rho_1$

$= A_1 v_1 dt \rho_1$

* It is based on "law of conservation of mass"

i.e, $\frac{dm}{dt} = const$

$$\frac{dm_1}{dt} = \frac{dm_2}{dt}$$

$\rho_1 = \rho_2$ for incompressible

$$A_1 V_1 \rho_1 = A_2 V_2 \rho_2$$

$$\boxed{A_1 V_1 = A_2 V_2}$$

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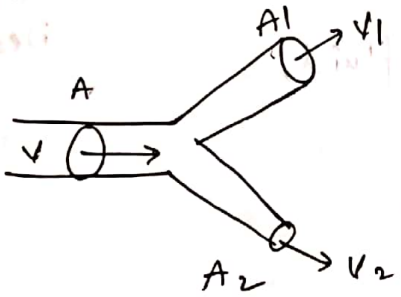
* As Area is more, velocity is less.

$\boxed{AxV = \text{constant}}$

Applications:

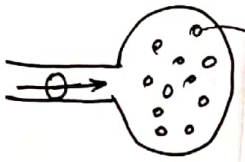
1)




$$\frac{dm}{dt} = \frac{dm_1}{dt} + \frac{dm_2}{dt}$$

$$AV = A_1V_1 + A_2V_2$$

2)




$$AV = nA_1V_1$$

3)

① $A_1V_1 = A_2V_2$

② $V_2^2 - V_1^2 = 2gh$

* Volume rate of flow of fluid:

$$\frac{V}{t} = \frac{Al}{t} = AV$$

Types of Energies in flowing fluid:

① Kinetic Energy:

$$K.E = \frac{1}{2}mv^2$$

* K.E per unit volume of fluid,

$$\frac{K.E}{V} = \frac{1}{2}\frac{m}{V}v^2 = \frac{1}{2}\rho v^2$$

* K.E per unit wt of fluid,

$$\frac{K.E}{mg} = \frac{\frac{1}{2}mv^2}{mg} = \frac{v^2}{2g} = \text{velocity head.}$$

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② Potential Energy:

P.E = mgh

* P.E per unit wt,

$$\frac{P.E}{mg} = \frac{mgh}{mg} = h = \text{gravitational head}$$

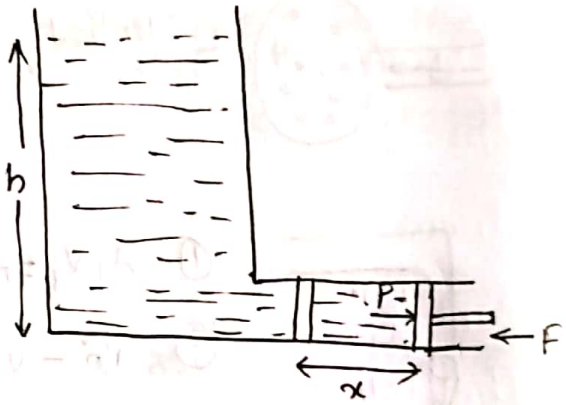
* P.E per unit volume of fluid,

$$\frac{P.E}{V} = \frac{mgh}{V} = \rho gh.$$

③ Pressure Energy:

* pressure at the piston,

$$P = \rho gh$$




* Let the piston is moved by (x) slowly as shown.

∴ work done against the pressure.

$$W = Fx$$

$$W = (PA)x$$

$$\boxed{W = PV}$$

This work done is stored as pressure energy in a liquid of volume V.

* pressure energy per unit volume $= \frac{PV}{V} = P$

* pressure energy per unit wt $= \frac{PV}{mg} = \frac{P}{\rho g} =$ pressure head.

Bernoulli's Theorem:-

Statement:- when a non-viscous, incompressible & irrotational fluid flowing steadily then at pt

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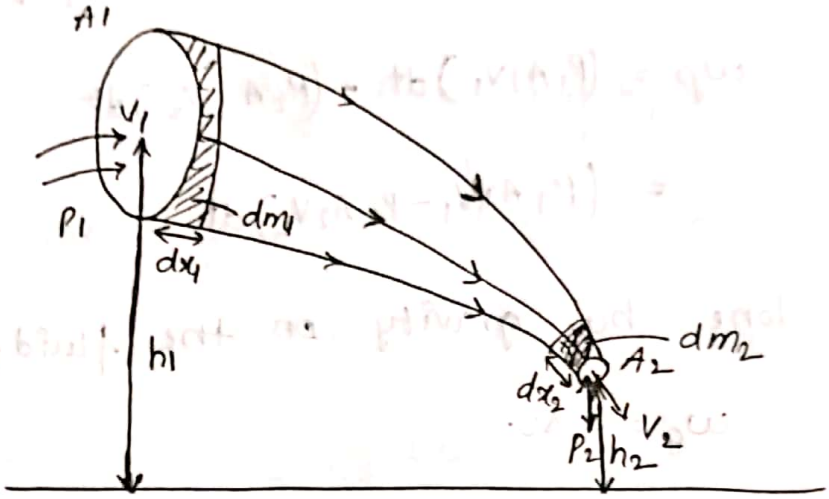
of it's path, total energy per unit volume is constant.

i.e, $\boxed{P + \frac{1}{2}\rho v^2 + \rho g h = \text{const}}$

proof:




Here $dm_1 = (A_1 v_1 dt)\rho$

& $dm_2 = (A_2 v_2 dt)\rho$

from eqn of continuity,

$$dm_1 = dm_2$$

$$A_1 v_1 \rho dt = A_2 v_2 \rho dt$$

$$\boxed{A_1 v_1 = A_2 v_2}$$

work done by fluid pressure on the fluid while entering the tube,

$$W_{P_1} = F_1 dx_1 \cos\theta \quad (\theta = 0)$$

$$= P_1 A_1 v_1 dt$$

and while leaving the tube,

$$W_{P_2} = F_2 dx_2 \cos 180^\circ$$

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$= -P_2 A_2 V_2 dt$

$\therefore$ net work done on fluid by pressure,

$$W_p = W_{p_1} + W_{p_2} = P_1 A_1 V_1 dt - P_2 A_2 V_2 dt$$

$$W_p = (P_1 A_1 V_1) dt - (P_2 A_2 V_2) dt$$

$$= (P_1 A_1 V_1 - P_2 A_2 V_2) dt$$

* work done by gravity on the fluid,

$$W_g = -\Delta U$$

$$W_g = U_1 - U_2 \qquad (dm_1 = dm_2)$$

$$= (dm_1) g h_1 - (dm_2) g h_2$$

$$W_g = (dm) g (h_1 - h_2)$$

* From work- energy theorem,

$$W_g + W_p = \Delta K$$

$$dm(h_1 - h_2) g + (P_1 A_1 V_1 - P_2 A_2 V_2) dt = K_2 - K_1$$

$$dm (h_1 - h_2) g + (P_1 A_1 V_1 - P_2 A_2 V_2) dt = \frac{1}{2} dm (V_2^2 - V_1^2)$$

$$P_1 A_1 V_1 dt + dm h_1 g + \frac{1}{2} dm V_1^2 = P_2 A_2 V_2 dt + dm g h_2 + \frac{1}{2} dm V_2^2$$

$$\frac{P_1 A_1 V_1 dt}{dm} + h_1 g + \frac{1}{2} V_1^2 = \frac{P_2 A_2 V_2 dt}{dm} + h_2 g + \frac{1}{2} V_2^2$$

$$\frac{P_1 A_1 V_1 dt}{\rho A_1 V_1 dt} + h_1 g + \frac{1}{2} V_1^2 = \frac{P_2 A_2 V_2 dt}{\rho A_2 V_2 dt} + h_2 g + \frac{1}{2} V_2^2$$

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$$\frac{P_1}{\rho} + h_1 g + \frac{1}{2} v_1^2 = \frac{P_2}{\rho} + h_2 g + \frac{1}{2} v_2^2$$

$$P_1 + \rho g h_1 + \frac{1}{2} \rho v_1^2 = P_2 + h_2 g \rho + \frac{1}{2} \rho v_2^2$$

implies, —(a)

$$\boxed{P + \rho g h + \frac{1}{2} \rho v^2 = \text{const}}$$

hence proved //.

Points:

① For a horizontal flow,

$$h_1 = h_2$$

∴ eqn (a) becomes,

$$\boxed{P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2} \quad ***$$

i.e, $P + \frac{1}{2} \rho v^2 = \text{const}$.

* As v is more ⇒ Pressure is less.

* As v is less ⇒ pressure is more.
(velocity)

② If fluid is at rest, (static fluid)

$$\boxed{P_1 + \rho g h_1 = P_2 + \rho g h_2}$$

i.e, $P + \rho g h = \text{const}$.

$$\Rightarrow \boxed{P_1 - P_2 = \rho g (h_2 - h_1)} \quad **$$

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③ we have,

$$p + \rho gh + \frac{1}{2}\rho v^2 = \text{const}$$

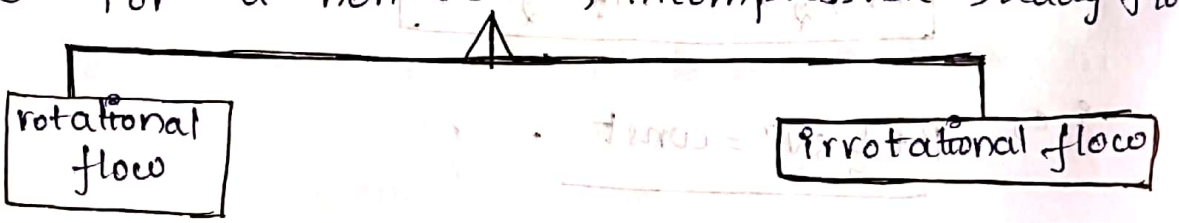
here,

$p \longrightarrow$ absolute pressure but not guage pressure

$v \longrightarrow$ velocity of fluid w.r.to container.

* $p + \rho gh \longrightarrow$ static pressure
* $\frac{1}{2}\rho v^2 \longrightarrow$ dynamic pressure
* $p + \rho gh + \frac{1}{2}\rho v^2 =$ total pressure = stagnation pressure.
* $p + \rho gh + \frac{1}{2}\rho v^2 = \text{const}$ is always written with respect to the container.

④ For a non-viscous, incompressible steady flow




bernoulli's eqn is applicable b/w two pts on the same streamline.

bernoullis eqn is applicable b/w two pts on the diff stream lines.

Eg: A flat closed tube of uniform cross-section has inner radius 'a' & outer radius 'b', it is filled completely with water and is kept on a flat horizontal surface. Now, entire water in the tube is rotated with a uniform and steady angular velocity ($\omega$) about the vertical

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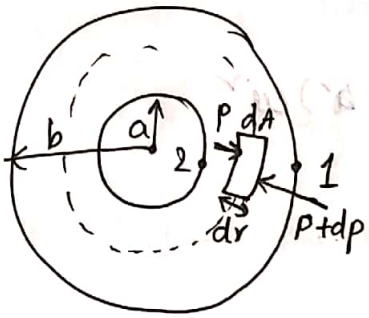
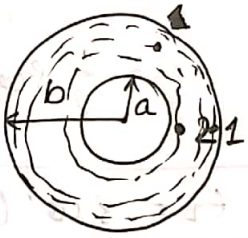
axis of the tube. Viscous forces b/w the tube & water is neglected. density of water is $\rho$.

1 & 2 are 2 pts in water on the same horizontal plane. let 'p' be the pressure and 'e' be the energy of the fluid particle per unit volume. Then find (1) $P_1 - P_2$ (2) $e_1 - e_2$.




$$(p+dp)\,dA - p\,dA = (dm)\,r\omega^2$$

$$dp \cdot dA = (dA)\,dr \cdot \rho\, r\omega^2$$

$$\int_{P_2}^{P_1} dp = \rho\omega^2 \int_{r=a}^{r=b} r \cdot dr$$

$$\boxed{P_1 - P_2 = \frac{1}{2}\rho\omega^2(b^2 - a^2)}$$

since $b > a \Rightarrow \boxed{P_1 > P_2}$

again we have,

$$V_1 = b\omega$$

& $V_2 = a\omega$

since $b > a \Rightarrow \boxed{V_1 > V_2}$

for horizontal flow,

$$E = P + \frac{1}{2}\rho V^2$$

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now,

$$E_1 = P_1 + \frac{1}{2}\rho v_1^2$$

& $$E_2 = P_2 + \frac{1}{2}\rho v_2^2$$

since $P_1 > P_2$ & $v_1 > v_2$ $\Rightarrow$ $\boxed{E_1 > E_2}$

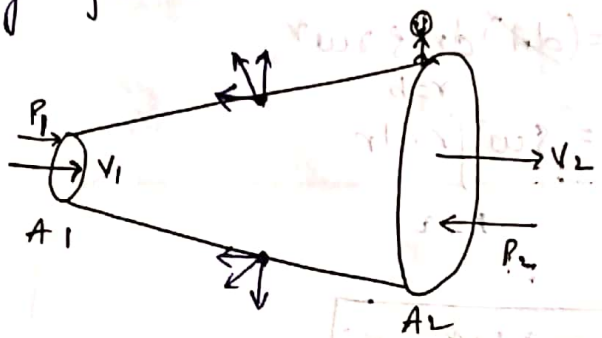
$$E_1 - E_2 = P_1 - P_2 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= \frac{1}{2}\rho\omega^2(b^2 - a^2) + \frac{1}{2}\rho(b^2 - a^2)\omega^2$$

$$\boxed{E_1 - E_2 = \rho\omega^2(b^2 - a^2)}$$

⑤ bernoulli's eqn is not applicable as seen from moving frame.

Ex:

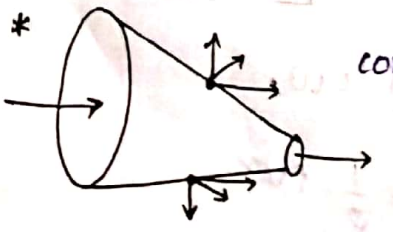



(horizontal flow)

w.r.to container,

$$\boxed{P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2} \Rightarrow \boxed{P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)}$$

work done by the fluid force on the walls of container is zero. bcoz container is at rest. i.e, pt of application of force is at rest.

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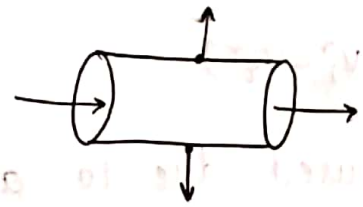
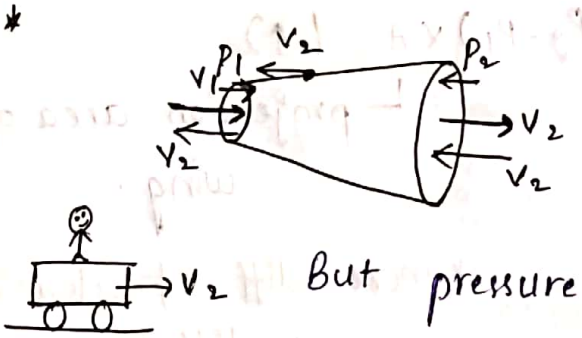
*



container doesnot move.

*







But pressure is invasient.

$$L.H.S = P_1 + \frac{1}{2} \rho (V_1 - V_2)^2$$

$$R.H.S = P_2$$

$$\therefore L.H.S \neq R.H.S$$

* hence proved that bernoulli's eqn is not written from moving frame.

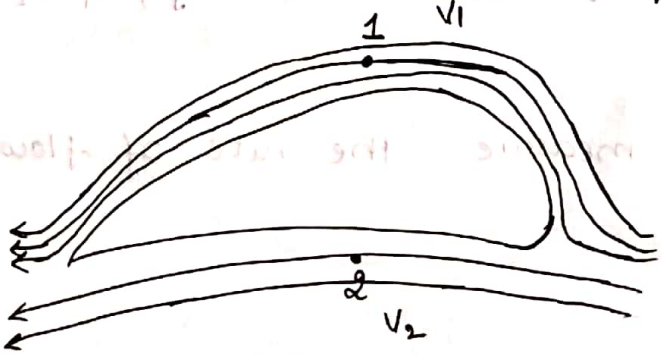
* here energy conservation is valid when observing from inertial frame bcoz no pseudo forces acting on the fluid.

Applications:

① Aerodynamic Lift:

$V_1 > V_2$

$\Rightarrow P_1 < P_2$




for horizontal flow, $P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$

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$$\therefore P_2 - P_1 = \frac{1}{2} \rho (V_1^2 - V_2^2)$$

dynamic lift - lift caused due to air.

$$\text{dynamic lift} = (P_2 - P_1) \times A \quad (\uparrow)$$

A → projection area of wing.

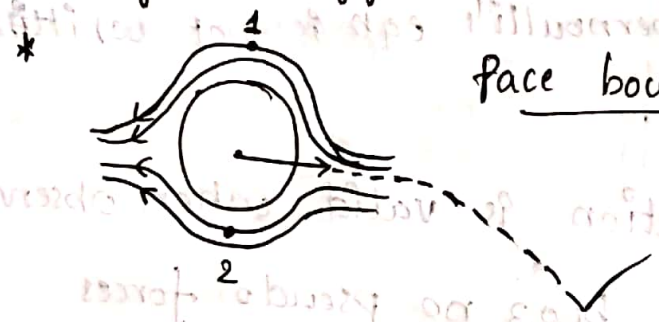
* for a level flight,
* mg is balanced by upward force.

$$(P_2 - P_1) A = mg$$

$$\frac{1}{2} \rho (V_1^2 - V_2^2) A = mg$$

* more diff of velocities, so more diff in pressure. So more force acts in upward direction.

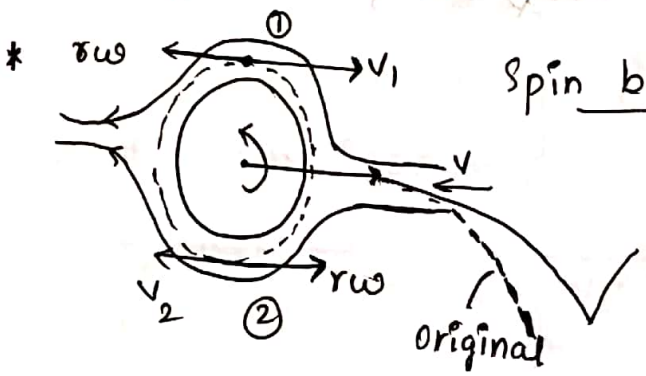
## ② Magnus Effects




Pace bowling

$V_1 = V_2$

$P_1 = P_2$




Spin bowling

$V_1 > V_2$

$P_1 < P_2$

$V_1 = V + r\omega$, $V_2 = V - r\omega$ ⇒ clearly, $V_1 > V_2 \Rightarrow P_1 < P_2$

Venturimeter

It is used to measure the rate of flow of fluid.

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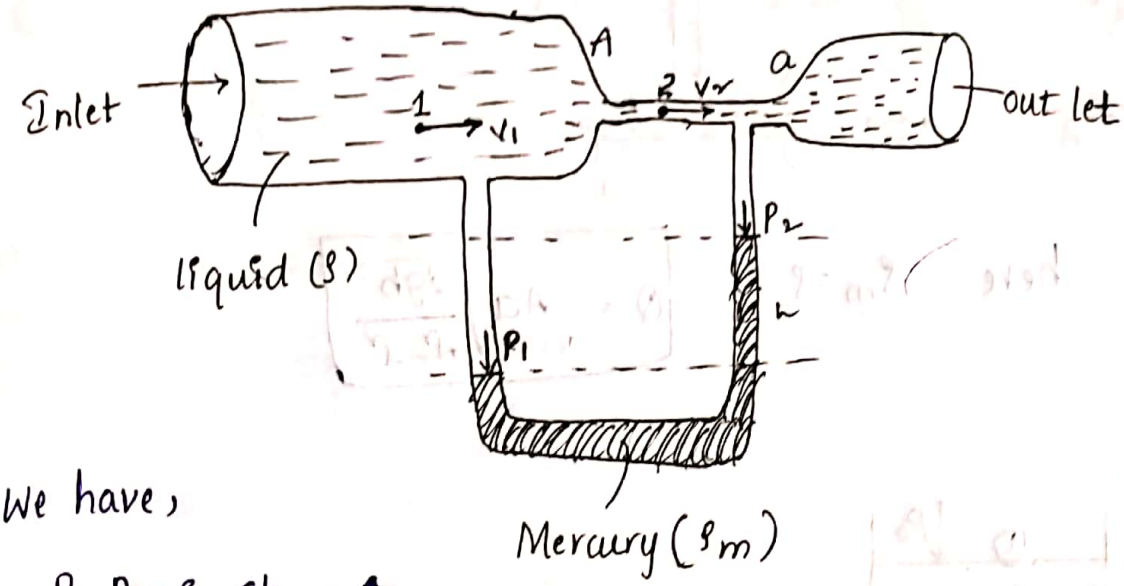


We have,

$$P_1 - P_2 = s_m g h \quad - (1)$$

eqn of continuity,

$$Q = AV_1 = aV_2 \quad - (2)$$

since $A > a \Rightarrow V_1 < V_2 \Rightarrow P_1 > P_2$.

Applying bernoulli's theorem b/w points (1) & (2).

$$P_1 + \frac{1}{2} s V_1^2 = P_2 + \frac{1}{2} s V_2^2$$

$$P_1 - P_2 = \frac{1}{2} s (V_2^2 - V_1^2) \quad - (3)$$

from eqn (1) & (3)

$$s_m g h = \frac{1}{2} s (V_2^2 - V_1^2)$$

$$s_m g h = \frac{1}{2} s \left(\frac{Q}{a}\right)^2 - \left(\frac{Q}{A}\right)^2$$

$$\frac{2 s_m g h}{s} = Q^2 \left(\frac{A^2 - a^2}{a^2 A^2}\right)$$

$$\boxed{Q = Aa \sqrt{\frac{2 s_m g h}{s(A^2 - a^2)}}}$$

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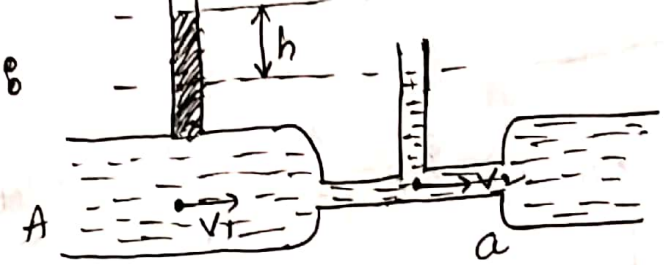
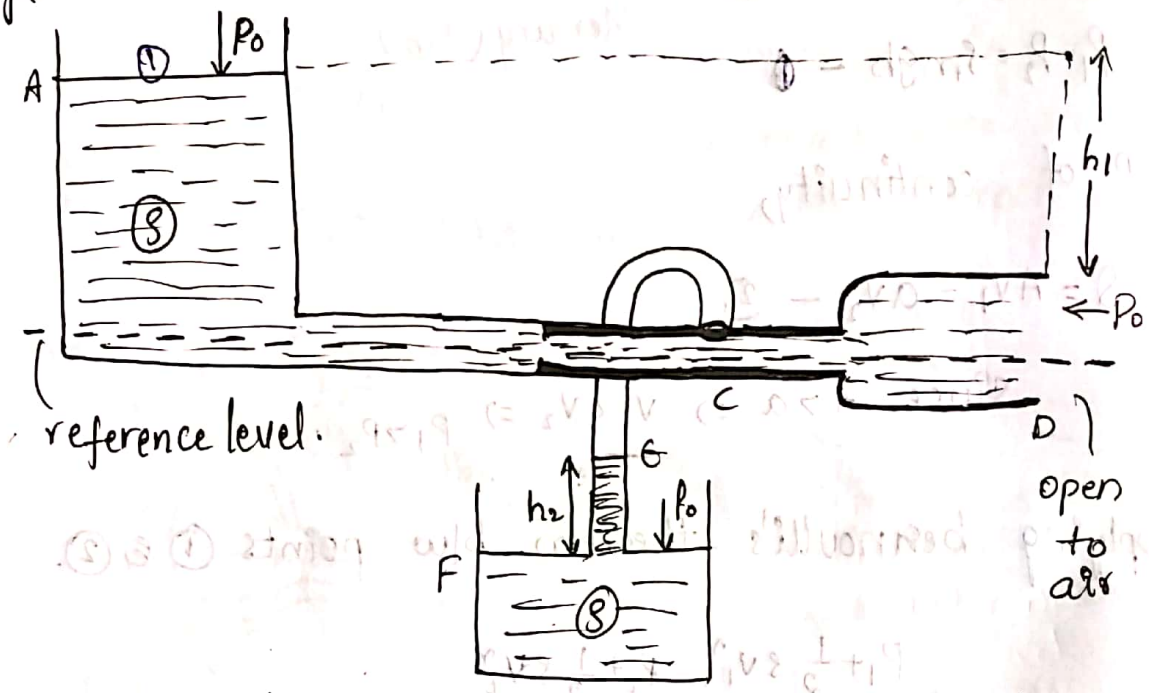
Eg:




here $\rho_m = \rho$

$$Q = Aa\sqrt{\frac{2gh}{A^2 - a^2}}$$

Eg:




A & F are large tanks contains same liquid. If the cross-section area at C is half the area at D & if D is a distance $h_1$ below the level of the liquid in A. To what height $h_2$ will liquid rise in pipe E.

Given,

$$A_C = \frac{1}{2} A_D$$

$$V_C = 2 V_D$$

let $V_D = V$

$$\Rightarrow V_C = 2V$$

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bernoulli's theorem b/w C & D.

$$P_C + \frac{1}{2} \rho v_C^2 = P_D + \frac{1}{2} \rho v_D^2$$

$$P_C + \frac{1}{2} \rho (2v)^2 = P_D + \frac{1}{2} \rho v_D^2$$

$$P_C = P_0 + \frac{1}{2} \rho v_0^2 - 2 \rho v^2$$

$$P_C = P_0 - \frac{3}{2} \rho v^2 \quad -(1)$$

Again,

$$P_0 - \rho g h_2 = P_C \quad -(2)$$

$$P_0 + \rho g h_2 = P_0 + \frac{3}{2} \rho v^2$$

$$\boxed{h_2 = \frac{3v^2}{2g}} \quad -(3)$$

bernoulli's theorem b/w (1) & D

$$P_0 + \frac{1}{2} \rho (0)^2 + \rho g h_1 = P_0 + \frac{1}{2} \rho v_D^2 + \rho g (0)$$

$$\rho g h_1 = \frac{1}{2} \rho v^2$$

$$\boxed{h_1 = \frac{v^2}{2g}} \quad -(4)$$

from eqn (3) & (4)

$$\boxed{h_2 = 3h_1}$$

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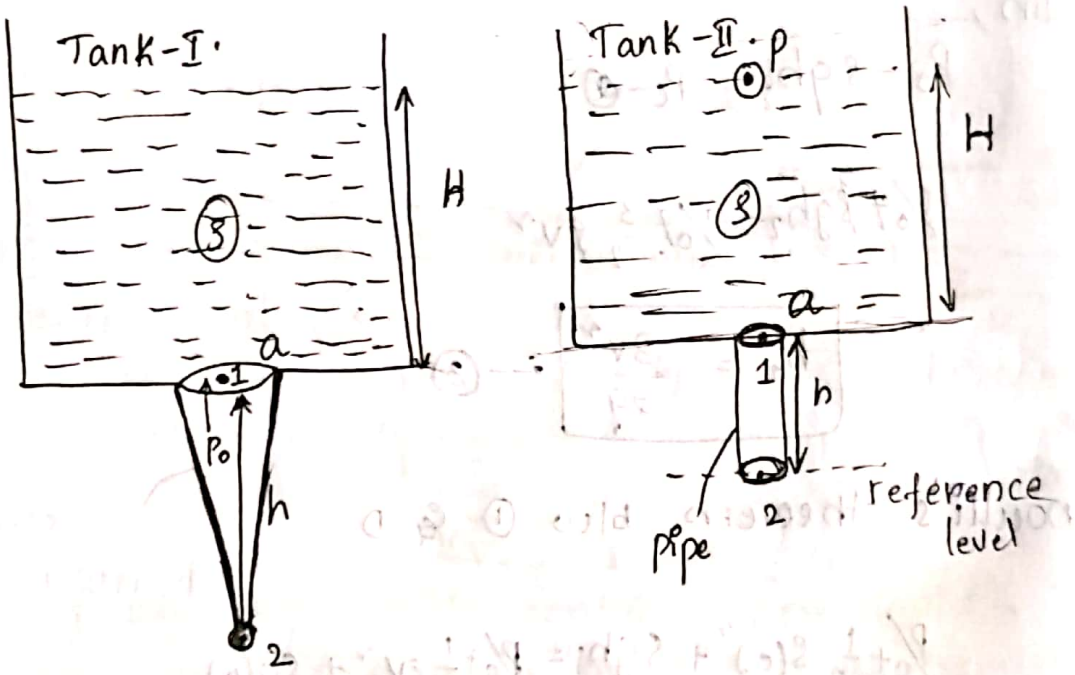
Eg: ① Find speed of the flow at a pt 2 in both the figures.

② Ratio of speed of flow at pt in 1st fig to that of 2nd figure.

③ Find difference in pressure at pt 1 in both figures.




① Tank-1:

bernoulli's eqn just above and bellow the tube.

$$P_0 + \rho g H + \frac{1}{2}\rho(0)^2 + \rho g(0) = P_0 + \frac{1}{2}\rho v_1^2 + \rho g(0)$$

$$\rho g H = \frac{1}{2}\rho v_1^2$$

$$\boxed{v_1 = \sqrt{2gH}}$$

now, $v^2 - u^2 = 2as$

$$\boxed{v_2 = \sqrt{2g(H+h)}}$$

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tank - 2 :

bernoulli's theorem b/w P & 2

$$P_0 + \rho g(H+h) + \frac{1}{2}\rho(0)^2 = P_0 + \frac{1}{2}\rho V_2^2 + \rho g(0)$$

$$\rho g(H+h) = \frac{1}{2}\rho V_2^2$$

$$\boxed{V_2 = \sqrt{2g(H+h)}}$$

(2) for tank - 1 :

$$V_1 = \sqrt{2gH}$$

for tank - 2 :

$$V_2 = \sqrt{2g(H+h)}$$

but $A_1 V_1 = A_2 V_2$ (for pipe)

$$V_1 = V_2 = \sqrt{2g(H+h)}$$

now, ratio, $\frac{\sqrt{2gH}}{\sqrt{2g(H+h)}} = \sqrt{\frac{H}{H+h}}$

(3) for tank - 1 : $P_1 = P_2$

for tank - 2 :

bernoulli's theorem b/w 1, 2.

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g h = P_0 + \frac{1}{2}\rho V_2^2 + \rho g(0)$$

$$\boxed{P_1 = P_0 - \rho g h}$$

now, pressure difference at pt-1 in both tanks

$$P_i - P_f = P_0 - (P_0 - \rho g h) \Rightarrow \boxed{\Delta P = \rho g h}$$

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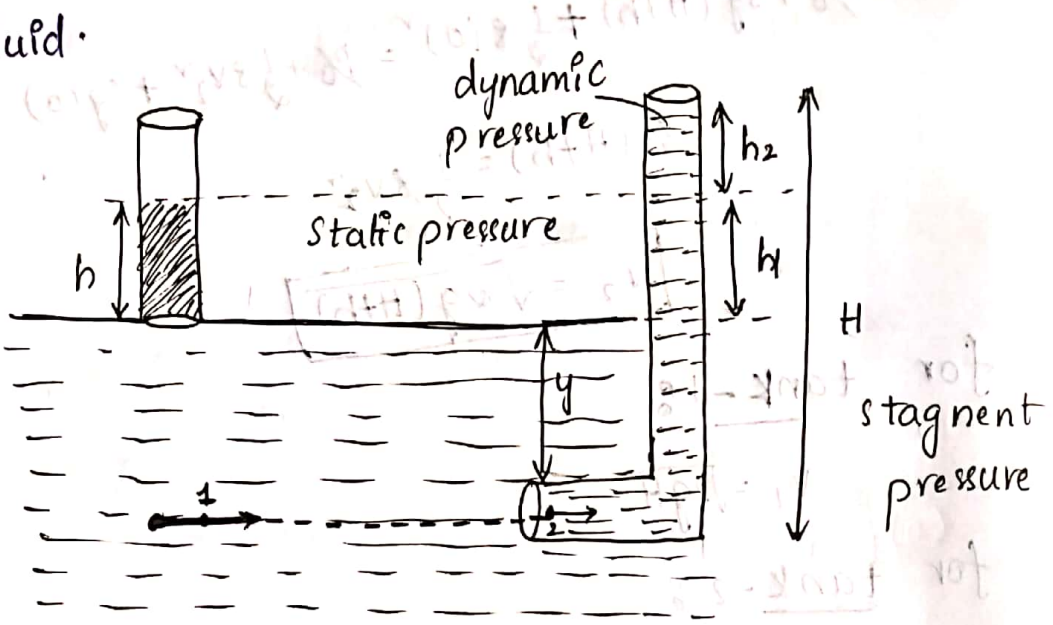
## Pitot tube:

It is used to measure speed of flow of fluid.



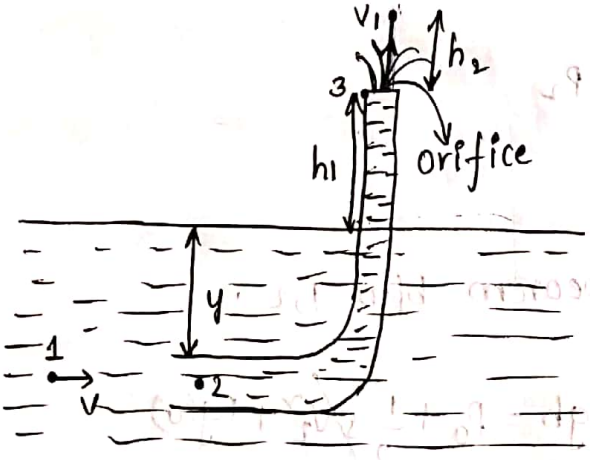

Applying bernoulli's theorm b/w ① & ②

$$P_0 + \rho g y_0 + \frac{1}{2}\rho v^2 + \rho g(0) = P_0 + \rho g(h_2 + h_1 + y) + \frac{1}{2}\rho(0)^2 + \rho g(0)$$

$$\frac{1}{2}\rho v^2 = \rho g(h_2 + h_1)$$

$$\boxed{V = \sqrt{2g(h_1 + h_2)}}$$

Eg:




pt 3 is just outside the orifice.

Bernoulli's theorem b/w ① & ③.

$$P_0 + \rho g y + \frac{1}{2}\rho v^2 + \rho g(0) = P_0 + \rho g(y + h_1) + \frac{1}{2}\rho v_1^2$$

$$\frac{1}{2}\rho v^2 = \rho g h_1 + \frac{1}{2}\rho v_1^2$$

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again, $h_2 = \frac{v_1^2}{2g}$ (vertically projected)

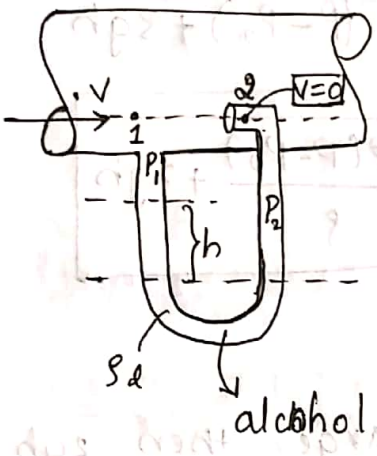
$$\frac{1}{2} \rho v^2 = g h_1 + \frac{1}{2} \times \rho g h_2$$

$$\frac{v^2}{2} = g(h_1 + h_2)$$

$$\frac{v^2}{2g} = h_1 + h_2$$

$$\boxed{h_2 = \frac{v^2}{2g} - h_1} \quad *$$

** pitot tube -




V = Velocity of plane w.r.to air.

bernoulli's theorem b/w 1 & 2.

$$P_1 + \frac{1}{2} \rho_{air} v^2 = P_2$$

$$P_2 - P_1 = \frac{1}{2} \rho_{air} v^2$$

Again,

$$P_2 - P_1 = \rho_a g h$$

$$\rho_a g h = \frac{1}{2} \rho_{air} v^2 \Rightarrow \boxed{v = \sqrt{\frac{2 \rho_a g h}{\rho_{air}}}} \quad **$$

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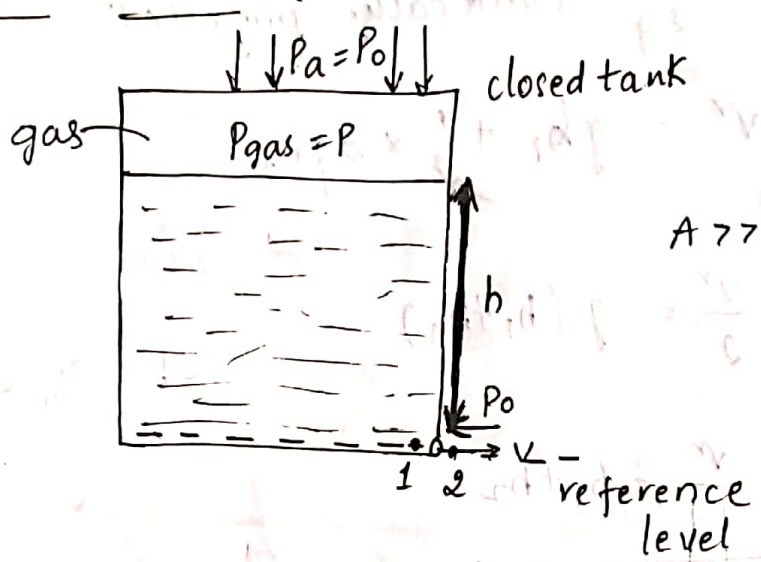
# Toricelli's Theorem




bernoull's theorem,

$$P + \rho gh + \frac{1}{2}\rho(0)^2 + \rho g(0) = P_o + \frac{1}{2}\rho v^2 + \rho g(0)$$

$$\frac{1}{2}\rho v^2 = (P - P_o) + \rho gh$$

$$\boxed{v = \sqrt{\frac{2(P-P_o)}{\rho} + 2gh}}$$

points:

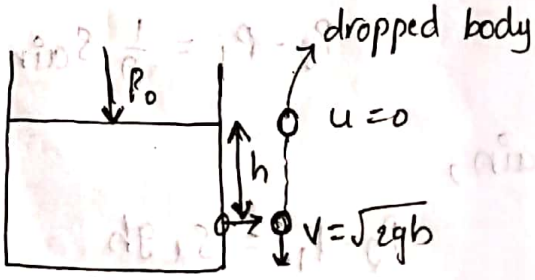
① If $P - P_o$ is very large then $2gh$ can be (rejected) neglected.

$$\boxed{v = \sqrt{\frac{2(P-P_o)}{\rho}}}$$

② $P - P_o = 0$

$$\boxed{P = P_o}$$

then $v = \sqrt{2gh}$.

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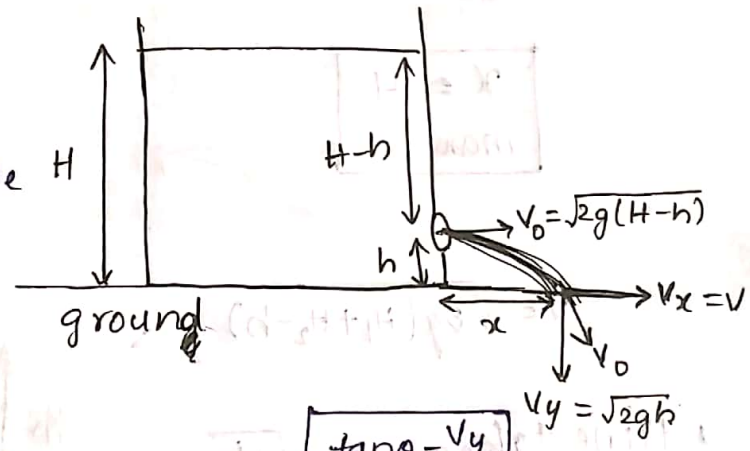
## Different Cases :

① * Speed of liquid while striking the ground,




$$V_0 = \sqrt{V_x^2 + V_y^2}$$

$$\tan\theta = \frac{V_y}{V_x}$$

* time taken by liquid to reach the ground,

$$h = \frac{1}{2} g t^2 \Rightarrow \boxed{t = \sqrt{\frac{2h}{g}}}$$

* Range of liquid,

$$x = vt$$

$$x = \sqrt{2g(H-h)}\, t$$

$$= \sqrt{2g(H-h)} \left(\sqrt{\frac{2h}{g}}\right.$$

$$= \sqrt{2(H-h)h \times 2}$$

$$\boxed{x = 2\sqrt{h(H-h)}}$$

for max range, $\frac{dx}{dh} = 0 \quad \frac{dx^2}{dh} = 0$

$$\frac{d}{dh}[h(H-h)] = 0$$

$$h(-1) + (H-h) = 0$$

$$\boxed{\frac{H}{2} = h}$$

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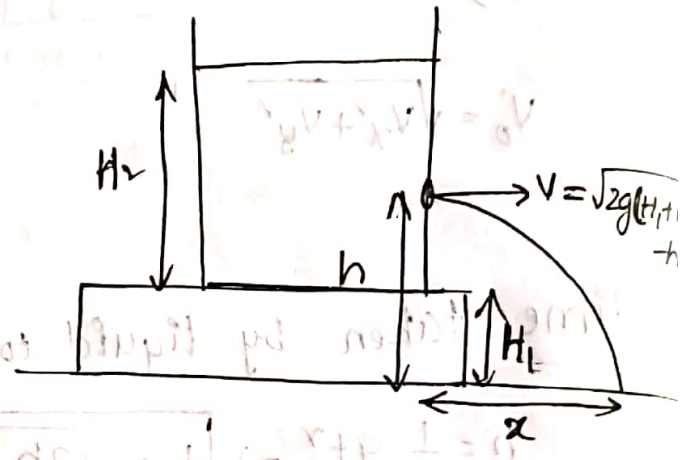
$$x_{max} = 2\sqrt{\frac{H}{2} \times \frac{H}{2}}$$

$$\boxed{x_{max} = H}$$

(2)

$$V = \sqrt{2g(H_1 + H_2 - h)}$$




* time taken, $t = \sqrt{\frac{2h}{g}}$

* range, $x = Vt$

$$= \sqrt{2g(H_1 + H_2 - h)} \times \sqrt{\frac{2h}{g}}$$

$$x = 2\sqrt{h(H_1 + H_2 - h)}$$

$$\frac{dx}{dh} = 0 \Rightarrow h(-1) + (H_1 + H_2 - h)(1) = 0$$

$$\boxed{\frac{H_1 + H_2}{2} = h}$$

$$x_{max} = 2\sqrt{\frac{H_1 + H_2}{2} \times \frac{H_1 + H_2}{2}}$$

$$\boxed{x_{max} = H_1 + H_2}$$

*** If $h < H_1$ then liquid coming from bottom of tank has max range.

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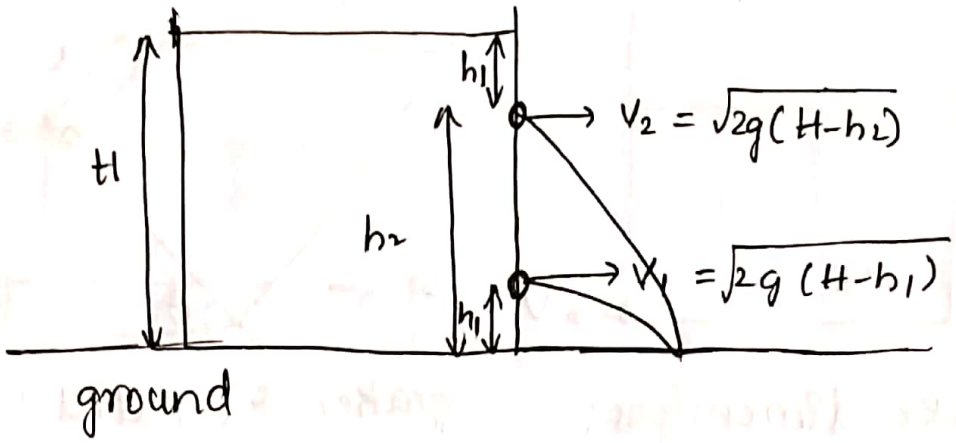
Eg:




for same range,

$$x_1 = x_2$$
$$V_1 t_1 = V_2 t_2$$

$$\sqrt{2g(H-h_1)} \times \sqrt{\frac{2h_1}{g}} = \sqrt{2g(H-h_2)} \times \sqrt{\frac{2h_2}{g}}$$

$$2\sqrt{(H-h_1)(h_1)} = 2\sqrt{(H-h_2)(h_2)}$$

$$Hh_1 - h_1^2 = Hh_2 - h_2^2$$

$$H(h_1-h_2) = h_1^2 - h_2^2$$

$$\boxed{H = h_1 + h_2}$$

for max range,

$$h = H/2$$

$$\boxed{H = \frac{h_1+h_2}{2}}$$

* Rate of flow of liquid,

$$\boxed{Q = AV}$$

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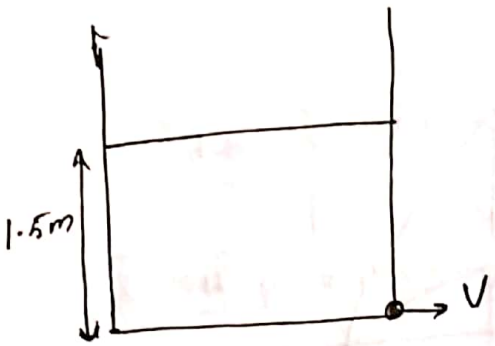
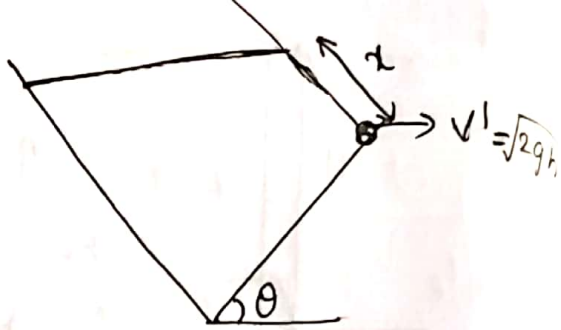
Eg:



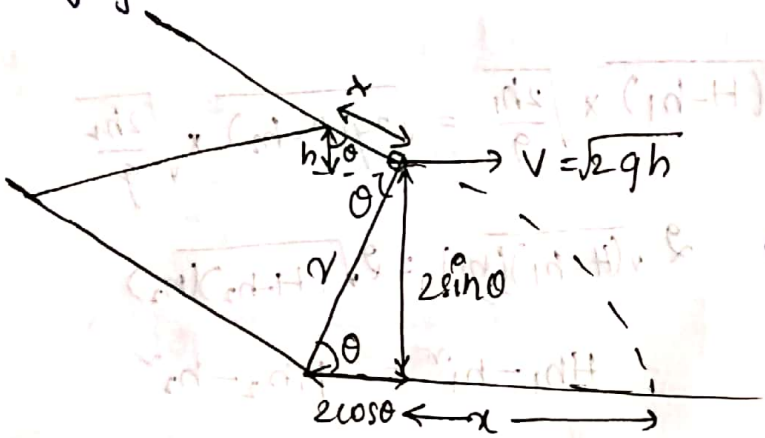




Take dimensions:

$$V = \sqrt{2g(1.5)}$$

$$= \sqrt{3g}$$

beaker is fitted so that liquid is not spplitting. hole is kept at bottom. Find velocity.




$$\cos\theta = \frac{h}{x}$$

$$\boxed{h = x\cos\theta}$$

volum of liquid is const.

$$V_1 = V_2$$

$$2 \times 1.5 = \frac{1}{2} \times (2+x) \times x \times x$$

$$3 = 2 + x$$

$$\boxed{x = 1}$$

$$\therefore h = \cos\theta \quad , \quad t = \sqrt{\frac{2(2\sin\theta)}{g}} = 2\sqrt{\frac{\sin\theta}{g}}$$

$$x = vt$$

$$v = \frac{1}{t}$$

$$\boxed{v = \frac{1}{2}\sqrt{\frac{g}{\sin\theta}}}$$

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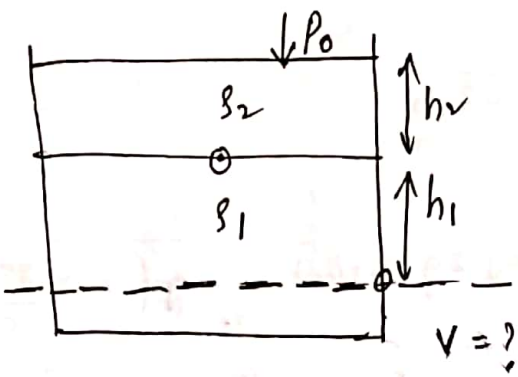
Eg:




M-I :

$$P_0 + \rho_2 g h_2 + \rho_1 g h_1 + \frac{1}{2}\rho (v)^2 + \rho_1 g (v) = P_0 + \rho_1 g (v) + \frac{1}{2}\rho_1 V^2$$

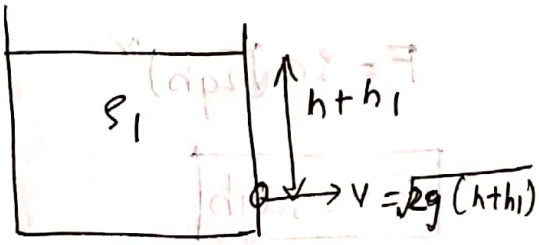
$$\frac{2(\rho_2 h_2 + \rho_1 h_1) g}{\rho_1} = V^2$$

$$\boxed{V = \sqrt{\frac{2g(\rho_2 h_2 + \rho_1 h_1)}{\rho_1}}}$$

M-II :

$$\rho_2 g h_2 = \rho_1 g h$$

$$\boxed{h = \frac{\rho_2}{\rho_1} h_2}$$




$$V = \sqrt{2g\left(\frac{\rho_2 h_2 + \rho_1 h_1}{\rho_1}\right)}$$

$$\boxed{V = \sqrt{\frac{2g(\rho_1 h_1 + \rho_2 h_2)}{\rho_1}}}$$

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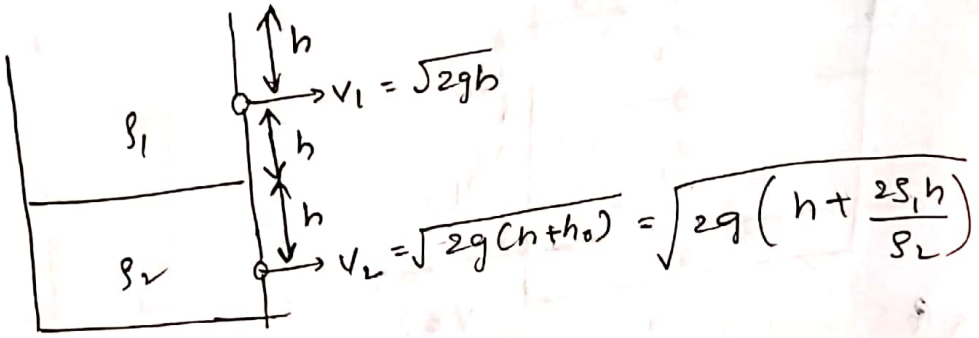
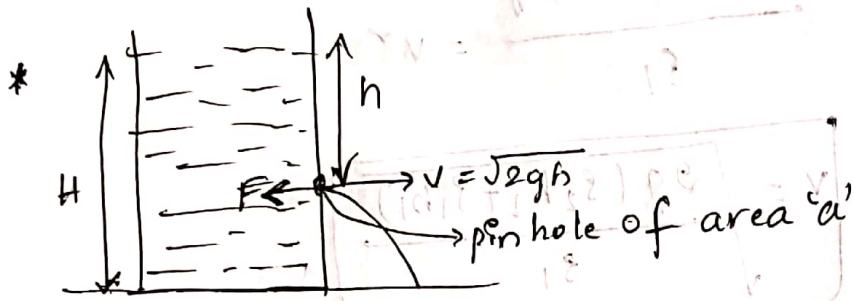
Eg:




$$\rho_1 g(2h) = \rho_2 g h_0$$

$$h_0 = \frac{2\rho_1 h}{\rho_2}$$

Force on the vessel due to ejecting liquid:




Force on the ejecting fluid,

$$F = v\left(\frac{dm}{dt}\right)$$

$$F = v(a v \rho)$$

$$\boxed{F = \rho a v^2}$$

$$F = \rho a(\sqrt{2gh})^2$$

$$\boxed{F = 2\rho agh}$$

∴ force on vessel due to ejecting liquid,

$$\boxed{F = \rho a v^2 = 2\rho agh}$$

⇒ acceleration of vessel, $\boxed{a_0 = \frac{F}{M}}$

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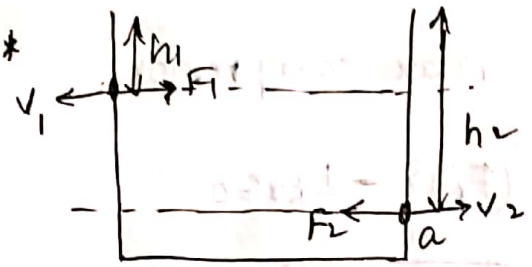



$$v_1 = \sqrt{2gh_1} \; ; \; v_2 = \sqrt{2gh_2}$$

∴ net force, $F = F_2 - F_1$

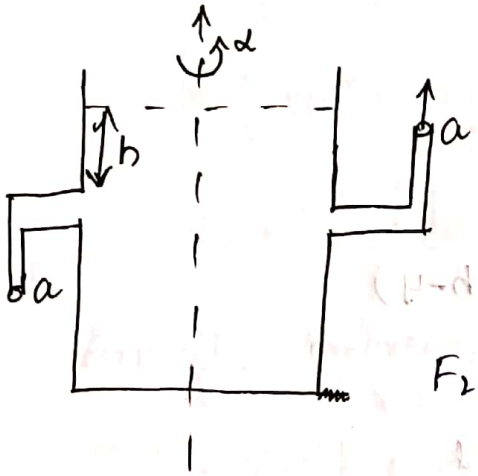
$$= \rho a (v_2^2 - v_1^2)$$

$$\boxed{F = 2\rho a g (h_2 - h_1)}$$

* If $h = 0$,

$$\boxed{F = 0}$$

*

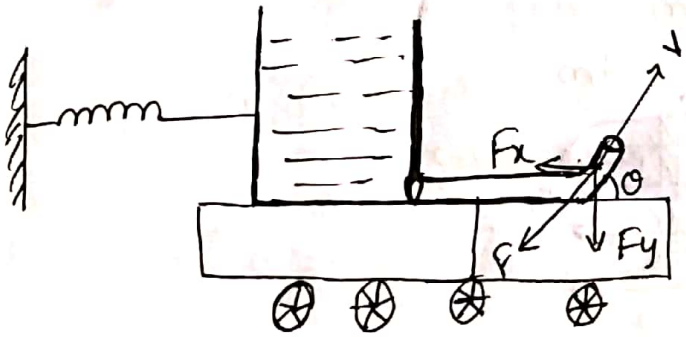



$F_2 = \rho a v^2 \odot$

$F_1 = \rho a v^2 \otimes$

$$\boxed{\begin{aligned} \bar{\tau} &= \bar{r}_1 \times \bar{F}_1 + \bar{r}_2 \times \bar{F}_2 \\ \bar{\tau} &= I\bar{\alpha} \end{aligned}}$$

*

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In eqm,

$$kx = F_x$$

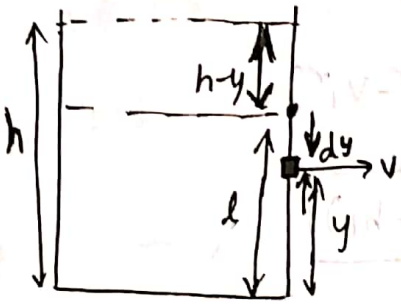
$$\boxed{x = \frac{F_x}{k}}$$

for max compression,

$$(F_x)x - \frac{1}{2}kx^2 = 0$$

$$\Rightarrow \boxed{x = \frac{2F_x}{k}}$$

for rectangular slit:



rectangular slit.

$$v = \sqrt{2g(h-y)}$$

now force,

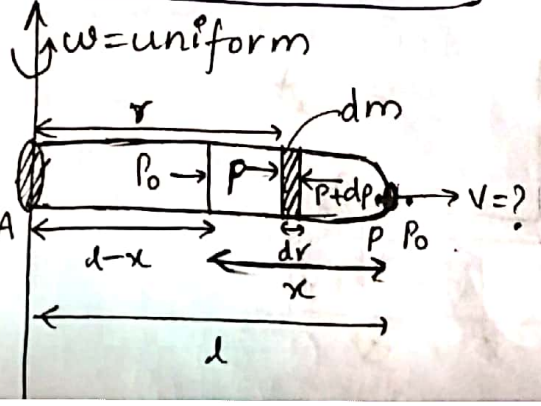
$$dF = \rho(dA)v^2$$

$$dF = \rho(b\,dy)\,2g(h-y)$$

∴ Total force,

$$F = \int dF = \int_{y=0}^{y=l} dF$$

$$\boxed{F = 2\rho g b\left[hl - \frac{l^2}{2}\right]}$$

Eg: 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net force,

$$(P+dp)A - PA = (dm)r\omega^2$$

$$(dp)A = A\,dr\,\rho r(\omega^2)$$

$$\int_{P_0}^{P} dp = \int_{(\ell-x)}^{\ell} \rho\omega^2 r \cdot dr$$

$$P - P_0 = \frac{1}{2}\rho\omega^2 [r^2]_{\ell-x}^{\ell}$$

$$P_0 - P_0 = \frac{1}{2}\rho\omega^2 (\ell^2 - (\ell-x)^2)$$

$$= \frac{1}{2}\rho\omega^2 (x(2\ell - x))$$

$$P - P_0 = \frac{1}{2}\rho\omega^2 (2\ell x - x^2)$$

$$P = P_0 + \frac{1}{2}\rho\omega^2 (2\ell x - x^2)$$

now bernoull's theorem,

$$P + \frac{1}{2}\rho(0)^2 = P_0 + \frac{1}{2}\rho V^2$$

$$\boxed{P = P_0 + \frac{1}{2}\rho V^2}$$

$$P_0 + \frac{1}{2}\rho\omega^2 (2\ell x - x^2) = P_0 + \frac{1}{2}\rho V^2$$

$$\omega^2 (2\ell x - x^2) = V^2$$

$$\boxed{V = \omega\sqrt{2\ell x - x^2}}$$

$x \longrightarrow$ from hole.

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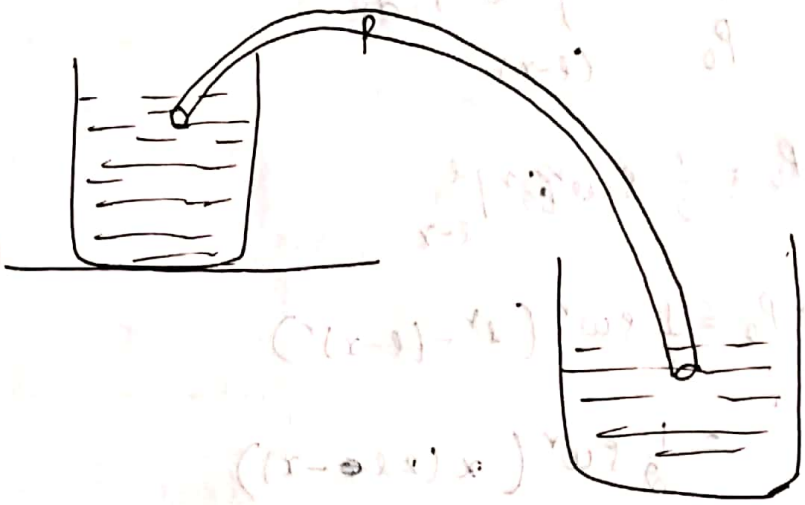
# Siphon tube

* It works under "gravity".
* It pumps the liquid from higher level to lower level without any help of machin.



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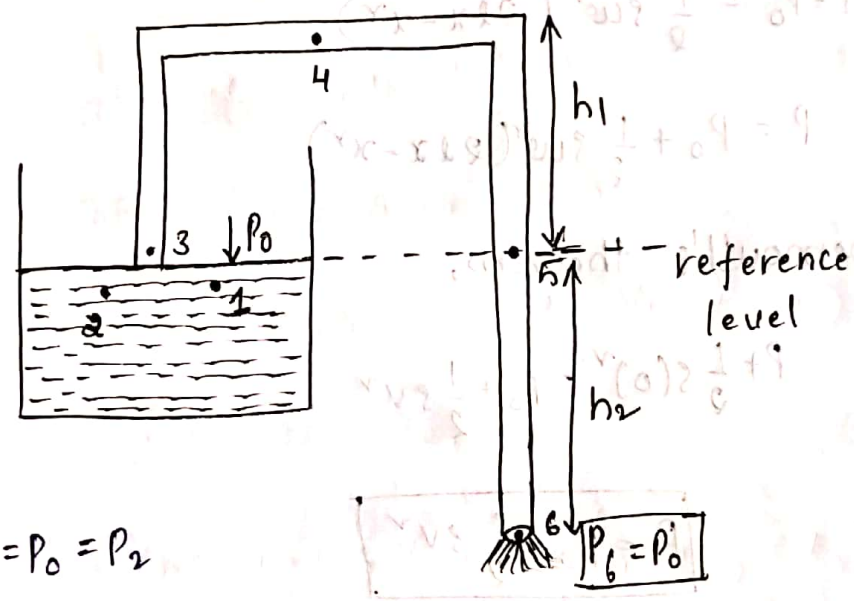


$P_1 = P_0 = P_2$

$V_2 = 0$

$V_3 = V_4 = V_5 = V_6 = V$ (say)

Bernoulli's eqn,

$$P_0 = P_3 + \frac{1}{2} \rho V_3^2 = P_4 + \frac{1}{2} \rho V_4^2 + \rho g h_1 = P_5 + \frac{1}{2} \rho V_5^2 + \rho g(0) =$$

$$P_6 + \frac{1}{2} \rho V_6^2 - \rho g h_2$$

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Clearly, $\boxed{P_3 = P_5}$

$$P_4 = P_0 - \frac{1}{2}\rho v^2 - \rho g h_1 \quad \text{— (1)}$$

[or] $\boxed{P_4 = P_3 - \rho g h_1}$

as $h_1 \uparrow \Rightarrow P_4 \downarrow$

and $P_0 = P_6 + \frac{1}{2}\rho v^2 - \rho g h_2$

$\boxed{v = \sqrt{2gh_2}}$ & $\boxed{h_2 = +ve}$

and $P_5 = P_0 - \rho g h_2$

i.e, $\boxed{P_5 < P_0}$

Again, $P_4 + \frac{1}{2}\rho v^2 + \rho g h_1 = P_0 + \frac{1}{2}\rho v^2 - \rho g h_2$

$$P_4 = (P_0 - \rho g (h_1 + h_2)$$

$$\boxed{h_1 + h_2 = \frac{P_0 - P_4}{\rho g}}$$

clearly, $h_1 + h_2 < \frac{P_0}{\rho g}$

& $\boxed{(h_1 + h_2)_{max} = \frac{P_0}{\rho g}}$ [$P_4 = 0$]

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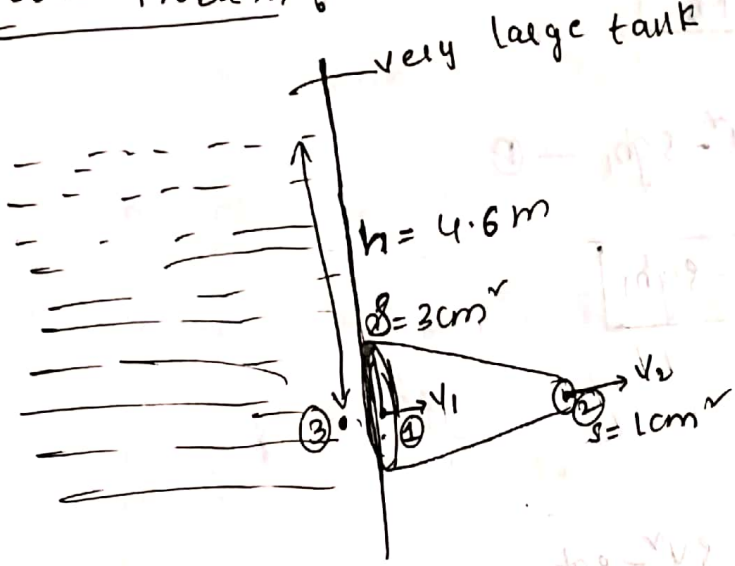
Irodov Problem 6




Find the horizontal force required to pull the tube.

Eqn of continuity,

$$A V_1 = s V_2 \quad -(1)$$

Bernoullis eqn b/w 1 & 2,

$$P_{in} + \frac{1}{2} \rho V_1^2 = P_{atm} + \frac{1}{2} \rho V_2^2$$

$$\Rightarrow P_{in} - P_{atm} = \frac{1}{2} \rho (V_2^2 - V_1^2) \quad -(2)$$

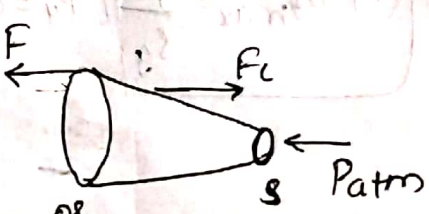
Again b/w (3) & (2),

$$P_{atm} + \rho g h + \frac{1}{2} \rho (0)^2 = P_{atm} + \frac{1}{2} \rho V_2^2$$

$$\boxed{V_2 = \sqrt{2gh}} \quad -(3)$$

Now,

FBD of tube:



projection area = A − s

$$F = F_c - P_{atm}(A - s) \quad -(4)$$

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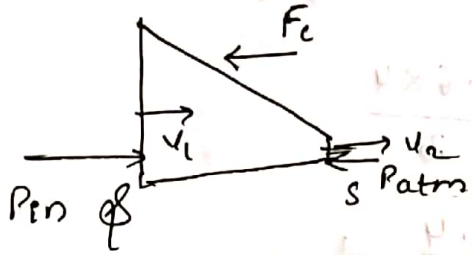
FBD of liquid




on liquid,

when $\frac{dm}{dt} = \rho A V_1 = s \rho V_2$

$$F_{net} = \frac{d\bar{P}}{dt}$$

$$P_{in} A - P_{atm} s = \frac{dm}{dt}(V_2 - V_1)$$

$$F_c = P_{in} A - P_{atm} s = \frac{dm}{dt}(V_2 - V_1) \quad -(5)$$

from eqns (4) & (5)

$$F = P_{in} A - P_{atm} s - s \rho V_2 (V_2 - V_1) - P_{atm} A + P_{atm} s$$

$$= (P_{in} - P_{atm}) A - s \rho V_2^2 \left(1 - \frac{V_1}{V_2}\right)$$

$$= \frac{1}{2} \rho (V_2^2 - V_1^2) A - s \rho V_2^2 \left(1 - \frac{V_1}{V_2}\right)$$

$$= \frac{1}{2} \rho V_2^2 \left(1 - \frac{V_1^2}{V_2^2}\right) A - s \rho V_2^2 \left(1 - \frac{V_1}{V_2}\right)$$

$$= \frac{1}{2} \rho V_2^2 \left(1 - \frac{V_1}{V_2}\right)\left(\left(1 + \frac{V_1}{V_2}\right) A - 2s\right)$$

$$= \frac{1}{2} \rho (2gh)\left(1 - \frac{s}{A}\right)\left(\left(1 + \frac{s}{A}\right) A - 2s\right)$$

$$= \frac{1}{2} \rho (2gh)\left(\frac{A - s}{A}\right)(A + s - 2s)$$

$$= \rho g h \left(\frac{A - s}{A}\right)(A - s) \Rightarrow \boxed{F = \frac{\rho g h (A - s)^2}{A}}$$

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Ans:

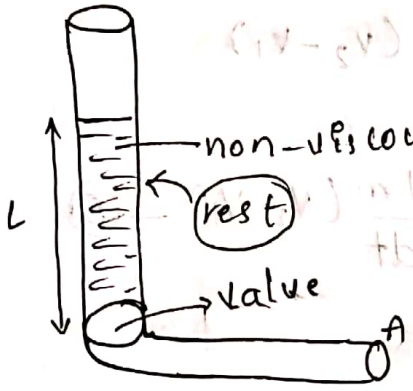
$$F = \frac{10^3 \times 10 \times 4.6 \times (2)^2 \times (10^{-4})^2}{3 \times 10^{-4}}$$

$$= \frac{4.6 \times 4}{2}$$

$$= \frac{18.4}{3}$$

$$\boxed{F = 6.13N}$$

Eg:




When valve is opened, Find the speed of fluid when all of it is in horizontal section.

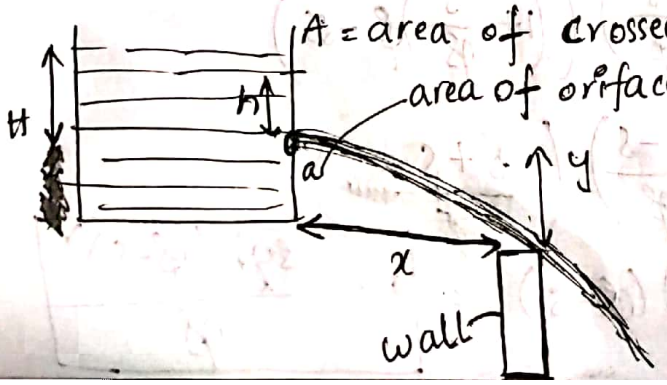
$$TE = \text{const}$$

$$P \cdot E = K \cdot E$$

$$Alg\left(\frac{L}{2}\right) = \frac{1}{2} AL \rho v^2$$

$$\boxed{v = \sqrt{gL}}$$

Eg:

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Find the height of liquid above the origin so that the jet just ceases to cross the wall. Also find the "time interval".

$$v = \sqrt{2gh}$$

& time taken by the liquid to hit the wall.

$$= \sqrt{\frac{2y}{g}}.$$

$$\& \quad x = \sqrt{2gh}\sqrt{\frac{2y}{g}}$$

$$\Rightarrow x = 2\sqrt{hy}$$

$$\boxed{h = \frac{x^2}{4y}}$$

* Now, time taken by liquid to fall from $H_1$ to $H_2$ is, (in tank)

$$t = \frac{A}{a}\sqrt{\frac{2}{g}}\left(\sqrt{H_1} - \sqrt{H_2}\right)$$

$$t = \frac{A}{a}\sqrt{\frac{2}{g}}\left(\sqrt{H} - \sqrt{h}\right)$$

$$\boxed{t = \frac{A}{a}\sqrt{\frac{2}{g}}\left(\sqrt{H} - \sqrt{\frac{x^2}{4y}}\right)}$$

Eg: At a height h below the bottom, a small height is made if the stream coming from hole strikes inclined plane normally. Calculate h. 8.33 m

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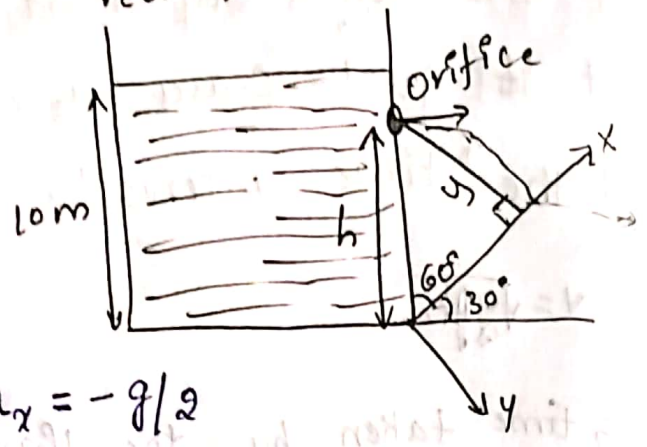
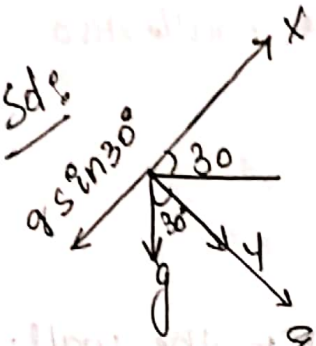
rectangular tank

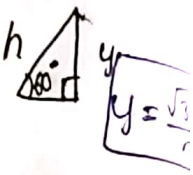



Sol:

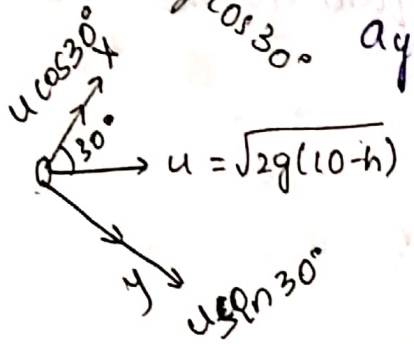



$a_x = -g/2$

$a_y = \sqrt{3}/2\, g$




$y = \sqrt{3}$




$u_x = u\cos 30°$

$= \sqrt{2g(10-h)} \times \frac{\sqrt{3}}{2}$

$= \sqrt{\frac{3g(10-h)}{2}}$

$u_y = u \sin 30° = \sqrt{\frac{g(10-h)}{2}}$

$v_x = 0$

$u_x + a_x t = 0$

$\sqrt{\frac{3g(10-h)}{2}} - \frac{g}{2} t = 0 \Rightarrow t = \sqrt{\frac{3g(10-h)}{2}} \times \frac{4}{g} = \sqrt{\frac{6(10-h)}{g}}$

$y = u_y t + \frac{1}{2} a_y t^2 \Rightarrow \frac{\sqrt{3}h}{2} = \sqrt{\frac{g(10-h)}{2}} \times \frac{6(10-h)}{g} + \frac{1}{2} \times \frac{\sqrt{3}}{2} \times \frac{3}{g} 6(10$

$\frac{\sqrt{3}h}{2} = 2\sqrt{3}(10-h) + \frac{3\sqrt{3}}{2}(10-h)$

$h = 2(10-h) + 3(10-h)$

$h = 20 - 2h + \frac{3}{10}(10-h)$

~~$20h = 200 + 30 - 3h$~~

~~$23h = 230$~~

$3h = 20 + 30 - 3h$

$6h = 50$

$h = \frac{50}{6}$

$h = 8.33$

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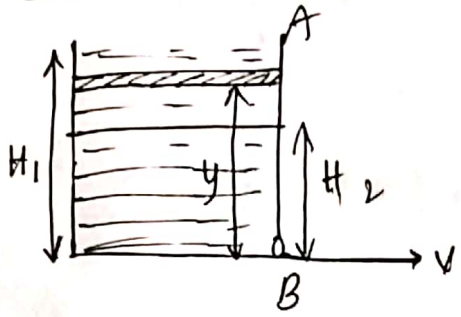
## Time taken to empty the tank:




In intermediate state,

$$v = \sqrt{2gy}$$

∴ rate of flow of liquid through hole.

$$\frac{dv}{dt} = av = a\sqrt{2gh}$$

We have,

$$-A \cdot \frac{dy}{dt} = a\sqrt{2gh}$$

$$-\int_{H_1}^{H_2} \frac{dy}{\sqrt{y}} = \int_0^t \frac{a}{A}\sqrt{2g}\,(dt)$$

$$-2\left[\sqrt{y}\right]_{H_1}^{H_2} = \frac{a}{A}\sqrt{2g}\,t$$

$$\frac{2}{\sqrt{2}}\left(\sqrt{H_1} - \sqrt{H_2}\right) = \frac{a}{A}\sqrt{g}\,t$$

$$\boxed{\frac{A}{a}\sqrt{\frac{2}{g}}\left(\sqrt{H_1} - \sqrt{H_2}\right) = t}$$

Clearly, $t \propto \sqrt{H_1} - \sqrt{H_2}$

time taken to empty first half of the tank,

here, $H_1 = H$ & $H_2 = H/2$.

$$t_1 = \frac{A}{a}\sqrt{\frac{2}{g}}\sqrt{H}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

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Time taken to empty the next half of the tank,

here $H_1 = \frac{H}{2} \Rightarrow H_2 = 0$

$$t_2 = \frac{A}{a}\sqrt{\frac{2}{g}}\left(\frac{\sqrt{H}}{\sqrt{2}}\right)$$

Clearly $t_1 < t_2$

$$\boxed{\frac{t_1}{t_2} = \sqrt{2}-1}$$

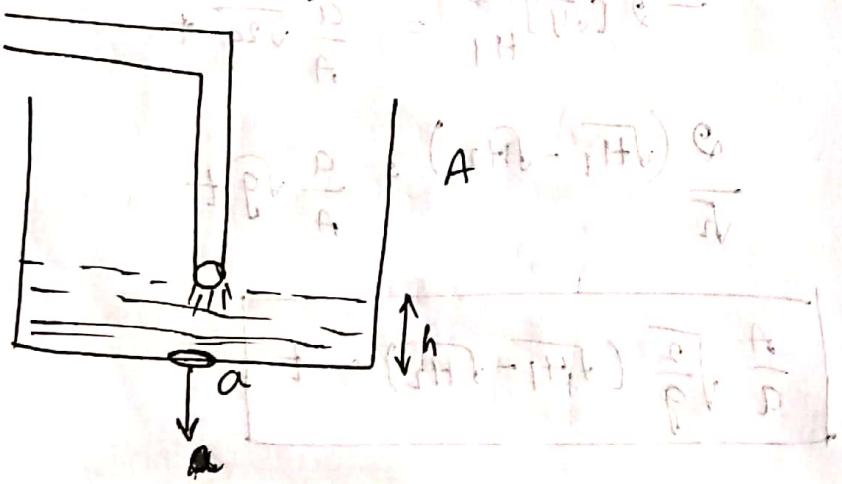
time taken to empty the $\frac{3}{4}$th of tank.

here $H_1 = H$, $H_2 = H/4$

$$t = \frac{A}{a}\sqrt{\frac{2}{g}}\left[\sqrt{H} - \sqrt{\frac{H}{4}}\right]$$

$$= \frac{A}{a}\sqrt{\frac{2}{g}}\sqrt{H}\left(\frac{1}{2}\right)$$

Eg:




∴ when liquid level remains same,

$$Q = av$$

$$Q = a\sqrt{2gh}$$

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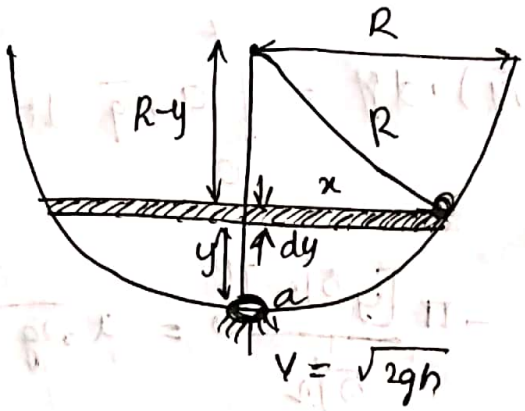
$$h_{max} = \frac{Q^2}{2a^2 g h}$$

Eg: spherical tank:




rate of flow of liquid through hole,

$$\frac{dV}{dt} = a\sqrt{2gy}$$

$$-\pi x^2 \left(\frac{dy}{dt}\right) = a\sqrt{2gy}$$

If $-\frac{dy}{dt}$ = const (rate of fall in liquid level is const)

$$x^2 \propto \sqrt{y}$$

$$\boxed{y = kx^4}$$

Again from eqn. ①,

$$-\pi x^2 \cdot \frac{dy}{dt} = a\sqrt{2gy}$$

from figure, $(R-y)^2 + x^2 = R^2$

$$\boxed{x^2 = 2Ry - y^2}$$

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$$-\int_{y=R}^{y=0} \pi \frac{(2Ry - y^2)\, dy}{\sqrt{y}} = \int_0^t a\sqrt{2g}\, dt$$

after solving we get 't'.

$$\int_{y=0}^{R} \pi (2R\sqrt{y} - y\sqrt{y}) \cdot dy = \int_0^t a\sqrt{2g} \cdot dt$$

$$\pi 2R \frac{[y^{3/2}]_0^R}{3/2} - \pi \frac{[y^{5/2}]_0^R}{5/2} = a\sqrt{2g} \cdot t$$

$$\frac{4\pi}{3} R \cdot R^{3/2} - \frac{2}{5}\pi R^{5/2} = a\sqrt{2g}\, t$$

$$a\sqrt{2g}\, t = \left(\frac{4\pi}{3} - \frac{2\pi}{5}\right) R^{5/2}$$

$$a\sqrt{2g}\, t = \frac{20\pi - 6\pi}{15} \times R^{5/2}$$

$$a\sqrt{2g}\, t = \frac{14\pi}{15} \times R^{5/2}$$

$$\boxed{t = \frac{14\pi R^{5/2}}{15a\sqrt{2g}}}$$

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